

MA2860 Elementary Differential Equations Notes and Problems

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*This text provides notes and problems that build on concepts in calculus for MA2860.
It is open source and meant for students and instructors to use, adapt, and contribute.
The computational component primarily consists of Sage code, and online symbolic
calculators are provided for supplemental study.*

Contents

Preface

THIS TEXT PROVIDES NOTES AND PROBLEMS that build on concepts in calculus for *MA2860 Elementary Differential Equations* at the University of Toledo. The book is a companion to various textbooks including:

- Zill, Dennis G. *A First Course in Differential Equations with Modeling Applications*, 10th ed. Cengage Learning, 2012.
- Boyce, William E., DiPrima Richard C. *Elementary Differential Equations*, 8th ed. John Wiley & Sons Inc., 2004.

Although it can be used with other texts as well.

IN MAKING THIS STARTING RESOURCE OF LESSON PLANNING **open source**, I hope that each user (with code access) will contribute with notes, corrections, and exercises (with solutions). Users without code access may also email me any corrections. *Professors may email to request code access*. Each week has a portion of the assignment reserved for preparatory homework and another portion for further discussion or exam preparation.

KNOWING THAT ALL PEOPLE ARE MADE IN THE IMAGE OF GOD WITH A POTENTIAL FOR THE HIGHEST GOOD, I hope that all users of this material will work to make this work of highest quality for all educators.

The idea for this work came from my professor in Dynamical Systems during my grad school years, Dr. Marko Budišić at Clarkson University. He encouraged students to compile class notes into one document for easy review. All credit to him.

This text uses a Differential Equations workbook compiled by Clarkson University faculty as a starting point, although the content is mostly different: K. Black, D. White, G. Yao, J. Martin, and M. Budišić.

THE COMPUTATIONAL COMPONENT of this book primarily consists of **SAGEMATH** code. No prerequisite experience is needed, as problems can be completed by following links and editing code in the browser. Essential manual for **SAGE-MATH** is the textbook:

- Bard, G.V. *Sage for Undergraduates*. American Mathematical Society, 2015. Available at no cost in PDF form: <http://www.gregorybard.com/Sage.html>.

ONLINE TOOLS for studying involve symbolic calculators at links: **CAUTION: do not rely heavily on them while studying!**

- <https://www.symbolab.com/solver/>
- <http://www.emathhelp.net/calculators/>

THE GOAL of this class is to teach the **principles** behind the calculations, so they can be built on and extended in future courses or practice, not convert people into human calculators. Therefore, emphasis is on algorithms, and memorization or **reliance on formulas and calculators is strongly discouraged**.

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Chapter 1: Introduction to Differential Equations

We begin our journey into the world of elementary differential equations by looking at some review.

Section 1.1: Review: Precalculus and Calculus

In this section, you will review the techniques from Calculus classes that you should absolutely know in and out: sketching functions, implicit derivatives, and three integration techniques: substitution, by-parts, and partial fractions.

a) Properties of exponents and logarithms For $a > 0$, $a \neq 1$, and integers m, n and $x, y > 0$. Then

- $a^m \cdot a^n = a^{m+n}$
- $a^{-n} = \frac{1}{a^n}$
- $\ln x^r = r \cdot \ln x$
- $\frac{a^m}{a^n} = a^{m-n}$, ($m > n$)
- $\ln 1 = 0$
- $\ln(e^x) = x, e^{\ln(x)} = x$
- $(a^m)^n = a^{mn}$
- $\ln e = 1$
- Should be familiar with graphs of the natural exponential and natural logarithm.
- $(ab)^n = a^n b^n$
- $\ln xy = \ln x + \ln y$
- $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
- $\ln\left(\frac{x}{y}\right) = \ln x - \ln y$
- $a^0 = 1$

b) Trigonometric Properties

- $\sin^2 \theta + \cos^2 \theta = 1$
- $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
- $1 + \tan^2 \theta = \sec^2 \theta$
- $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$
- $1 + \cot^2 \theta = \csc^2 \theta$
- $\sin(2\theta) = 2 \sin \theta \cos \theta$
- $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$
- $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$
- $\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$
- Need to know the values of the trig functions for multiples of $\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$
- Need to be familiar with polar coordinates: $x = r \cos(\theta), y = r \sin(\theta)$.

c) **Product rule:** If $f(x)$ and $g(x)$ are differentiable functions:

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$$

Example:

$$f(x) = x^2 \sin x$$

$$f'(x) = (2x)(\sin x) + (x^2)(\cos x) = 2x \sin x + x^2 \cos x$$

d) **Chain rule:** If $y = f(g(x))$, then:

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

Example:

$$y = (3x^2 + 1)^5$$

$$\frac{dy}{dx} = 5(3x^2 + 1)^4 \cdot (6x) = 30x(3x^2 + 1)^4$$

e) **Integration by Substitution** If $u = g(x)$, then

$$\int f(g(x))g'(x) dx = \int f(u) du$$

Example:

$$\int 2x \cos(x^2) dx$$

Let $u = x^2$, so $du = 2x dx$:

$$\int 2x \cos(x^2) dx = \int \cos(u) du = \sin(u) + C = \sin(x^2) + C$$

f) **Integration by Parts** From the product rule:

$$\int u dv = uv - \int v du$$

Example:

$$\int xe^x dx$$

Let $u = x \implies du = dx$ and $dv = e^x dx \implies v = e^x$:

$$\int xe^x dx = xe^x - \int e^x dx = xe^x - e^x + C = (x - 1)e^x + C$$

g) **Partial Fraction Decomposition:** If the integrand is a rational function $\frac{P(x)}{Q(x)}$, decompose into simpler fractions.

Example:

$$\int \frac{1}{x^2 - 1} dx$$

Factor denominator:

$$\frac{1}{x^2 - 1} = \frac{1}{(x - 1)(x + 1)} = \frac{A}{x - 1} + \frac{B}{x + 1}$$

Multiply through:

$$1 = A(x + 1) + B(x - 1)$$

Setting $x = 1 \implies 1 = 2A \implies A = \frac{1}{2}$ Setting $x = -1 \implies 1 = -2B \implies B = -\frac{1}{2}$

So:

$$\frac{1}{x^2 - 1} = \frac{1}{2} \cdot \frac{1}{x - 1} - \frac{1}{2} \cdot \frac{1}{x + 1}$$

Now integrate:

$$\begin{aligned}\int \frac{1}{x^2 - 1} dx &= \frac{1}{2} \ln |x - 1| - \frac{1}{2} \ln |x + 1| + C \\ &= \frac{1}{2} \ln \left| \frac{x - 1}{x + 1} \right| + C\end{aligned}$$

In-Class Exercise

Exercise 1.1.1

Simplify the following expressions:

(a) $e^{4 \ln |x+1|} =$

(b) $\exp \left[-\frac{1}{2} \ln(x^2) \right] =$

Exercise 1.1.2

Evaluate the following integrals using substitution, integration by parts, and partial fractions, as appropriate.

(a) $\int_0^x x e^{x^2} dx$ ¹

¹ Hint: Substitution

(b) $\int_1^2 x \sin(\pi x) dx$ ²

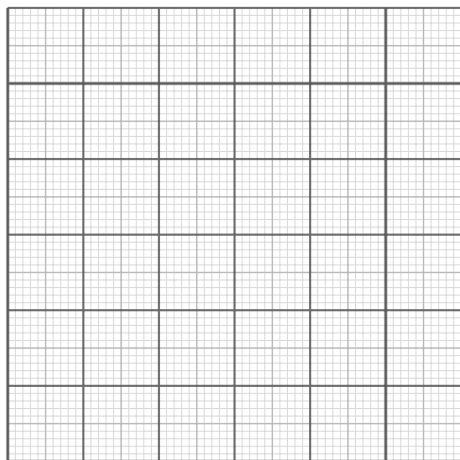
² Hint: Integration by parts

(c) $\int \frac{x}{x^2 + x - 6} dx$

Exercise 1.1.3

Sketch the graph of the function $f(x) = (x^2 - 4x + 3)e^{-2x}$ ³

³ Determine any minima/maxima, x-intercepts and y-intercepts, and asymptote $x \rightarrow +\infty$ for an accurate sketch.

**Exercise 1.1.4**

If $y(x) = e^x \sin(2x)$, evaluate the expression $y''(x) - 2y'(x) + 5y(x)$. Compute the required derivatives⁴ first, and then substitute.

⁴ We use y' to mean the derivative with respect to the argument, in this case dy/dx . When the function depends on time, say $x(t)$, then we sometimes additionally use the dot-notation $\dot{x} = dx/dt$.

Discussion Problems**Exercise 1.1.5**

Compute the derivative $y'(x)$ if the following relation between y and x holds:

$$\ln|y^2 + 2| + x^3 \sin(2x) = C, \text{ a constant.}$$

Exercise 1.1.6

What is the domain of definition of each of the two functions:

(a) $\ln(2x + 1)$

(b) $\ln|2x + 1|$

Exercise 1.1.7

Split the following rational functions into partial fractions:

(a) $\frac{1}{x^2+4x+3}$

(b) $\frac{2x}{x^2+4x+3}$

(c) $\frac{x^2+4x}{x^2+4x+3}$

Exercise 1.1.8

A curve is defined by the implicit expression $\cos(2x) + \frac{1}{2}y^2 = 1$. Calculate **all** the points at which the tangent⁵ to this curve is

⁵ Hint: find the implicit derivative $y'(x)$

(a) $y'(x) \rightarrow \infty$, (b) $y'(x) \rightarrow 0$

(a) Vertical.

(b) Horizontal.

Exercise 1.1.9

Evaluate the integrals:⁶

(a) $\int \ln(x) dx$

(c) $\int \frac{x}{x+2} dx$

(e) $\int \frac{x^2}{x^2+x-6x} dx$

(b) $\int xe^x dx$

(d) $\int \frac{2x}{x^2-4} dx$

(f) $\int e^x \sin(\pi x) dx$

⁶ Hint: (a) integration by parts, (b) substitution, (d) substitution/partial fraction, (f) by parts

Section 1.2: Basic Terminology & History**Basic Terminology**

Definition 1.2.1. A differential equation is an equation that contains one or more derivatives of an unknown function.

A differential equation is an ordinary differential equation (ODE) if it involves an unknown function of only one independent variable. Example:

$$\frac{dy}{dt} = -ky.$$

A differential equation involving partial derivatives of a function of two or more independent variables is a partial differential equation. Example:

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}.$$

Differential equations describe how one quantity changes in relation to other variables.

For a standard algebraic equation, the solution is a number, but for a differential equation, the solution is a function.

- $y(x)' = y + x^2 + 3$ is a differential equation. The unknown function is y depending on the variable x and the derivative of y is involved in the equation.
- $\cos x = -1$ is not a differential equation. There is no unknown function.
- $y = x^2 - 2x + 1$ is not a differential equation since there is no derivative involved.
- $y'(x) = y$ is a differential equation, with unknown function y with its derivative depending on the variable x .
- $\frac{d}{dx}(\tan x) = \sec^2 x$ is not a differential equation since there is no unknown function

Definition 1.2.2. The order of a differential equation is the highest derivative that appears in it.

- $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$ is a second-order ODE.
- $y' + 3x^2 = 1$ is a first-order ODE.
- $y'' - x^2y' + y = \cos x$ is a second-order ODE.
- $y^4 - xy' + y^2 = 3$ is a fourth-order ODE.

Definition 1.2.3 (Linearity). An ODE is linear if the dependent variable y and its derivatives $y', y'', \dots, y^n$ all appear only to the first power and not multiplied together. Example:

$$a_2(x)y'' + a_1(x)y' + P(x)y = Q(x).$$

An ODE is nonlinear if it contains products, powers, or nonlinear functions of the dependent variable or its derivatives. Example:

$$\frac{dy}{dx} = y^2 + \sin(y).$$

Definition 1.2.4. A differential equation is homogeneous if all terms depend on the unknown function or its derivatives. If there is a term that does not, it is nonhomogeneous.

- **Homogeneous:** $\frac{dy}{dx} + p(x)y = 0$
- **Nonhomogeneous:** $\frac{dy}{dx} + p(x)y = q(x)$

Definition 1.2.5. An Initial Value Problem (IVP) of an n -th order differential equation is the differential equation together with an initial condition of y and the first $n - 1$ derivatives specified at a single point. Example:

$$\frac{dy}{dt} = -ky, \quad y(0) = y_0.$$

Definition 1.2.6. A Boundary Value Problem (BVP): is a differential equation together with conditions specified at two or more points. Example:

$$\frac{d^2y}{dx^2} + y = 0, \quad y(0) = 0, \quad y(\pi) = 0.$$

Definition 1.2.7. A solution of a differential equation is any function that satisfies the differential equation and its initial conditions on some open interval.

Note that solutions should be accompanied by intervals, since they can provide important information about the solution.⁷ The solution is general if no initial condition is provided.

⁷ If y is a solution to the differential equation, differentiate to obtain y' , then substitute y and y' into the differential equation to verify.

Example 1.2.8.

$$\frac{dy}{dx} = 2x \quad \Rightarrow \quad y = x^2 + C$$

is the general solution. If $y(0) = 3$, then $y = x^2 + 3$ is the particular solution.

Exercise 1.2.1

Determine whether the given function is a solution of the differential equation.

- a) Determine whether $y = e^{2x} + 1$ is a solution of

$$\frac{dy}{dx} - 2y = -2.$$

b) Determine whether $y = \sin(x) + \cos(x)$ is a solution of

$$\frac{d^2y}{dx^2} + y = 0.$$

c) Determine whether $y = x^2$ is a solution of

$$\frac{dy}{dx} - 2x = 0.$$

d) Determine whether $y = e^{2x}$ is a solution of

$$\frac{dy}{dx} - 2y = 0.$$

e) Is $x = e^{4t}$ a solution to

$$x''' - 12x'' + 48x' - 64x = 0?$$

f) Is $y = \sin t$ a solution to

$$\left(\frac{dy}{dt}\right)^2 = 1 - y^2?$$

g) Verify that $x = Ce^{-2t}$ is a solution to $x' = -2x$. Find C to solve for the initial condition $x(0) = 100$.

h) Determine whether $y = x^{-\frac{3}{2}}$, $y = x^{-\frac{1}{2}}$, $y = -9x^{-\frac{3}{2}}$, and $y = 7x^{-\frac{1}{2}} - 9x^{-\frac{3}{2}}$ are solutions of the differential equation $4x^2y'' + 12xy' + 3y = 0$, for $x > 0$.

Historical Background

The study of ODEs is closely tied to the development of calculus and mathematical physics:

- *17th Century (Foundations)*: The birth of differential equations coincides with the invention of calculus by *Isaac Newton* and *Gottfried Wilhelm Leibniz* in the late 1600s. Newton formulated equations of motion, such as

$$F = ma \quad \Rightarrow \quad m \frac{d^2x}{dt^2} = F(x, t),$$

which is a second-order ODE.

The Bernoulli brothers also developed methods to solve differential equations to extend the range of their applications

- *18th Century (Development)*: Mathematicians like *Leonhard Euler*, a student of Johann Bernoulli and *Joseph-Louis Lagrange* developed systematic methods for solving first- and second-order ODEs, as well as studying oscillations, planetary motion, and mechanical systems.
- *19th Century (Theory)*: With contributions from *Augustin-Louis Cauchy*, *Sophie Kovalevskaya*, and others, the rigorous theory of existence and uniqueness of solutions was established. The study of linear differential equations and series solutions flourished.

- *20th Century (Modern Applications):* Differential equations became essential in physics, engineering, biology, and economics. The qualitative theory of dynamical systems (initiated by *Henri Poincaré*) emphasized studying the behavior of solutions without necessarily finding exact formulas.
- *Today:* ODEs remain central in mathematical modeling across disciplines: population dynamics, control systems, epidemiology, neuroscience, climate modeling, finance, etc.

Ordinary differential equations provide the mathematical language for describing change. From Newton's laws of motion to modern systems biology, ODEs are the foundation of mathematical modeling in the natural and applied sciences.

Section 1.3: Mathematical modelling with Differential Equations

Example 1.3.1 (Newton's Law of Cooling). *Newton's law of cooling states that the rate of change of temperature $T(t)$ of an object is proportional to the difference between the object's temperature and the ambient temperature T_a .*

$$\frac{dT}{dt} = -k(T - T_a), \quad k > 0.$$

Solution: This is a first-order linear ODE. We solve it using separation of variables:

$$\frac{dT}{T - T_a} = -k dt.$$

Integrating both sides:

$$\ln |T - T_a| = -kt + C.$$

Exponentiating:

$$T - T_a = Ce^{-kt}.$$

Hence the general solution is

$$T(t) = T_a + (T_0 - T_a)e^{-kt},$$

where $T_0 = T(0)$ is the initial temperature.

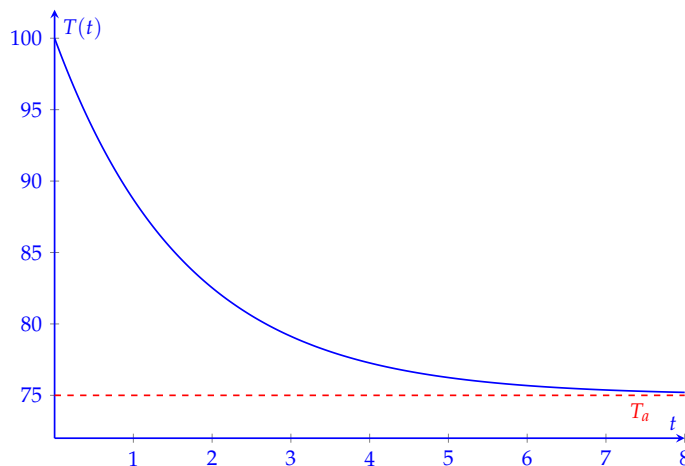


Figure 1.1: Exponential decay of temperature in Newton's Law of Cooling.

Example 1.3.2 (Simple Harmonic Oscillator). The motion of a mass attached to a spring (without damping) is governed by

$$m \frac{d^2x}{dt^2} + kx = 0,$$

where m is the mass and k is the spring constant.

Solution: Dividing through by m gives

$$\frac{d^2x}{dt^2} + \omega^2 x = 0, \quad \text{where } \omega = \sqrt{\frac{k}{m}}.$$

This is a second-order linear homogeneous ODE with constant coefficients. The characteristic equation is

$$r^2 + \omega^2 = 0 \quad \Rightarrow \quad r = \pm i\omega.$$

Hence, the general solution is

$$x(t) = C_1 \cos(\omega t) + C_2 \sin(\omega t),$$

which describes oscillatory motion with angular frequency ω .

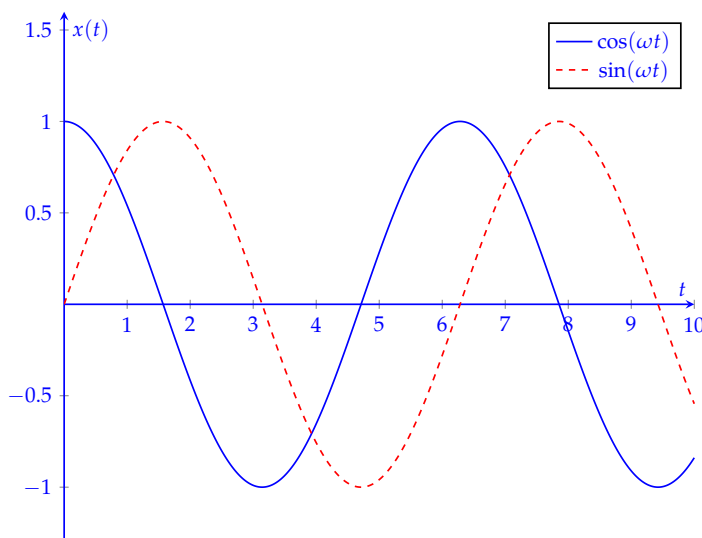


Figure 1.2: Solutions of the simple harmonic oscillator.

Example 1.3.3 (Logistic Population Growth). A simple model for population growth is given by the logistic equation:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K}\right),$$

where r is the intrinsic growth rate and K is the carrying capacity.

Solution: Separating variables:

$$\frac{dP}{P(1 - P/K)} = r dt.$$

Using partial fractions:

$$\frac{1}{P(1 - P/K)} = \frac{1}{P} + \frac{1}{K - P} \cdot \frac{1}{K}.$$

So,

$$\int \left(\frac{1}{P} + \frac{1}{K - P} \right) dP = r \int dt.$$

This gives

$$\ln |P| - \ln |K - P| = rt + C.$$

Simplifying:

$$\ln \left| \frac{P}{K - P} \right| = rt + C.$$

$$\Rightarrow P(t) = \frac{K}{1 + \left(\frac{K}{P_0} - 1 \right) e^{-rt}},$$

where $P_0 = P(0)$.

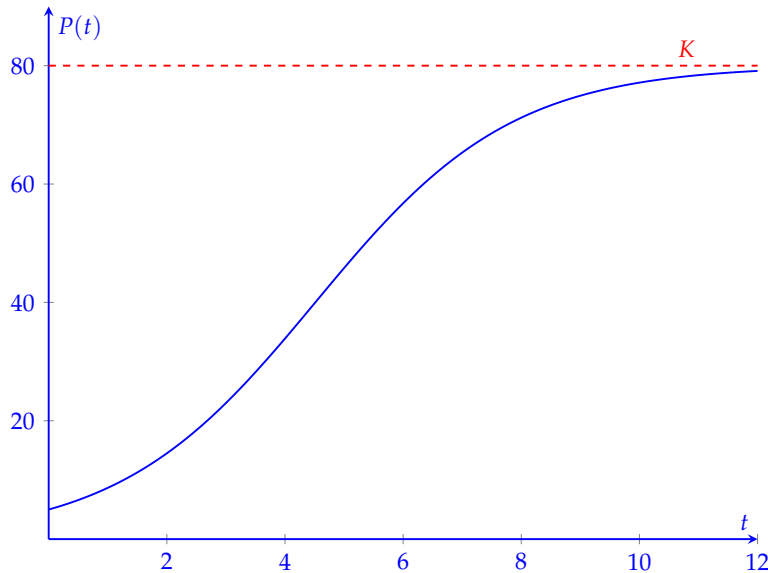


Figure 1.3: Logistic growth approaching the carrying capacity $K = 80$.

Section 1.4: Slope (Direction) Fields

Definition 1.4.1. A **slope field** is a graphical representation of a first-order differential equation

$$\frac{dy}{dx} = f(x, y).$$

At each point (x, y) in the plane, a small line segment is drawn with slope $f(x, y)$.

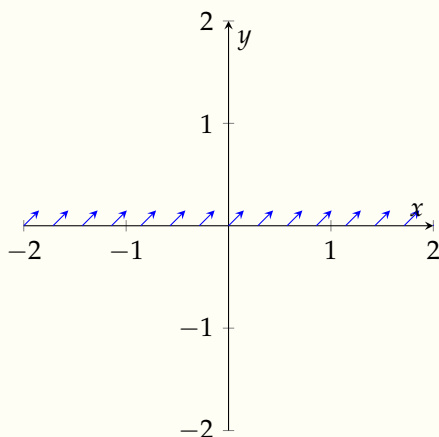
Definition 1.4.2. The graph of a solution of a differential equation is called a **solution curve**. A curve C is called an **integral curve** of a differential if every solution curve is a part of it.

- The slope field provides a qualitative view of solutions without explicitly solving the ODE.
- Solution curves are tangent everywhere to the slope field.

Example 1.4.3 (Constant Slope Field). Consider

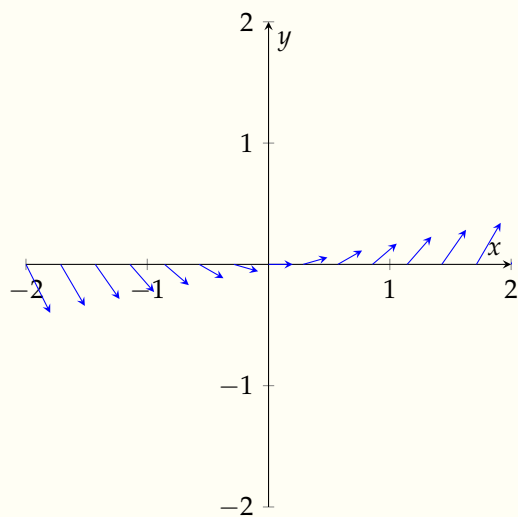
$$\frac{dy}{dx} = 1.$$

Every line segment has slope 1 (a 45° line). Solution curves are lines of the form $y = x + C$.



Example 1.4.4 (Non-constant Slope Field). We visualize the slope field for

$$\frac{dy}{dx} = x.$$



The general solution is:

$$y = \frac{x^2}{2} + C$$

which agrees with the slope field.

Example 1.4.5 (Logistic Growth). Consider the logistic ODE:

$$\frac{dy}{dx} = y(1 - y).$$

- When $0 < y < 1$, slopes are positive (growth).
- When $y > 1$, slopes are negative (decay).
- $y = 0$ and $y = 1$ are equilibrium solutions.

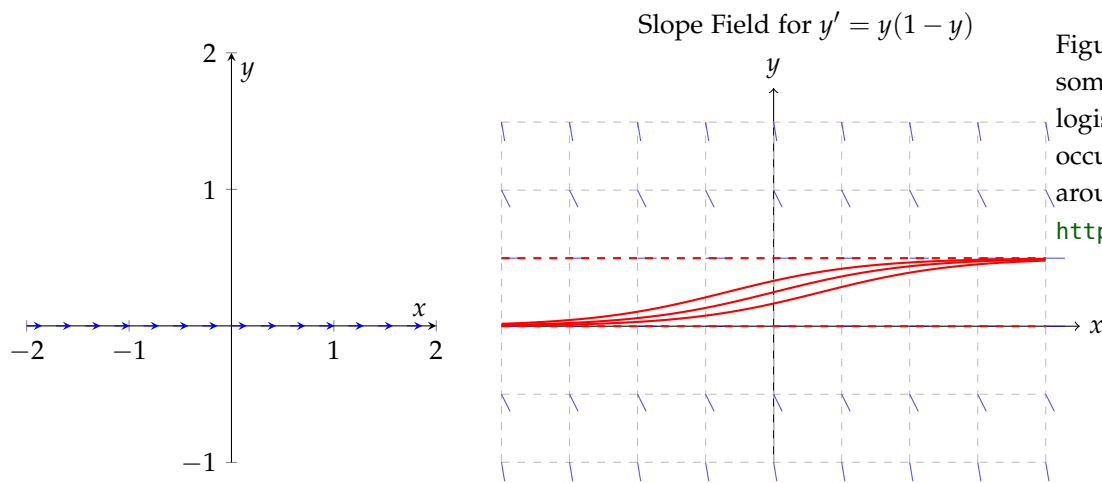


Figure 1.4: Slope field and some solution curves for the logistic equation. Equilibria occur at $y = 0$ and $y = 1$. Play around with the sage code here <https://shorturl.at/K9f7l>

Example 1.4.6. For the differential equation,

$$\frac{dy}{dx} = \frac{x}{y},$$

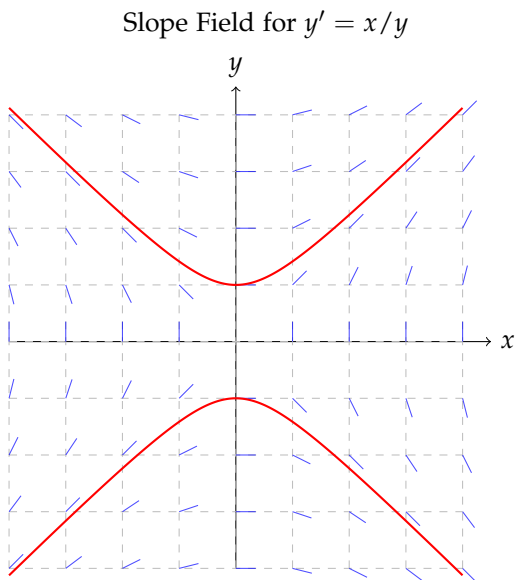


Figure 1.5: An *integral curve* is more general term for the graph of a solution, including those that are not functions. For example, the full hyperbola $y^2 - x^2 = K$ is an integral curve, but the upper branch ($y = \sqrt{x^2 + K}$) and lower branch ($y = -\sqrt{x^2 + K}$) are the individual solution curves.

- Slope fields help visualize the qualitative behavior of solutions.
- Equilibrium solutions correspond to horizontal slopes ($f(x, y) = 0$).
- They are especially useful when an ODE cannot be solved analytically.

Practice Problems

Exercise 1.4.1

Give definitions of the following terms and/or answer the questions⁸:

- Define an equilibrium of a ODE.
- Explain the procedure for calculating equilibria.

⁸ research it if we did not discuss it in class.

(c) **Explain** the difference between a *general* and *particular* solution.

Exercise 1.4.2

Show that out of three functions Provided two are solutions of the ODE

$$x^2 y''(x) - x y'(x) + y(x) = 0, \text{ and the third is not.}$$

(a) $y_1(x) = \ln(x)$ (b) $y_2(x) = x \ln(x)$ (c) $y_3(x) = x - x \ln(x)$

Summary

1. A differential equation relates a function to its derivatives.
2. The order of a differential equation is the highest derivative present.
3. A solution of a differential equation is a function that satisfies the equation.
4. Initial value problems specify conditions at a single point, while boundary value problems specify conditions at multiple points.
5. Mathematical modeling uses differential equations to describe real-world phenomena.
6. Historical development of differential equations is closely tied to calculus and physics.
7. Slope fields visualize solutions qualitatively.

Chapter 2: First Order Ordinary Differential Equations (ODEs)

Section 2.1: Introduction

As mentioned in the previous chapter, the order is determined by the highest derivative appearing in the equation. For a first order differential equation, only y' appears.

A first-order ordinary differential equation is an equation involving an unknown function $y(x)$ and its first derivative $y'(x)$. The general form is:

$$y' = f(x, y).$$

We will explore several common types of first-order ODEs and the methods used to solve them.

First-order differential equations can be classified into several types:

Type	General Form	Method
Separable	$\frac{dy}{dx} = g(x)h(y)$	Rewrite as $\frac{dy}{h(y)} = g(x) dx$ and integrate both sides.
Linear	$y' + P(x)y = Q(x)$	Use an integrating factor $\mu(x) = e^{\int P(x) dx}$. Multiply the equation by $\mu(x)$, then integrate.
Exact	$M(x, y) dx + N(x, y) dy = 0$	Check if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. If so, find a potential function $\Phi(x, y)$ such that $d\Phi = M dx + N dy$. The solution is $\Phi(x, y) = C$.
Special Forms		
Bernoulli	$y' + P(x)y = Q(x)y^n$	Substitute $u = y^{1-n}$ to convert it to a linear equation.
Homogeneous	$y' = f(y/x)$	Substitute $v = y/x$ to convert it to a separable equation.

Section 2.2: Existence and Uniqueness

Before we proceed, we need to establish the conditions under which a differential equation will have a solution.

Theorem 2.2.1 (Existence and Uniqueness for 1st Order Nonlinear Equations (Picard–Lindelöf)). Suppose $f(x, y)$ and $\frac{\partial f}{\partial y}$ are continuous in a rectangle $R : \alpha < x < \beta, \delta < y < \gamma$ containing (x_0, y_0) . Then the initial value problem

$$y' = f(x, y), \quad y(x_0) = y_0$$

has a unique solution in some interval around x_0 .

Theorem 2.2.2 (Existence and Uniqueness for 1st Order Linear Equations). *If the functions P and Q are continuous on an open interval $I : \alpha < x < \beta$ containing the point $x = x_0$, then there exists a unique function $y = \phi(x)$ that satisfies the differential equation*

$$y' + P(x)y = Q(x) \quad (2.1)$$

for each $x \in I$, and that also satisfies the initial condition $y(x_0) = y_0$.

Section 2.3: Separable Equations

A first-order differential equation is **separable** if it can be written in the form:

$$\frac{dy}{dx} = g(x)h(y)$$

where $g(x)$ is a function of x alone and $h(y)$ is a function of y alone.

Solution Method

1. Separate the variables by rearranging the equation:

$$\frac{1}{h(y)} dy = g(x) dx$$

2. Integrate both sides of the equation:

$$\int \frac{1}{h(y)} dy = \int g(x) dx + C$$

3. Solve the resulting algebraic expression for y to get the general solution.

Example 2.3.1. Solve:

$$y' = xy.$$

Solution:

$$\frac{dy}{dx} = xy \Rightarrow \frac{dy}{y} = x dx.$$

Integrate:

$$\ln |y| = \frac{1}{2}x^2 + C \Rightarrow y(x) = Ce^{x^2/2}.$$

Example 2.3.2. Solve $y' = \frac{x^2}{y^2}$.

Solution:

$$y^2 dy = x^2 dx \Rightarrow \int y^2 dy = \int x^2 dx$$

Integrate both sides

$$\frac{y^3}{3} = \frac{x^3}{3} + C \Rightarrow y^3 = x^3 + 3C$$

Since $3C$ is an arbitrary constant, we can write $y^3 = x^3 + K$.

$$y = \sqrt[3]{x^3 + K}$$

We should note that this solution method for certain kinds of differential equations may lead to an Integral-defined function (i.e the integral cannot be written in terms of any elementary function).

Practice Problems

Exercise 2.3.1

Solve the following separable ODEs:

- | | | |
|--|---------------------------------|------------------------------------|
| (a) $(2y) e^{y^2} y' = 1$ | (f) $y' = \frac{e^{-y}}{y}$ | (k) $y' = \frac{3x+5}{(x+1)(x-2)}$ |
| (b) $y' = \frac{\cos(3x)}{\sin^2(3x)}$ | (g) $y' = \frac{1}{y^2 \ln(y)}$ | (l) $y' = \frac{2x}{x^2 + 3x + 2}$ |
| (c) $y' = \frac{1}{1+x^2}$ | (h) $y' = e^x \cos(x)$ | (m) $y' = \frac{x^2+1}{x(x^2+1)}$ |
| (d) $y' = \frac{2x}{x^2+4}$ | (i) $y' = -\ln(y)$ | (n) $y' = \frac{1}{(x+1)^2(x-2)}$ |
| (e) $y' = \frac{1}{\sqrt{1-x^2}}$ | (j) $y' = \frac{1}{x^2-1}$ | |

Exercise 2.3.2

Solve the following separable ODE

- (a) $y' = xe^{x^2}$.
- (b) $y' = \frac{\ln(x)}{x^2}$.
- (c) $y' = \sqrt{4-y^2}$.
- (d) $y' = \frac{\arctan(x)}{1+x^2}$.

Exercise 2.3.3

Solve the following separable ODEs:

- | | | |
|------------------------|----------------------------|-------------------------------|
| (a) $y' = y^2 \sin(x)$ | (e) $y' = (y-1)(y-3)x^2$ | (i) $y' = (y-1)(y+3) \sin(x)$ |
| (b) $y' = y^3 x$ | (f) $y' = (y+1)(y-2)xe^x$ | (j) $y' = (y+4)(y-2)x^3$ |
| (c) $y' = (y^2-1)x$ | (g) $y' = (y^2-4)xe^{x^2}$ | (k) $y' = (y-5)(y+1)e^{-x}$ |
| (d) $y' = (y+2)(y-1)$ | (h) $y' = (y^2-9) \cos(x)$ | (l) $y' = (y^2-16)x^4$ |

Exercise 2.3.4

Identify the separable ODEs out of those given below. Demonstrate that it can be written in the standard form $y'(x) = F(x)G(y)$:

(a) $y' + xy - x = 0$

(b) $y' - 3(y^2 - 5y - 6) = 0$

Exercise 2.3.5Solve $\frac{dy}{dx} = y^2 - 4$.**Exercise 2.3.6**Find the implicit formula $F(x, y) = \text{const.}$ for the **solution** of the following separable ODE:

$$y'(x) = (y^2 - 6y + 8)xe^x$$

Exercise 2.3.7Find the solution to the separable ODE $y'(x) = y(3 - y)$.**Exercise 2.3.8**

Consider a population of bacteria that grows at a rate proportional to the size of the population.

- (a) Determine a ODE to describe the growth of bacteria at any time t .
- (b) Sketch the direction field, and make a prediction of what the solution is based on the sketch.
- (c) Determine the solution analytically.
- (d) The number of bacteria in a colony at 10:00am is approximately five million. At noon the number is approximately seven million. How many bacteria were in the colony at 9:00am?

Exercise 2.3.9Determine the equilibrium solutions for each ODE and decide if they are stable (an attractor), unstable (a repeller) or semistable¹. Draw a 1-D phase portrait for each.¹ Do some research if you need to.

(a) $y' = (y - 2)(y + 1)(y + 2)$

(c) $y' = y^3 + 2y^2 + y$

(b) $y' = y^3 - 9y$

(d) $y' = -y - y^3$

Exercise 2.3.10A model for the population $P(t)$ in a suburb of a large city is given by the Initial Value Problem (IVP)²

$$\frac{dP}{dt} = P(10^{-1} - 10^{-7}P), \quad P(0) = 5000,$$

² IVPs are ODEs with additional condition that the solution or its derivatives has to take certain values at a specific point.where t is measured in months. What is the limiting value of the population³? At what time will the population be equal to one-half of this limiting value?³ Hint: recall equilibrium solutions**Exercise 2.3.11**

Solve these IVPs:

(a) $e^x \frac{dy}{dx} = xy^2, y(0) = 2$

(b) $\frac{dy}{dx} = y^2 + 1, y(0) = 3$

Exercise 2.3.12

Find the 1-parameter family of solutions for each ODE:

(a) $xy' + 2y = e^{-x}$

(b) $x \frac{dy}{dx} = 2y + x^4 \sin(x^2)$

(c) $\frac{dy}{dx} + xy = xy^3$

(d) $(1 + x^2)y' + 4xy = \frac{4x}{(1 + x^2)^3}$

Exercise 2.3.13Solve $y' = e^{-x^2}$, $y(3) = 5$.See how to solve here⁴⁴ <https://math.stackexchange.com/questions/154968/is-there-really-no-way-to-integrate-e>**Section 2.4: Linear Equations**Linear⁵ ODEs are those that can be written in the form

$$\underbrace{a_0(x)y(x) + a_1(x)\frac{dy}{dx} + a_2(x)\frac{d^2y}{dx^2} + \dots + a_n(x)\frac{d^ny}{dx^n}}_{\text{Linear combination of derivatives}} = g(x)$$

are called **linear** ODEs. Notice: the **only way** the **dependent** variable appears in the equation is in expressions of the sort⁶

$$\dots + a_k(x)y^{(k)}(x) + \dots,$$

that is y or its derivatives are never squared, multiplied with each other, or appear in more complicated formulas. The **coefficient functions** $a_k(x)$ may be very complicated themselves, but they never involve y or its derivatives.⁷The **standard form** of a linear ODE is obtained by dividing through by the highest coefficient, to result in expression that has a constant ($= 1$) as its highest coefficient:

$$\underbrace{P_0(x)y(x) + P_1(x)\frac{dy}{dx} + P_2(x)\frac{d^2y}{dx^2} + \dots + 1 \cdot \frac{d^ny}{dx^n}}_{\text{Linear combination of derivatives}} = \underbrace{Q(x)}_{\text{Input or Inhomogeneity}}.$$

The **standard form** is the starting point for many solution algorithms.The standard⁸ form for **1st order linear ODEs** is:

$$P(x)y(x) + y'(x) = Q(x)$$

where $P(x)$ and $Q(x)$ are continuous functions of x .Here are several examples of 1st order linear ODEs. Rearrange them into a standard form and identify their coefficient function $P(x)$ and input function $Q(x)$ for each of them.

(a) $y'(x) = -0.5y(x) + 8$

(b) $4y'(x) = -2y(x) + 32 + 4 \sin(x)$

(c) $x^2y'(x) + 0.5y(x) = x^3$

(d) $y'(x) = 3y(x)$

Exercise 2.4.1

Which of the following differential equations are linear?

(a) $y' + 2xy = 0$

(b) $y' = y + 5$

(c) $y' + y = x$

Is any of them separable?

⁵ The name comes from the **linear combination**, which is a more-general expression that appears in various parts of mathematics.⁶ $y^{(k)}$ denotes the k -th derivative. So $y^{(2)}(x) = y''(x)$.⁷ For more on linear equations, see the beginning of Zill §2.3.⁸ If you want to show that an ODE is **linear**, rearrange the terms to match the standard form. If you can find such functions $P(x)$ and $Q(x)$ that **do not depend on** y, y' then you are looking at a linear ODE.

Solution Method using Variation of parameters

To solve 1st order linear ODEs, we will use the **variation of parameters** method:⁹

Step (1) Convert a linear ODE into its standard form $y' + P(x)y = Q(x)$.

Step (2) Create the homogeneous subproblem by zeroing out the input function:

$$y'_H + P(x)y_H = 0$$

Step (3) Find the general solution of the homogeneous subproblem.¹⁰

Step (4) To form a **candidate solution** y_I for the inhomogeneous problem

$$y'_I + P(x)y_I = Q(x), \quad (4.1)$$

replace the integration parameter C in y_H by an unknown function $I(x)$.¹¹

Step (5) **Substitute the candidate**¹² y_I into the **full (original)** ODE. You will need to take its derivative first, which will contain the unknown functions $I(x)$ and $I'(x)$. Simplify until you get an expression $I'(x) = \dots$

$I(x)$ will not appear on the right hand side (and neither will y).

Step (6) Integrate $I'(x)$ to calculate $I(x)$ function¹³, which then gives you formula for $y_I(x)$.

Step (7) Now you have two functions: y_H and y_I . y_H contains a constant C , while y_I has no free constant. The general solution to the original problem is:

$$y(x) = y_H(x) + y_I(x)$$

⁹ Textbook explains the **integrating factor** technique instead. The two methods are different, but yield equivalent results. Through variation of parameters we will learn skills that will transfer over to higher-order ODEs.

¹⁰ For 1st order ODEs, this is always possible by separation of variables.

¹¹ Variation of constant $C \rightarrow$ function $I(x)$ is what gives the method its name.

¹² Anytime something is called "solution candidate" your instinct should be to "plug it into" the ODE.

¹³ The integration constant inside $I(x)$ is not important, so pick a constant that makes $I(x)$ simple. Say 0.

Example 2.4.1. Find general solution to

$$xy' - 4y = x^6 e^x$$

(this is the same problem as textbook §2.3.Ex.3). Additionally, find the particular solution that has $y(2) = 0$.

General solution:

Step (1) Standard form: $y' + \underbrace{\frac{-4}{x}}_{P(x)} y = \underbrace{x^5 e^x}_{f(x)}$

Step (2) Homogeneous subproblem:

$$y'_H + \frac{-4}{x} y_H = 0 \quad (4.2)$$

Step (3) Solving for y_H :

$$\begin{aligned} \frac{dy_H}{dx} &= \frac{4y_H}{x} && \text{(Separate variables)} \\ \frac{dy_H}{y_H} &= \frac{4dx}{x} \\ \int \frac{dy_H}{y_H} &= \int \frac{4dx}{x} \\ \ln|y_H| &= 4\ln|x| + K && \text{(Exponentiate both sides)} \\ y_H(x) &= Cx^4 \end{aligned}$$

Step (4) **Candidate solution** for the inhomogeneous problem: $y_I(x) = I(x)x^4$

Step (5) Substitute $y_I = I(x)x^4$ into $y'_I + \frac{-4}{x}y_I = x^5e^x$

$$\begin{aligned}y'_I(x) &= I'(x)x^4 + 4I(x)x^3 \\I'(x)x^4 + 4I(x)x^3 + \frac{-4}{x}I(x)x^4 &= x^5e^x \\I'(x)x^4 &= x^5e^x \Rightarrow I'(x) = xe^x\end{aligned}$$

Step (6) Integrate $I'(x)$ (in this case, integration by parts):

$$\begin{aligned}I(x) &= xe^x - e^x + K \quad (\text{in this step only, constant is arbitrary}) \\I(x) &= xe^x - e^x \\y_I(x) &= I(x)x^4 = (xe^x - e^x)x^4\end{aligned}$$

Step (7) General solution:

$$y(x) = \underbrace{Cx^4}_{y_H(x)} + \underbrace{x^5e^x - x^4e^x}_{y_I(x)}$$

(Particular) solution to the IVP: Start with the general solution (terms collected):

$y(x) = x^4(C + xe^x - e^x)$, and plug into it the time-point $x = 2$:

$$y(2) = 16C + 32e^2 - 16e^2 = 16(C + e^2)$$

At the same time, Initial Value says $y(2) = 0$, therefore

$$16(C - e^2) = y(2) = 0 \Rightarrow C = e^2.$$

So the particular solution to the IVP is:

$$y(x) = x^4(e^2 + xe^x - e^x)$$

Example 2.4.2. Solve

$$y' - 2y = e^x.$$

Step (1) Standard form: $y' - 2y = e^x$, with $P(x) = -2$, $Q(x) = e^x$.

Step (2) Homogeneous subproblem: $y' - 2y = 0$.

Step (3) Homogeneous solution:

$$y_H(x) = Ce^{-\int -2dx} = Ce^{2x}.$$

Step (4) Candidate solution (Variation of parameters): let

$$y_I(x) = I(x)e^{2x}.$$

Step (5) Substitute into the full ODE:

$$I'(x)e^{2x} + 2I(x)e^{2x} - 2I(x)e^{2x} = e^xe^{2x} \Rightarrow I'(x) = e^{-x}.$$

Step (6) Integrate $I'(x)$:

$$I(x) = \int e^{-x} dx = -e^{-x}.$$

Thus,

$$y_I(x) = -e^{-x}e^{2x} = -e^x.$$

Step (7) General solution:

$$y(x) = Ce^{2x} - e^x.$$

Exercise 2.4.2

Find general solution to

$$\dot{y} = -\frac{1}{2}y + 8 + \sin(t)$$

Exercise 2.4.3

Solve:

$$y' + y = e^x.$$

Exercise 2.4.4

Solve

$$y' - y = e^{2x}.$$

Exercise 2.4.5

Solve the IVP:

$$y' + \frac{2}{x}y = \frac{\cos(x)}{x^2}, \quad y(\pi) = 0.$$

Solution Method using an Integrating Factor

The method of **integrating factor** is another way to solve 1st order linear ODEs. The goal of this method is to rewrite the differential equation in a separable form so that we can integrate both sides.

Multiplying the entire equation by a certain function $\mu(x)$ (the integrating factor), will allow us to rewrite the left-hand side as a single derivative of a product.

Step (1) Rewrite the linear ODE in its standard form: $y' + P(x)y = Q(x)$.

Step (2) Using $P(x)$ from the standard form, compute the **integrating factor**, $\mu(x)$:

$$\mu(x) = e^{\int P(x) dx}$$

Step (3) Multiply the entire differential equation by the integrating factor:

$$\mu(x) \frac{dy}{dx} + \mu(x)P(x)y = \mu(x)Q(x)$$

The left side will become the derivative of a product:

$$\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$$

To see why, apply the product rule to the left-hand side:

$$\frac{d}{dx} \left[\overbrace{\mu(x)y}^{\text{product}} \right] = \overbrace{\mu(x) \frac{dy}{dx} + \frac{d\mu}{dx} y}^{\text{product rule}} = \overbrace{\mu(x) \frac{dy}{dx} + \mu(x)P(x)y}^{\text{LHS of linear eq multiplied by } \mu(x)}.$$

This is only true if $\frac{d\mu}{dx} = \mu(x)P(x)$.

Thus, solving for $\mu(x)$ gives

$$\mu(x) = e^{\int P(x) dx}$$

Step (4) Integrate both sides with respect to x :

$$\mu(x)y = \int \mu(x)Q(x) dx + C$$

Step (5) Solve for y to find the general solution:

$$y = \frac{1}{\mu(x)} \left(\int \mu(x)Q(x) dx + C \right)$$

Example 2.4.3. Solve:

$$y' + y = e^x.$$

Solution: Here $P(x) = 1$, so

$$\mu(x) = e^{\int 1 dx} = e^x.$$

Multiply through:

$$e^x y' + e^x y = e^{2x}.$$

LHS is derivative:

$$\frac{d}{dx}(e^x y) = e^{2x}.$$

Integrate:

$$e^x y = \frac{1}{2}e^{2x} + C \Rightarrow y(x) = \frac{1}{2}e^x + Ce^{-x}.$$

Example 2.4.4. Solve

$$y' - y = e^{2x}.$$

Solution: Here, $P(x) = -1$ and $Q(x) = e^{2x}$. So integrating Factor,

$$\mu(x) = e^{\int -1 dx} = e^{-x}$$

Multiply by $\mu(x)$:

$$e^{-x} \frac{dy}{dx} - e^{-x} y = e^{-x} e^{2x} = e^x \Rightarrow \frac{d}{dx}(e^{-x} y) = e^x$$

Integrating,

$$\int \frac{d}{dx}(e^{-x} y) dx = \int e^x dx \Rightarrow e^{-x} y = e^x + C$$

Thus,

$$y = e^x(e^x + C) = e^{2x} + Ce^x.$$

Practice Problems

Exercise 2.4.6

Each 1st order linear ODEs can be rewritten in the standard form with components

- input function;
- coefficient function;
- linear combination of derivatives.

- (a) Rewrite the following linear ODE in the standard form and identify the listed components:

$$t^2 \dot{x} + \sin(t)x + \cos(t) = 0.$$

(b) Write out the corresponding homogeneous subproblem.

Exercise 2.4.7

Solve

$$y' - 3y = 6$$

Exercise 2.4.8

Solve

$$xy' - 4y = x^6 e^x.$$

Exercise 2.4.9

Given a linear ODE.

$$t^2 \frac{dx}{dt} + x(t) - 3 = 0. \quad (4.3)$$

(a) Use *variation of parameters* technique to find the general solution to the ODE.

Do not use any formulas from memory or the textbook. Show every step of the process.

(b) Find the solution that additionally satisfies the Initial Value $x(1) = 1$.

Exercise 2.4.10

Find the **general** solution of the equation $\dot{x} + \sin(t)x = 3$.

Exercise 2.4.11

Solve the IVP $y' + \frac{2}{x}y = \frac{\cos(x)}{x^2}$, $y(\pi) = 0$.

Exercise 2.4.12

The equation below could be linear, separable, or exact. Identify its type and support your answer by either:

- putting the linear into its standard form,
- performing the exactness test,
- identifying the separated F and G components.

Do not erase attempts that are unsuccessful. Showing that an ODE is **not** of a certain class is still a useful result.

$$(y - \ln x) \cdot y'(x) = 1 + \ln x + \frac{y}{x}$$

Exercise 2.4.13

A tank initially contains 2,000 litres of water with a concentration of mercury of 6.0×10^{-5} grams per litre. Water that contains 3.0×10^{-5} grams per litre is pumped into the tank at 100 litres per hour. The well mixed solution is pumped out of the tank at 100 litres per hour. How long will it take for the concentration in the tank to reach 4.5×10^{-5} grams per litre?

Exercise 2.4.14

A vat with capacity of 500 gallons initially has 10 lbs. of salt dissolved in 100 gallons of brine. A brine solution with 0.5 lbs. of salt per gallon is pumped into the tank at a rate of 4 gallons per minute, and the well mixed solution is pumped out at a rate of 2 gallons per minute. How much salt will be in the vat when the vat is full? (Hint: first find the solution for the volume of the fluid in the tank, before tackling the amount of salt in it.)

Exercise 2.4.15

A car's fuel tank holds 15 gallons of gasoline. When the tank is full, the gasoline contains 0.015 gallons of ethanol. Suppose that pure gasoline is pumped into the tank at a rate of 1 gallon per minute, and the well mixed solution is pumped out at a rate of 1 gallon per minute. How long will it take for the amount of ethanol in the tank to be 0.003 gallons?

Exercise 2.4.16

Solve the IVP $y' + y \tan(x) = \sin(2x)$, $y(0) = 1$.

Exercise 2.4.17

Solve the linear ODE $y' + y \cot(x) = 2x \csc(x)$.

Exercise 2.4.18

Solve the ODE, $y' + y \ln(x^2) = 1$.

Exercise 2.4.19

Solve the following linear IVPs:

(a) $(t+1)\dot{x}(t) + x(t) = \ln t$, $x(1) = 10$

(b) $\dot{P} + 2tP = P + 4t - 2$, $P(1) = 1$

Section 2.5: Exact Equations

Definition 2.5.1. A differential expression written in the form $M(x, y) dx + N(x, y) dy$ is an **exact differential** in a region R of the xy -plane if it corresponds to the differential of some function $F(x, y)$, that is defined in R . A first order differential equation of the form

$$M(x, y) dx + N(x, y) dy = 0$$

is exact if the following condition holds in R :

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Theorem 2.5.2. If $M(x, y)$ and $N(x, y)$ have continuous first partial derivatives in a simply connected region R of the xy -plane, then the differential equation

$$M(x, y) dx + N(x, y) dy = 0$$

is exact if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

for all points (x, y) in R .

Solution Method

Step (1) Write the differential equation in the form $M(x, y) dx + N(x, y) dy = 0$:

Step (2) Check the exactness condition: $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Step (3) If exact, find a function $F(x, y)$ such that $\frac{\partial F}{\partial x} = M(x, y)$ and $\frac{\partial F}{\partial y} = N(x, y)$.

Step (4) Integrate $M(x, y)$ with respect to x , treating y as a constant:

$$F(x, y) = \int M(x, y) dx + g(y)$$

The function $g(y)$ is a "constant" of integration that depends on y .

Step (5) Differentiate this expression for $F(x, y)$ with respect to y and set it equal to $N(x, y)$:

$$\frac{\partial F}{\partial y} = \frac{\partial}{\partial y} \left(\int M(x, y) dx \right) + g'(y) = N(x, y)$$

Step (6) Solve for $g'(y)$ and integrate to find $g(y)$.

Step (7) Substitute $g(y)$ back into the expression for $F(x, y)$. The general solution is $F(x, y) = C$.

Example 2.5.3. Solve $(2x + y) dx + (x + 2y) dy = 0$.

Solution:

Step (1) Write the differential equation in the form $M(x, y) dx + N(x, y) dy = 0$:

$$(2x + y) dx + (x + 2y) dy = 0$$

Step (2) Check Exactness: $M(x, y) = 2x + y$ and $N(x, y) = x + 2y$. $\frac{\partial M}{\partial y} = 1$ and $\frac{\partial N}{\partial x} = 1$. The equation is exact.

Step (3) Integrate M w.r.t x :

$$F(x, y) = \int (2x + y) dx + g(y) = x^2 + xy + g(y)$$

Step (4) Differentiate w.r.t y and compare to N :

$$\frac{\partial F}{\partial y} = x + g'(y)$$

We set this equal to $N(x, y) = x + 2y$:

$$x + g'(y) = x + 2y \implies g'(y) = 2y$$

Step (5) Integrate to find $g(y)$:

$$g(y) = \int 2y dy = y^2$$

Step (6) General Solution: $F(x, y) = x^2 + xy + y^2 = C$.

Example 2.5.4. Solve:

$$(2xy + 3) dx + (x^2 + 4y) dy = 0.$$

Check exactness:

$$M = 2xy + 3, \quad N = x^2 + 4y, \\ \frac{\partial M}{\partial y} = 2x, \quad \frac{\partial N}{\partial x} = 2x \implies \text{Exact}.$$

Find potential $\Phi(x, y)$:

$$\frac{\partial \Phi}{\partial x} = M = 2xy + 3 \implies \Phi(x, y) = x^2y + 3x + g(y).$$

Differentiate wrt y :

$$\frac{\partial \Phi}{\partial y} = x^2 + g'(y) = N = x^2 + 4y \implies g'(y) = 4y.$$

So $g(y) = 2y^2$. Solution:

$$\Phi(x, y) = x^2y + 3x + 2y^2 = C.$$

Integration Factor for Non-Exact Equations

Recall that the left-hand side of a linear ODE can be made into a product derivative by multiplying it by an integrating factor $\mu(x)$. Similarly, a non-exact equation can sometimes be made exact by multiplying it by an appropriate integrating factor $\mu(x, y)$.

Sometimes it is possible to find an integrating factor $\mu(x, y)$, a function of only one variable (either x or y) such that

$$\mu(x, y)M(x, y) dx + \mu(x, y)N(x, y) dy = 0$$

is exact.

In this case, the exactness condition becomes:

$$\frac{\partial}{\partial y}[\mu M] = \frac{\partial}{\partial x}[\mu N].$$

By the product rule, this expands to:

$$\mu M_y + M\mu_y = \mu N_x + N\mu_x.$$

Rearranging gives:

$$N\mu_x - M\mu_y = \mu(M_y - N_x). \quad (5.1)$$

If μ is a function of x only, then $\mu_y = 0$ and (5.1) becomes:

$$\frac{\mu_x}{\mu} = \frac{M_y - N_x}{N}.$$

If the right-hand side is a function of x only, then we can integrate to find $\mu(x)$:

$$\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}.$$

Example 2.5.5. The following ODE is not exact. Find an integrating factor that makes it exact, and then solve it.

$$xy \, dx + (x^2 - y^2) \, dy = 0.$$

Solution:

Step (1) Write in standard form:

$$xy \, dx + (x^2 - y^2) \, dy = 0.$$

Step (2) Check exactness:

$$M = xy, \quad N = x^2 - y^2, \\ \frac{\partial M}{\partial y} = x, \quad \frac{\partial N}{\partial x} = 2x \quad \Rightarrow \text{Not exact.}$$

Step (3) Find integrating factor:

$$\frac{M_y - N_x}{N} = \frac{x - 2x}{x^2 - y^2} = \frac{-x}{x^2 - y^2}.$$

This is not a function of x only, but it is a function of y only if we divide by x :

$$\frac{M_y - N_x}{-M} = \frac{x - 2x}{-xy} = \frac{1}{y}.$$

So we can take $\mu(y) = e^{\int \frac{1}{y} dy} = e^{\ln|y|} = |y|$.

Step (4) Multiply the original equation by $\mu = y$:

$$y(xy \, dx + (x^2 - y^2) \, dy) = 0 \quad \Rightarrow \quad (xy^2) \, dx + (x^2y - y^3) \, dy = 0.$$

Step (5) Check exactness:

$$M = xy^2, \quad N = x^2y - y^3, \\ \frac{\partial M}{\partial y} = 2xy, \quad \frac{\partial N}{\partial x} = 2xy \quad \Rightarrow \text{Exact.}$$

Step (6) Find potential function:

$$\frac{\partial F}{\partial x} = M = xy^2 \Rightarrow F(x, y) = \frac{1}{2}x^2y^2 + g(y).$$

$$\frac{\partial F}{\partial y} = x^2y + g'(y) = N = x^2y - y^3 \Rightarrow g'(y) = -y^3.$$

$$g(y) = -\frac{1}{4}y^4.$$

Step (7) General solution:

$$F(x, y) = \frac{1}{2}x^2y^2 - \frac{1}{4}y^4 = C.$$

Exercise 2.5.1

The following ODE is not exact. Find an integrating factor that makes it exact, and then solve it.

$$xy \, dx + (2x^2 + 3y^2 - 20) \, dy = 0.$$

Practice Problems

Exercise 2.5.2

Determine whether the following ODE is exact. If it is, find its general solution.

$$(3x^2y - 6x) \, dx + (x^3 + 2y) \, dy = 0.$$

Exercise 2.5.3

Determine whether the following ODE is exact. If it is, find its general solution.

$$y'(x) = \frac{3x + 4y}{x}.$$

Exercise 2.5.4

Solve the following IVPs:

1. $(y + x)^2 \, dx + (2xy + x^2 - 1) \, dy = 0, \quad y(1) = 1.$

2. $(e^x + y) \, dx + (x + \cos(y)) \, dy = 0, \quad y(0) = 0.$

3. $(e^x + 2y) \, dx + (2 + x + ye^y) \, dy = 0, \quad y(0) = 1.$

4.

$$\left(\frac{3y^2 - t^2}{y^5} \right) \frac{dy}{dt} + \frac{t}{2y^4} = 0, \quad y(1) = 1.$$

Exercise 2.5.5

Determine whether the following ODE is exact. If it is, find its general solution.

$$(2xy + 3) \, dx + (x^2 + 4y) \, dy = 0.$$

Exercise 2.5.6

Find the solution and interval of validity for the IVP:

$$\frac{2ty}{t^2 + 1} - 2t - (2 - \ln(t^2 + 1))y' = 0, \quad y(5) = 0.$$

Exercise 2.5.7

Determine whether the following ODE is exact. If it is, find its general solution:

$$y'(x) = \frac{xy^2 - \cos(x)\sin(x)}{y(1 - x^2)}$$

Section 2.6: Substitution

Many first-order ODEs can be solved by making an appropriate substitution that transforms the equation into a separable or linear form. Two common substitution methods are for Homogeneous equations and Bernoulli equations.

Homogeneous Equations

Definition 2.6.1. A function $F(x, y)$ is **homogeneous of degree** $n \in \mathbb{R}$ if for all t :

$$F(tx, ty) = t^n F(x, y)$$

For example, $F(x, y) = x^3 + y^3$ is a homogeneous function of degree 3 because

$$F(tx, ty) = (tx)^3 + (ty)^3 = t^3(x^3 + y^3) = t^3 F(x, y).$$

whereas $F(x, y) = x + y^3$ is not homogeneous because

$$F(tx, ty) = tx + (ty)^3 = tx + t^3 y^3 \neq t^n F(x, y)$$

for any n .

Definition 2.6.2. A first-order ODE of the form

$$M(x, y) dx + N(x, y) dy = 0 \quad \text{or} \quad \frac{dy}{dx} = F(x, y) = -\frac{M(x, y)}{N(x, y)} \quad (6.1)$$

is **homogeneous of degree** n if both $M(x, y)$ and $N(x, y)$ are homogeneous functions of the same degree n .

So we can perform the substitution $v = \frac{y}{x}$ to transform the equation into a separable form. Since $M(x, y)$ and $N(x, y)$ are homogeneous of degree n , we have:

$$M(x, y) = x^n M(1, v) \quad \text{and} \quad N(x, y) = x^n N(1, v).$$

And equation (6.1) becomes:

$$x^n M(1, v) dx + x^n N(1, v) dy = 0 \quad \Rightarrow \quad M(1, v) dx + N(1, v) dy = 0 \quad (\text{for } n \text{ degree}) \quad (6.2)$$

where $v = \frac{y}{x} \rightarrow y = vx \Rightarrow dy = v dx + x dv$ which reduces equation (??) to a separable form in v and x

$$\frac{dx}{x} = -\frac{N(1, u)}{M(1, u) + uN(1, u)} du.$$

Solution Method for Homogeneous Equations

Step (1) Make the substitution $v = \frac{y}{x}$, which implies $y = vx$.

Step (2) Differentiate $y = vx$ using the product rule to get $\frac{dy}{dx} = v + x \frac{dv}{dx}$.

Step (3) Substitute v and the new expression for $\frac{dy}{dx}$ into the original equation. The result will be a separable equation in v and x .

Step (4) Solve the separable equation for v .

Step (5) Substitute back $v = \frac{y}{x}$ to get the solution in terms of x and y .

Example 2.6.3. Solve $\frac{dy}{dx} = \frac{x+y}{x}$.

Solution:

1. **Check Homogeneity:** Divide numerator and denominator by x :

$$\frac{dy}{dx} = \frac{1 + y/x}{1} = 1 + \frac{y}{x}$$

This is a homogeneous equation.

2. **Substitution:** Let $v = y/x$. Then $\frac{dy}{dx} = v + x \frac{dv}{dx}$.

3. **New Equation:** $v + x \frac{dv}{dx} = 1 + v$. This simplifies to $x \frac{dv}{dx} = 1$, which is separable.

4. **Solve Separable Equation:** $dv = \frac{1}{x} dx$.

$$\int dv = \int \frac{1}{x} dx \Rightarrow v = \ln |x| + C$$

5. **Substitute back:** $\frac{y}{x} = \ln |x| + C$.

6. **General Solution:** $y = x(\ln |x| + C)$.

Example 2.6.4. Solve $\frac{dy}{dx} = \frac{x+y}{x}$.

Answer:

$$y = x \ln |x| + Cx$$

Example 2.6.5. Solve:

$$\frac{dy}{dx} = \frac{x+y}{x}$$

Answer:

$$\frac{y}{x} = \ln |x| + C \Rightarrow y = x \ln |x| + Cx$$

Example 2.6.6. Solve:

$$\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$$

Answer:

$$\left(\frac{y}{x}\right)^2 = 2 \ln |x| + C$$

Bernoulli Equations

Definition 2.6.7. A first-order ODE of the form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

is a **Bernoulli equation** if n is any real number other than 0 or 1.

The substitution $v = y^{1-n} \Rightarrow \frac{dv}{dx} = (1-n)y^{-n} \frac{dy}{dx}$ transforms the equation into a linear ODE in v .

Solution Method for Bernoulli Equations

Step (1) Identify the Bernoulli equation in the form $\frac{dy}{dx} + P(x)y = Q(x)y^n$.

Step (2) Make the substitution $v = y^{1-n}$, which implies $\frac{dv}{dx} = (1-n)y^{-n} \frac{dy}{dx}$.

Step (3) Rearrange the original equation to express $\frac{dy}{dx}$ in terms of y , $P(x)$, and $Q(x)$.

Step (4) Substitute v and the new expression for $\frac{dy}{dx}$ into the rearranged equation. The result will be a linear equation in v and x .

Step (5) Solve the linear equation for v using an integrating factor if necessary.

Step (6) Substitute back $v = y^{1-n}$ to get the solution in terms of x and y .

Example 2.6.8. Solve $\frac{dy}{dx} + y = y^2$.

Solution:

Here $n = 2$, so $v = y^{-1}$. Compute $\frac{dv}{dx} = -y^{-2} \frac{dy}{dx}$. Original equation: $\frac{dy}{dx} = y^2 - y$.

Substitute:

$$\frac{dv}{dx} = -\left(1 - \frac{1}{y}\right) = -1 + v$$

So:

$$\frac{dv}{dx} - v = -1$$

This is linear. Integrating factor e^{-x} :

$$\frac{d}{dx}(ve^{-x}) = -e^{-x}$$

Integrate:

$$ve^{-x} = e^{-x} + C \implies v = 1 + Ce^x$$

Back-substitute $v = 1/y$:

$$y = \frac{1}{1 + Ce^x}$$

Example 2.6.9. Solve $\frac{dy}{dx} + \frac{2}{x}y = xy^3$.

Solution:

Here $n = 3$, so $v = y^{-2}$. Compute $\frac{dv}{dx} = -2y^{-3} \frac{dy}{dx}$.

⋮

$$y = \frac{1}{\sqrt{x^2 + Cx^4}}$$

Example 2.6.10. Solve:

$$\frac{dy}{dx} + y = y^2$$

Solution:

Here $P(x) = 1$, $Q(x) = 1$, $n = 2$.

Substitute: $v = y^{1-2} = y^{-1}$.

$\vdots$

$$\frac{1}{y} = 1 + Ce^x \Rightarrow y = \frac{1}{1 + Ce^x}$$

Example 2.6.11. Solve:

$$\frac{dy}{dx} + \frac{2}{x}y = xy^3$$

Solution:

Here $P(x) = \frac{2}{x}$, $Q(x) = x$, $n = 3$.

Substitute: $v = y^{1-3} = y^{-2}$.

$\vdots$

$$y^{-2} = x^2 + Cx^4 \Rightarrow y = \frac{1}{\sqrt{x^2 + Cx^4}}$$

Practice Problems

Exercise 2.6.1

Solve $(x^2 + y^2) dx + (x^2 - 2xy) dy = 0$.

Exercise 2.6.2

Solve $(x + y) dx - (x - y) dy = 0$.

Exercise 2.6.3

Solve the following homogeneous ODEs:

1. $\frac{dy}{dx} = \frac{x+3y}{3x+y}$

2. $(y^2 + xy) dx + x^2 dy = 0$

3. $-y dx + (x + \sqrt{xy}) dy = 0$

4. $xy' = y + \sqrt{x^2 - y^2}, \quad x > 0$

Exercise 2.6.4

Solve the following Bernoulli equations:

1. $xy' + y = y^{-2}$

2. $y' - y = e^x y^2$

3. $y' = y(xy^3 - 1)$

4. $xy' - (1+x)y = xy^2$

Exercise 2.6.5

Solve $\frac{dy}{dx} = \frac{x^2 - y^2}{2xy}$ using the homogeneous substitution.

Exercise 2.6.6

Solve $\frac{dy}{dx} + \frac{y}{x} = x^2y^3$ using the Bernoulli substitution.

Exercise 2.6.7

Solve $\frac{dy}{dx} = \frac{y^2 + xy}{x^2}$ using the homogeneous substitution.

Exercise 2.6.8

Solve:

$$\frac{dy}{dx} = \frac{x^2 - y^2}{2xy}$$

Exercise 2.6.9

Solve:

$$\frac{dy}{dx} = \frac{y+x}{y-x}$$

Exercise 2.6.10

Solve:

$$\frac{dy}{dx} + \frac{y}{x} = x^2y^3$$

Exercise 2.6.11

Solve:

$$\frac{dy}{dx} - 2y = 3y^2$$

Section 2.7: Qualitative Analysis of First-Order ODEs

A first-order ODE can be written in the form

$$\frac{dy}{dx} = f(x, y).$$

If f does not depend on x , then the equation is called **autonomous**:

$$\frac{dy}{dx} = f(y).$$

Autonomous equations are often used to model the evolution of a system over time, where x represents time and y represents the state of the system.

Equilibria and Stability

Definition 2.7.1. An *equilibrium* (or *fixed point*) of an autonomous ODE is a value $y = y_0$ such that $f(y_0) = 0$.

At an equilibrium, the state of the system does not change over time.

Definition 2.7.2. An equilibrium y_0 is **stable** if solutions that start close to y_0 remain close to y_0 as $x \rightarrow \infty$.

Example 2.7.3. Consider the autonomous ODE:

$$\frac{dy}{dx} = y(1 - y).$$

The equilibria are found by setting $f(y) = y(1 - y) = 0$, which gives $y_0 = 0$ and $y_0 = 1$.

If solutions that start close to y_0 move away from y_0 as $x \rightarrow \infty$, then the equilibrium is **unstable**.

Phase Portraits and Direction Fields

A **phase portrait** is a graphical representation of the trajectories of a dynamical system in the phase plane. Each point in the phase plane represents a state of the system, and the trajectories show how the state evolves over time. To construct a phase portrait for an autonomous ODE, we can use a **direction field** (or **slope field**). A direction field is a grid of arrows that indicate the slope of the solution curve at each point in the phase plane. The slope at a point (x, y) is given by $f(y)$. To sketch a direction field:

1. Choose a grid of points in the xy -plane.
2. At each point (x, y) , compute the slope $f(y)$
3. Draw a small line segment with the computed slope at each point.

Example 2.7.4. Consider the autonomous ODE:

$$\frac{dy}{dx} = y(1 - y).$$

The equilibria are $y_0 = 0$ and $y_0 = 1$. To sketch the direction field, we compute the slope at various points:

- For $y < 0$, $f(y) < 0$, so the slope is negative.
- For $0 < y < 1$, $f(y) > 0$, so the slope is positive.
- For $y > 1$, $f(y) < 0$, so the slope is negative.

The direction field shows that solutions starting below $y = 0$ move downward, solutions starting between $y = 0$ and $y = 1$ move upward, and solutions starting above $y = 1$ move downward. Thus, $y = 0$ is an unstable equilibrium and $y = 1$ is a stable equilibrium.

The direction field in Figure ?? illustrates the behavior of solutions near the equilibria $y = 0$ and $y = 1$.

The phase portrait in Figure ?? shows the stability of the equilibria. It is given in the diagram by arrows indicating the direction of movement of solutions near the equilibria.

Note: The stability of an equilibrium can often be determined by examining the

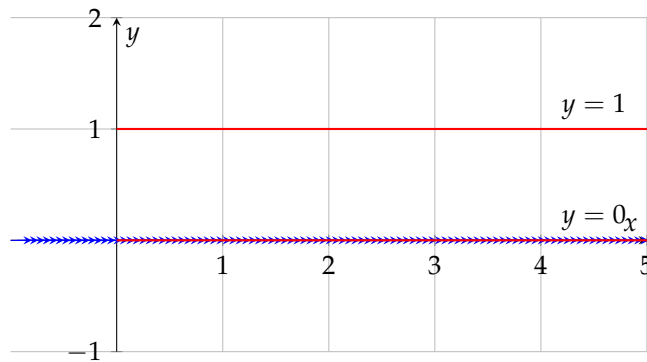


Figure 2.1: Direction field and equilibria for $\frac{dy}{dx} = y(1-y)$.

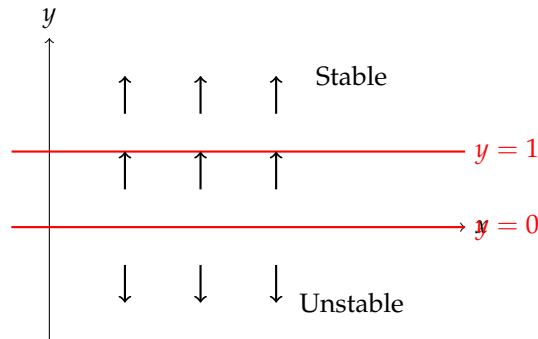


Figure 2.2: Phase portrait for $\frac{dy}{dx} = y(1-y)$.

sign of $f(y)$ near the equilibrium. If $f(y)$ changes from positive to negative as y increases through y_0 , then y_0 is stable. If $f(y)$ changes from negative to positive, then y_0 is unstable.

Section 2.8: A Numerical Method for ODEs

Euler's Method

Euler's method is a simple numerical technique for approximating the solution of an initial value problem (IVP) of the form:

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0.$$

The idea is to use the slope of the solution at a known point to estimate the value of the solution at a nearby point. The method proceeds as follows:

1. Choose a step size h .
2. Compute the next point using the formula:

3.

$$y_{n+1} = y_n + hf(x_n, y_n), \quad x_{n+1} = x_n + h.$$

4. Repeat the process to compute subsequent points.

Example 2.8.1. Solve the IVP:

$$\frac{dy}{dx} = y - x^2 + 1, \quad y(0) = 0.5$$

1. Using Euler's method with step size $h = 0.2$ to approximate $y(1)$.
2. Using an analytical method to find the exact solution and compare it with the approximation.

Example 2.8.2. Solve the IVP:

$$y' = 0.1\sqrt{y} + 0.4x^2, \quad y(2) = 4$$

1. Using Euler's method with step size $h = 0.1$ to approximate $y(2.5)$.
2. Using an analytical method to find the exact solution and compare it with the approximation.

Example 2.8.3. Solve the IVP:

$$y' = 0.2xy, \quad y(1) = 1$$

1. Using Euler's method with step size $h = 0.1$ to approximate $y(1.5)$.
2. Using Euler's method with step size $h = 0.05$ to approximate $y(1.5)$.
3. Using an analytical method to find the exact solution and compare it with the approximation.

Summary

- First-order ODEs appear in many forms; solution methods depend on type.
- Existence & uniqueness depend on continuity conditions.
- Separable $\Rightarrow$ integrate directly.
- Linear $\Rightarrow$ integrating factor.
- Exact $\Rightarrow$ potential function.
- Substitution methods (homogeneous, Bernoulli) transform to solvable forms.
- Autonomous ODEs model time-evolving systems.
- Equilibria are found by setting $f(y) = 0$.
- Stability determined by sign of $f(y)$ near equilibria.
- Phase portraits visualize system behavior.
- Euler's method approximates solutions numerically.

Section 2.9: Practice for Test — First-Order ODEs

You can use the following step-by-step calculator to help you study: <https://goo.gl/sZEyvL>. Do not rely on it too much, though, as you will not have such help in an exam.

Exercise 2.9.1

For each equation below,

- Determine whether the given ODE is exact, linear, or separable by identifying the functions that demonstrate the type: M, N for exact, standard form for linear, F, G for separable and perform the appropriate tests (if needed).
- Find the general solution of equations using the appropriate technique.

(a) $(4x - 8y^3)y'(x) + (5x + 4y) = 0$

(b) $y' - e^{3x+2y} = 0$

(c) $(x - y^3 + y^2 \sin(x))y'(x) - 3xy^2 - 2y \cos(x) = 0$

Exercise 2.9.2

Find the value of k such that the given ODE is exact:

$$y^3 + kxy^4 - 2x + (3xy^2 + 20x^2y^3)y'(x) = 0$$

Exercise 2.9.3

Conceptual questions and definitions:

- What is the equilibrium of an ODE?
- What is the difference between general and particular solutions of an ODE?
- What determines the angle of line segments used to plot direction fields?
- What is an equilibrium of an ODE?
- Define what it means for an equilibrium to be stable? Then explain how one can determine if an equilibrium is stable without finding a solution of an ODE.

Exercise 2.9.4

Find the fixed points, determine their stability, and sketch the phase portrait around them.

- $\dot{x} - (x - 2)(x + 2)^2(x - 7)$
- $\dot{x} = \cos(x)$
- $\dot{x} = \cos(x) + 1$
- $\dot{x} = \left(\frac{1}{5} - \frac{1}{x}\right) \left(\frac{1}{5} - \frac{2}{x}\right)^2$, for $x > 0$

Exercise 2.9.5

Use phase portrait information to determine what the solution does as $x \rightarrow \infty$ if $y(0) = 3$ for the ODE:

$$y'(x) + (y - 2)(y - 4)^2.$$

Exercise 2.9.6

Each of these ODEs could be linear, separable, or exact. Identify the type of each equation and support your answer by:

- putting the linear equation into its standard form and circling the input term,
- performing the exactness test on the exact equation,
- identifying the separated F and G components for the separable ODE.

- $(4y + 2t - 5) + (6y + 4t - 1)\dot{y} = 0$
- $xy'(x) + (3x + 1)y - e^{-3x} = 0$
- $\dot{N} + N - Nte^{t+2} = 0$
- $\sqrt{1 - y^2} - y'(x)\sqrt{1 - x^2} = 0$
- $(2xy + x^2 - 1)y'(x) + (x + y)^2 = 0$
- $(x + 2)^2y' - 5 + 8y + 4xy = 0$
- $\sin(3x) + 2[\cos(3x)]^2y' \cdot y = 0$
- $(e^x + y) + (2 + x + ye^y)y'(x) = 0$
- $y' + \tan(x)y - \cos^2(x) = 0$

Exercise 2.9.7

(Challenge) Give an example of a first-order ODE that is neither linear, separable, or exact. Perform the tests as in the previous question that demonstrate this.

Exercise 2.9.8

Find the **general solution** of each of the following equations using any technique that you find appropriate. Then, find the **particular solution** that satisfies the given initial condition.

(a) $x^2y'(x) - y + xy, y(-1) = -1$

(b) $xy'(x) + y - 4x - 1, y(1) = 8$

(c) $(x - y^3 + y^2 \sin(x)) - (3xy^2 + 2y \cos(x))y'(x) = 0, y(0) = 2$

Exercise 2.9.9

The following equation is not exact (it fails the exactness test). Nevertheless, apply the technique for solving exact equations. At which point does the technique break down?

$$(x^2 + 2xy - y^2)dx + (y^2 + 2xy - x^2)dy = 0$$

Chapter 3: Modeling with First Order ODEs

A **model** is a mathematical description of a real-world situation.

- Many processes in nature, engineering, economics, and biology change at a rate that depends on the current state of the system.
- These are often modeled by first-order differential equations:

$$\frac{dy}{dt} = f(y, t)$$

where

- t : independent variable (usually time),
- $y(t)$: dependent variable (population, temperature, amount, etc.),
- $f(y, t)$: rule describing the rate of change.

Steps in Modeling

1. Identify the system (population, chemical reaction, motion, etc.).
2. Describe the rate of change in words (e.g., “rate proportional to the population”).
3. Translate into a differential equation.
4. Solve the equation (analytically or numerically).
5. Interpret the solution in context (units, growth/decay behavior).

Section 3.1: linear Models

(a) Exponential Growth and Decay

Rate of change proportional to the current amount:

$$\frac{dy}{dt} = ky, \quad k \in \mathbb{R}, \quad y(t_0) = y_0.$$

Solution:

$$y(t) = y_0 e^{kt}.$$

Applications: radioactive decay ($k < 0$), population growth ($k > 0$), carbon dating.

Example 3.1.1. Exponential Growth (Population) A bacteria culture has initial population $P(0) = 500$. It doubles every 3 hours.

Solution:

$$\frac{dP}{dt} = kP, \quad P(t) = P_0 e^{kt}.$$

$$\text{At } t = 3, P(3) = 1000 = 500e^{3k}, \text{ so } e^{3k} = 2 \Rightarrow k = \frac{\ln 2}{3}.$$

$$P(t) = 500e^{(\ln 2/3)t}.$$

(b) Newton's Law of Cooling/Heating

The rate of cooling is proportional to the temperature difference between an object and its surroundings:

Rate proportional to difference from ambient temperature T_a :

$$\frac{dT}{dt} = -k(T - T_a).$$

Solution:

$$T(t) = T_a + (T_0 - T_a)e^{-kt}.$$

Exercise 3.1.1

Newton's Law of Cooling: Coffee at 90°C is placed in a 20°C room. After 10 minutes it cools to 60°C . Find T after 20 minutes.

Solution:

$$T(t) = 20 + 70e^{-kt}.$$

$$\text{At } t = 10, 60 = 20 + 70e^{-10k} \Rightarrow e^{-10k} = \frac{4}{7}. \text{ Thus } k = \frac{1}{10} \ln \frac{7}{4}.$$

$$T(20) = 20 + 70 \left(\frac{4}{7} \right)^2 \approx 42.9^\circ\text{C}.$$

Exercise 3.1.2

A blacksmith removes a metal from the forge at 750°C . 5 minutes later, the temperature is 683°C . How long would it take the metal to cool down to 500°C ?

(c) Mixing Problems

A mixing tank contains a volume of liquid with some solute (e.g., salt). If liquid flows in and out at certain rates, the concentration changes over time.

Let $Q(t)$ be the amount of solute (kg) at time t :

$$\frac{dQ}{dt} = \text{rate in} - \text{rate out}.$$

If inflow concentration is c_{in} and flow rate is r , and the tank volume is V , then:

$$\frac{dQ}{dt} = rc_{in} - \frac{r}{V}Q.$$

Example 3.1.2. Mixing Problem: Tank: 100 L water with 20 g salt. Brine with 2 g/L enters at 5 L/min, mixture flows out at 5 L/min.

Solution: Let $A(t)$ = salt amount.

$$\frac{dA}{dt} = 10 - \frac{A}{20}.$$

Solution:

$$A(t) = 200 - 180e^{-t/20}.$$

(d) Motion with Resistance

Falling object with air resistance proportional to velocity:

$$m \frac{dv}{dt} = mg - kv,$$

$$\frac{dv}{dt} = g - \frac{k}{m}v.$$

Terminal velocity occurs when $\frac{dv}{dt} \rightarrow 0$.

(e) Series Circuits

Series circuits are a common source of first-order linear differential equations in physics and engineering.

I - Components of a Series Circuit A typical circuit consists of:

- $i(t)$: current (A)
- A resistor with resistance R (measured in ohms Ω).
- An inductor with inductance L (measured in henries H).
- A capacitor with capacitance C (measured in farads F).
- A voltage source $E(t)$ (measured in volts).

II - Governing Equation (Kirchhoff's Voltage Law) Kirchhoff's Voltage Law states that the sum of voltage drops around a closed loop equals the applied voltage (The sum of potential differences around a closed loop is zero):

$$L \frac{di}{dt} + Ri + \frac{1}{C} \int i(t) dt = E(t).$$

Taking derivatives, this can also be expressed as:

$$L \frac{d^2i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i(t) = \frac{dE}{dt}.$$

III Special Cases

- **RL Circuit (no capacitor):**

$$L \frac{di}{dt} + Ri = E(t).$$

This is a first-order linear ODE in $i(t)$.

- **RC Circuit (no inductor):**

$$Ri + \frac{1}{C} \int i(t) dt = E(t),$$

or equivalently, in terms of charge $q(t)$ on the capacitor,

$$R \frac{dq}{dt} + \frac{1}{C} q = E(t).$$

Example 3.1.3. RL Circuit: Suppose $L = 1\text{ H}$, $R = 2\ \Omega$, and $E(t) = 10\text{ V}$. Then:

$$\frac{di}{dt} + 2i = 10.$$

Solution:

$$i(t) = 5 + Ce^{-2t}.$$

Practice Exercises

1. Solve the RL circuit:

$$L \frac{di}{dt} + Ri = E_0,$$

with $L = 2$, $R = 4$, $E_0 = 20$, and $i(0) = 0$.

2. A tank initially contains 100 L of pure water. Brine containing 0.5 kg/L of salt enters at 5 L/min, and the mixture leaves at the same rate. Find the amount of salt after t minutes.
3. A cup of coffee at 90°C is left in a room at 20°C . After 10 minutes the coffee has cooled to 70°C . Find the temperature after 20 minutes.

Section 3.2: Nonlinear Models

Nonlinear first-order differential equations arise in many real-world applications such as biology, physics, and chemistry. Unlike linear models, nonlinear models often exhibit more complex behavior such as saturation, thresholds, or blow-up in finite time.

General form:

$$\frac{dy}{dt} = f(t, y), \quad \text{where } f \text{ is nonlinear in } y.$$

(a) Population Models (Logistic Growth)

A refinement of the exponential growth model that accounts for limited resources:

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right),$$

where

- $P(t)$ is the population at time t ,
- $r > 0$ is the intrinsic growth rate,
- $K > 0$ is the carrying capacity.

Example 3.2.1. Suppose a population grows logistically with $r = 0.5$, $K = 1000$, and initial population $P(0) = 100$.

$$P(t) = \frac{1000}{1 + 9e^{-0.5t}}.$$

(b) Nonlinear Mixing Model

If concentration depends nonlinearly (e.g., saturation or reaction), the equation can become nonlinear.

Example 3.2.2.

$$\frac{dQ}{dt} = rc_{in} - \frac{r}{V}Q - kQ^2,$$

where kQ^2 models a nonlinear reaction term.

(c) Newton's Law of Cooling with Radiation

Newton's law (linear) is:

$$\frac{dT}{dt} = -k(T - T_s).$$

If radiation is included (Stefan-Boltzmann law), we obtain:

$$\frac{dT}{dt} = -k(T - T_s) - \sigma(T^4 - T_s^4),$$

which is nonlinear.

Example 3.2.3. An object at 500 K cools in surroundings at 300 K with:

$$\frac{dT}{dt} = -0.1(T - 300) - 10^{-8}(T^4 - 300^4).$$

This nonlinear ODE requires numerical methods (Euler, Runge-Kutta). The qualitative behavior: $T(t) \rightarrow 300$ as $t \rightarrow \infty$.

5. Practice Exercises

- Exponential Growth:** A population of 100 triples to 300 after 5 years. Find population after 10 years.
- Logistic Growth:** Fish population with carrying capacity 5000, initial 100, growth rate $k = 0.2$. Write explicit solution.
- Newton's Cooling:** A body at 37°C in a 20°C room cools to 30°C after 1 hour. Find temperature after 2 hours.
- Mixing:** Tank with 200 L pure water. Brine with 1 g/L enters at 4 L/min; mixture leaves at 2 L/min. Find salt content after 1 hour.
- Motion with Resistance:** A 2 kg mass falls under gravity ($g = 9.8$) with air resistance $k = 4$. Find terminal velocity as $t \rightarrow \infty$.

Exercise 3.2.1

A tank initially contains 2,000 litres of water with a concentration of mercury of 6.0×10^{-5} grams per litre. Water that contains 3.0×10^{-5} grams per litre is pumped into the tank at 100 litres per hour. The well mixed solution is pumped out of the tank at 100 litres per hour. How long will it take for the concentration in the tank to reach 4.5×10^{-5} grams per litre?

Exercise 3.2.2

A vat with capacity of 500 gallons initially has 10 lbs. of salt dissolved in 100 gallons of brine. A brine solution with 0.5 lbs. of salt per gallon is pumped into the tank at a rate of 4 gallons per minute, and the well mixed solution is pumped out at a rate of 2 gallons per minute. How much salt will be in the vat when the vat is full? (Hint: first find the solution for the volume of the fluid in the tank, before tackling the amount of salt in it.)

Exercise 3.2.3

A car's fuel tank holds 15 gallons of gasoline. When the tank is full, the gasoline contains 0.015 gallons of ethanol. Suppose that pure gasoline is pumped into the tank at a rate of 1 gallon per minute, and the well mixed solution is pumped out at a rate of 1 gallon per minute. How long will it take for the amount of ethanol in the tank to be 0.003 gallons?

Section 3.3: Modeling with Systems of First Order ODEs**Predator-Prey Model (Lotka-Volterra)**

A system of nonlinear equations describing interactions between two species:

$$\begin{aligned}\frac{dx}{dt} &= \alpha x - \beta xy, \\ \frac{dy}{dt} &= \delta xy - \gamma y,\end{aligned}$$

where

- $x(t)$ = prey population,
- $y(t)$ = predator population,
- α = prey growth rate,
- β = predation rate,
- γ = predator death rate,
- δ = predator reproduction rate from consuming prey.

Chapter 4: Higher (Second) Order Differential Equations ODEs

Higher-order ODEs often serve as models for mechanical and electrical systems, as relationships between position/velocity/acceleration and charge/current/voltage require multiple derivatives.

Section 4.1: Linear Equations

A **linear differential equation of order n** is an equation of the form

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g(x),$$

where the functions $a_0(x), a_1(x), \dots, a_n(x)$ and $g(x)$ are given, and y is the unknown function.

If $g(x) = 0$, the equation is called **homogeneous**. Otherwise, it is called **inhomogeneous/nonhomogeneous**.

Definition 4.1.1 (Initial Value Problem). A *higher-order initial value problem (IVP)* specifies the values of the function and its derivatives at a single point.

$$\begin{aligned} \text{That is, } & a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = g(x), \\ \text{Subject to } & y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1} \end{aligned}$$

For a 2nd-order equation, an IVP looks like

$$y'' + p(x)y' + q(x)y = g(x), \quad y(x_0) = y_0, y'(x_0) = y_1.$$

Theorem 4.1.2 (Existence of a Solution). Let $a_0(x), a_1(x), \dots, a_n(x)$ and $g(x)$ be continuous on an interval I , and let $a_n(x) \neq 0$ for all $x \in I$. If the initial condition $x = x_0 \in I$, then a unique solution $y(x)$ of the IVP exists.

Definition 4.1.3 (Boundary Value Problems). A second-order **boundary value problem (BVP)** specifies the values of the function at two different points.

$$\begin{aligned} \text{That is, } & y'' + p(x)y' + q(x)y = g(x) \\ \text{Subject to } & y(a) = y_0, y(b) = y_1. \end{aligned}$$

Homogeneous Equations

A linear homogeneous n -th order equation has the form

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_0(x)y = 0.$$

Theorem 4.1.4 (Superposition Principle). Let $y_1, y_2, \dots, y_k$ be solutions of the *homogeneous* n th-order differential equation on an interval I . Then the linear combination

$$y = c_1 y_1(x) + c_2 y_2(x) + \dots + c_k y_k(x),$$

where c_i are arbitrary constants, is also a solution on the I .

Example 4.1.5. Verify that the functions $y_1 = x^2$ and $y_2 = x^2 \ln x$ are both solutions of the homogeneous linear equation $x^3 y''' - 2xy' + 4y = 0$ on the interval $(0, \infty)$. Show by the superposition principle that the linear combination is also a solution on the same interval

Definition 4.1.6 (Linear Dependence/Independence). A set of functions $\{f_1(x), f_2(x), \dots, f_k(x)\}$ is said to be **linearly dependent** on an interval I if there exist constants $c_1 \neq c_2 \neq \dots \neq c_k \neq 0$ such that

$$c_1 f_1(x) + c_2 f_2(x) + \dots + c_k f_k(x) = 0$$

for every x in the interval. Otherwise, the set is **linearly independent**

When finding solutions to linear differential equations, we are interested in linearly independent functions. We can use a more "indirect" definition to check whether a set of functions is linearly independent.

Definition 4.1.7 (Wronskian). Suppose that the set of functions $\{f_1(x), f_2(x), \dots, f_k(x)\}$ are $k - 1$ times differentiable. The determinant

$$W(f_1, f_2, \dots, f_k)(x) = \begin{vmatrix} f_1(x) & f_2(x) & \dots & f_k(x) \\ f_1'(x) & f_2'(x) & \dots & f_k'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(k-1)}(x) & f_2^{(k-1)}(x) & \dots & f_k^{(k-1)}(x) \end{vmatrix}$$

is called the **Wronskian** of the functions.

Theorem 4.1.8 (Criterion for Linearly Independent Solutions). Let $y_1, y_2, \dots, y_k$ be k solutions of the homogeneous linear n th-order differential equation

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_0(x)y = 0.$$

on an interval I . Then the set of solutions is linearly independent on I if $W(y_1, y_2, \dots, y_k) \neq 0$ for every x in the interval.

Example 4.1.9. Verify that

$$y_1 = \frac{\cos(2 \ln x)}{x^3} \quad \text{and} \quad y_2 = \frac{\sin(2 \ln x)}{x^3}$$

are linearly independent solutions of the differential equation $x^2 y'' + 7xy' + 13y = 0$ on the interval $I = (0, \infty)$.

Definition 4.1.10 (Fundamental Set of Solutions). A fundamental set of solutions of a homogeneous linear k th-order differential equation

$$a_k(x)y^{(k)} + a_{k-1}(x)y^{(k-1)} + \cdots + a_0(x)y = 0.$$

on an interval I is any set of k linearly independent solutions $\{y_1, y_2, \dots, y_k\}$ on the interval I .

Theorem 4.1.11 (General Solution of Homogeneous Equations). Let $y_1, y_2, \dots, y_k$ be a fundamental set of solutions of the homogeneous linear k th-order differential equation on the interval I . Then the general solution of the equation on the interval is

$$y = c_1y_1 + c_2y_2 + \cdots + c_ky_k,$$

where c_i are arbitrary constants.

Example 4.1.12. Given that $y_1 = e^{3x}$ and $y_2 = e^{-3x}$ are solutions of the homogeneous linear equation $y'' - 9y = 0$ on the interval $I = (-\infty, \infty)$, construct a general solution to the ODE *if possible*.

Example 4.1.13. Given that $y_1 = e^{2x}$ and $y_2 = 5e^{2x}$ are solutions of the homogeneous linear equation $y'' - 4y = 0$ on the interval $I = (-\infty, \infty)$, construct a general solution to the ODE *if possible*.

Example 4.1.14. Given that $y_1 = e^x$, $y_2 = e^{2x}$ and $y_3 = e^{3x}$ are solutions to the homogeneous linear equation $y''' - 6y'' + 11y' + 6y = 0$ on the interval $I = (-\infty, \infty)$, Determine whether they form a fundamental set of solution. Construct a general solution for the ODE.

Nonhomogeneous/Inhomogeneous Equations

A nonhomogeneous linear equation has the form

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_0(x)y = g(x).$$

Recall from solving first order nonhomogeneous equations, that the solution is

$$y(x) = y_H(x) + y_I(x),$$

where $y_H(x)$ is the solution to the homogeneous part, and $y_I(x)$ is the solution to the nonhomogeneous/inhomogeneous part.

For the sake of consistency with most textbooks out there, we would write the solution as

$$y(x) = y_h(x) + y_p(x),$$

where y_h solves the homogeneous equation and y_p (**particular solution**) is a solution to the inhomogeneous part.

When y_h is a general solution, $c_1y_1 + c_2y_2 + \cdots + c_ky_k$, we call it the *complementary solution* y_c .

Theorem 4.1.15 (General Solution of Nonhomogeneous Equations). *Let y_p be any particular solution of the nonhomogeneous linear k th-order differential equation*

$$a_k(x)y^{(k)} + a_{k-1}(x)y^{(k-1)} + \cdots + a_0(x)y = g(x).$$

on an interval I , and let $y_1, y_2, \dots, y_k$ be a fundamental set of solutions of the associated homogeneous equation on the interval I . Then the general solution of the equation on the interval is

$$y = \underbrace{c_1y_1 + c_2y_2 + \cdots + c_ky_k}_{y_c} + y_p,$$

where c_i are arbitrary constants.

Methods for finding y_p include:

- Variation of parameters (general but more involved).
- Method of undetermined coefficients (for exponential, polynomial, sine, cosine $g(x)$).

Example 4.1.16. (a) Is $y_p = -4x^2$ a particular solution of the nonhomogeneous equation

$$y'' - 3y' + 4y = -16x^2 + 24x - 8?$$

(b) Is $y_p = -\frac{11}{12} - \frac{1}{2}x$ a particular solution of the nonhomogeneous equation

$$y''' - 6y'' + 11y' - 6y = 3x?$$

NOTE: there is an equivalent superposition principle for nonhomogeneous equations.

IT SAYS TO THE EFFECT THAT A NONHOMOGENEOUS PART CAN BE SPLIT UP BY SUMMATION AND SOLVED FOR INDEPENDENTLY BEFORE COMBINING.

Example 4.1.17. Given that

(i) $y_p = -4x^2$ is a particular solution of the nonhomogeneous equation

$$y'' - 3y' + 4y = -16x^2 + 24x - 8$$

(ii) $y_p = e^{2x}$ is a particular solution of the nonhomogeneous equation

$$y'' - 3y' + 4y = 2e^{2x}$$

(iii) $y_p = xe^x$ a particular solution of the nonhomogeneous equation

$$y'' - 3y' + 4y = 2xe^x - e^x$$

what is the particular solution of

$$y'' - 3y' + 4y = -16x^2 + 24x - 8 + 2e^{2x} + 2xe^x - e^x?$$

Section 4.2: Reduction of Order

The **reduction of order** method is a technique used to find a second linearly independent solution to a **second-order linear homogeneous** ODE when one solution is already known.

Consider the equation

$$y'' + p(x)y' + q(x)y = 0.$$

Suppose $y_1(x)$ is a known solution. To find another solution, assume

$$y_2(x) = v(x)y_1(x),$$

where $v(x)$ is an unknown function. Substituting into the equation gives a first-order equation for $v'(x)$, which can then be solved.

Example 4.2.1. Solve

$$y'' - y = 0,$$

given that $y_1(x) = e^x$ is a solution.

Solution. Assume $y_2(x) = v(x)e^x$. Then

$$y_2' = v'e^x + ve^x, \quad y_2'' = v''e^x + 2v'e^x + ve^x.$$

Substitute into the ODE:

$$(v''e^x + 2v'e^x + ve^x) - (ve^x) = e^x(v'' + 2v') = 0.$$

So

$$v'' + 2v' = 0.$$

Let $w = v'$, then $w' + 2w = 0$. Solution: $w = Ce^{-2x}$. Solving this 1st-order ODE:

$$v = \int Ce^{-2x} dx = -\frac{C}{2}e^{-2x} + D.$$

Thus

$$y_2(x) = v(x)e^x = -\frac{C}{2}e^{-x} + De^x.$$

A new independent solution is $y_2(x) = e^{-x}$.

So the general solution is

$$y(x) = C_1e^x + C_2e^{-x}.$$

Exercise 4.2.1

Given $y_1(x) = x$ is a solution of the ODE

$$x^2y'' - 2xy' + 2y = 0,$$

use reduction of order to find a second solution.

Exercise 4.2.2

Given $y_1(x) = x^2$ is a solution of the ODE

$$x^2y'' - 3xy' + 4y = 0,$$

Find the general solution of the ODE on the interval $(-\infty, \infty)$.

Section 4.3: Homogeneous Linear Equations with Constant Coefficients

Characteristic equations¹ help us solve higher-order linear ODEs whose parameters (coefficients) are constants (do not change).

The characteristic equation converts an ODE into a polynomial equation.

Given a **homogeneous 2nd order linear equations with constant coefficients**

$$ay'' + by' + cy = 0; \quad a, b, c \in \mathbb{R},$$

If we try a function of the form $y = e^{mx}$ as a solution, the ODE becomes

$$am^2e^{mx} + bme^{mx} + ce^{mx} = 0 \Rightarrow e^{mx}(am^2 + bm + c) = 0.$$

This forms the basis of the characteristic polynomial, $(am^2 + bm + c) = 0$ since $e^{mx} \neq 0$ for all values.

The roots of this characteristic polynomial help us write down the solution of the ODE.

As polynomials can have complex numbers as roots, we will have to know basic complex number arithmetic.

¹ “Characteristic” and “auxiliary” equations are used interchangeably in many texts. Note that another characteristic equations will appear in matrix ODEs, and its roots will carry a similar interpretation.

Complex Numbers

Complex numbers $z \in \mathbb{C}$ are specified using two (independent) real numbers (elements of $\mathbb{R}$). There are two different representations of any complex number z :

A complex number is of the form

$$z = x + iy, \quad x, y \in \mathbb{R}, \quad i^2 = -1.$$

$$\operatorname{Re}(z) = x, \quad \operatorname{Im}(z) = y, \quad \mathbb{C} = \{x + iy : x, y \in \mathbb{R}\}.$$

Notations

Cartesian form: $z = \operatorname{Re} z + i \operatorname{Im} z$. Components are called real and imaginary parts.

Polar form: $z = |z|e^{i \arg(z)} = r(\cos \theta + i \sin \theta)$, with $r = |z| = \sqrt{x^2 + y^2}$ and $\theta = \arg(z)$. Components are called magnitude (or modulus) and angle (or argument).

Exponential form: $z = re^{i\theta}$, using Euler’s formula $e^{i\theta} = \cos \theta + i \sin \theta$.

Trigonometric form: $z = r(\cos \theta + i \sin \theta)$.

Matrix form:

$$z \equiv \begin{bmatrix} x & -y \\ y & x \end{bmatrix}.$$

Vector form: $z = (x, y)$ in $\mathbb{R}^2$.

Operations

For $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$:

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2), \quad z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2).$$

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}, \quad \frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}.$$

$$\bar{z} = x - iy, \quad |z| = \sqrt{x^2 + y^2}.$$

Exponential Function:

The complex exponential function e^z can be defined in several equivalent ways:

1. Power Series Definition

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad z \in \mathbb{C}.$$

2. Limit Definition

$$e^z = \lim_{n \rightarrow \infty} \left(1 + \frac{z}{n}\right)^n.$$

3. Differential Equation Definition e^z is the unique function satisfying

$$\frac{d}{dz} e^z = e^z, \quad e^0 = 1.$$

4. Euler's Formula (for purely imaginary arguments)

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad \theta \in \mathbb{R}.$$

A consequence of Euler's formula is that the cos and sin functions can be defined using complex exponentials

Hyperbolic functions The hyperbolic sine and cosine are defined by:

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh^{-1} y = \ln \left(y + \sqrt{y^2 + 1} \right), \quad y \geq 1$$

$$\cosh^{-1} y = \ln \left(y + \sqrt{y^2 - 1} \right), \quad y \geq 1$$

Connection to Trigonometric Functions

$$\cosh(ix) = \cos x, \quad \sinh(ix) = i \sin x.$$

These definitions are consistent and extend the exponential function naturally from $\mathbb{R}$ to $\mathbb{C}$.

De Moivre's Theorem

$$(r(\cos \theta + i \sin \theta))^n = r^n(\cos(n\theta) + i \sin(n\theta)), \quad n \in \mathbb{Z}.$$

Roots of Complex Numbers The n th roots of $z = re^{i\theta}$ are

$$z_k = r^{1/n} e^{i(\frac{\theta + 2k\pi}{n})}, \quad k = 0, 1, \dots, n-1.$$

Exercise 4.3.1

Compute real and imaginary parts of each of the following numbers:

- | | | |
|----------------------|---------------------------|----------------------------|
| (a) $-2i(2 + 3i)$ | (c) $(2 - 3i)(2 + 3i)$ | (e) $e^{2+i\frac{\pi}{4}}$ |
| (b) $(1 + i)(2 + i)$ | (d) $e^{i\frac{\pi}{4}t}$ | (f) $\frac{1}{2+i}$ |

Exercise 4.3.2

Compute real and imaginary parts of each of the following expressions ²:

- | | | |
|----------------------|----------------------------|---------------------|
| (a) $(2 + i)(3 - i)$ | (c) $e^{2-i\frac{\pi}{2}}$ | (e) $\frac{1}{1-i}$ |
| (b) e^{3i} | (d) $\frac{1}{i}$ | |

² Product of exponentials: $e^{a \pm b} = e^a e^{\pm b}$
Euler's formula: $e^{\pm ix} = \cos(x) \pm i \sin(x)$

Homogeneous Linear Equations with Constant Coefficients

A homogeneous linear differential equation with constant coefficients has the form

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = 0,$$

where a_i are constants.

The method of solution:

Step (1) Form the **characteristic equation**:

$$a_n r^n + a_{n-1} r^{n-1} + \dots + a_0 = 0.$$

Step (2) Solve for the roots r .

Step (3) Build the general solution from the roots:

- Real distinct roots $r_1, \dots, r_n$:

$$y(x) = C_1 e^{r_1 x} + \dots + C_n e^{r_n x}$$

- Repeated root r of multiplicity m :

$$y(x) = (C_1 + C_2 x + \dots + C_m x^{m-1}) e^{rx}$$

- Complex roots $r = \alpha \pm i\beta$:

$$y(x) = e^{\alpha x} = e^{\alpha x \pm i\beta x} = e^{\alpha x} (C_1 \cos \beta x \pm C_2 \sin \beta x)$$

Definition of a concept and **calculation** are not always equivalent in mathematics. Definition speaks of *meaning*, while calculation tells us *how to compute* something. For example, *definition* of a solution of $ax^2 + bx + c = 0$ is "any number that satisfies the equation", and the calculation is given by the familiar quadratic formula. For quintic (5th order polynomial), there is no similar formula (calculation) of a solution, just a definition.

Example 4.3.1. Solve

$$y'' + 4y' + 4y = 0.$$

Solution. Characteristic equation:

$$r^2 + 4r + 4 = 0 \Rightarrow (r + 2)^2 = 0.$$

Repeated root: $r = -2$ of multiplicity 2. So the solution is

$$y(x) = (C_1 + C_2x)e^{-2x}.$$

Example 4.3.2. Solve

$$y''' - 3y'' + 3y' - y = 0.$$

Solution. Characteristic equation:

$$r^3 - 3r^2 + 3r - 1 = 0 \Rightarrow (r - 1)^3 = 0.$$

So root $r = 1$ with multiplicity 3. General solution:

$$y(x) = (C_1 + C_2x + C_3x^2)e^x.$$

Example 4.3.3. Solve the IVP

$$y'' - 5y' + 6y = 0, \quad y(0) = 4, \quad y'(0) = 7.$$

Solution. The characteristic equation is

$$r^2 - 5r + 6 = 0 \Rightarrow (r - 2)(r - 3) = 0,$$

so $r = 2, 3$.

Thus the general solution is

$$y(x) = C_1e^{2x} + C_2e^{3x}.$$

Using $y(0) = 4$ gives $C_1 + C_2 = 4$. Differentiating:

$$y'(x) = 2C_1e^{2x} + 3C_2e^{3x}, \quad y'(0) = 2C_1 + 3C_2 = 7.$$

Solve the system:

$$C_1 = 5, \quad C_2 = -1.$$

Final solution:

$$y(x) = 5e^{2x} - e^{3x}.$$

Example 4.3.4. Solve the IVP

$$y'' + 4y' + 13y = 0, \quad y(0) = 1, \quad y'(0) = 0.$$

Solution. Characteristic equation:

$$r^2 + 4r + 13 = 0, \quad \Delta = -36.$$

Thus $r = -2 \pm 3i$.

So

$$y(x) = e^{-2x} (C_1 \cos 3x + C_2 \sin 3x).$$

Initial condition $y(0) = 1$ gives $C_1 = 1$.

Now

$$y'(x) = e^{-2x} [-2(C_1 \cos 3x + C_2 \sin 3x) - 3C_1 \sin 3x + 3C_2 \cos 3x].$$

At $x = 0$:

$$y'(0) = -2C_1 + 3C_2 = 0.$$

Substitute $C_1 = 1$: $-2 + 3C_2 = 0 \Rightarrow C_2 = \frac{2}{3}$.

Final solution:

$$y(x) = e^{-2x} \left(\cos 3x + \frac{2}{3} \sin 3x \right).$$

Example 4.3.5. Solve $y'' + 4y' + 7y = 0$.

Example 4.3.6. Solve $3y''' + 5y'' + 10y' - 4y = 0$ assuming that the characteristic polynomial has at least one rational root.

Hint: It has one rational root and two complex roots

remember how to find rational roots from high school algebra.

Example 4.3.7. The complex-valued function $f(t)$ depends on real variable t :

$$f(t) = \exp [(-1 + i2\pi)t]$$

(Notation $e^x = \exp(x)$)

(a) Compute the real part of the function $\operatorname{Re} f(t)$ and the imaginary part $\operatorname{Im} f(t)$.

(b) Sketch the real and imaginary parts for $0 < t < 3$. Make sure to accurately represent growth/decay of each function, their value at $t = 0$, and points at which the functions intersect with the horizontal axis.

(c) For what values t is the function $f(t)$ purely real (imaginary part is 0) and for what values is it purely imaginary (real part is 0)?



Example 4.3.8. Compute solutions to the following linear homogeneous ODEs. In all cases, use real-valued formulas.

(a) $y'' + 4y = 0$

(b) $y'' + 2y' + y = 0$

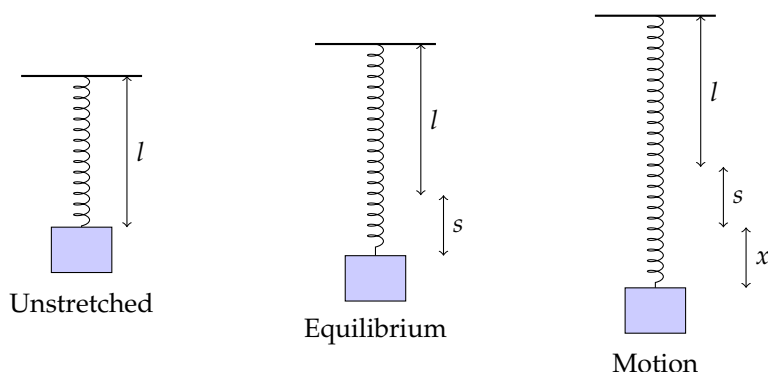


Figure 4.1: Vertical spring-mass system:

(a) natural length l ,

(b) equilibrium $l + s$

Hooke's law: $(mg = ks)$,

(c) motion $l + s + x$.

Example 4.3.9. An unforced response of an oscillator can be modeled using the ODE $m\ddot{y}(t) + b\dot{y}(t) + ky(t) = 0$, where m is the mass, b is the damping coefficient, and k the spring stiffness.

- (i) For $m = 4\text{kg}$ and $k = 9\text{N/m}$ and undetermined $b \geq 0$, find the roots of the characteristic equation.
- (ii) Find the ranges of values of $b\text{N/m}$ such that the general solution is: (a) decaying and shows no oscillation, (b) decaying and has oscillations. and write the general solution for each
- (iii) For $m = 2\text{kg}$, $b \geq 0$, and k such that the spring is stretched 5m at equilibrium under gravity, state the general solution to the new differential equation. (Take $g = 10\text{m/s}^{-2}$)

Solution:

The unforced oscillator is

$$m\ddot{y} + b\dot{y} + ky = 0.$$

- (i) **Characteristic roots for $m = 4$, $k = 9$.**

The characteristic equation is

$$4r^2 + br + 9 = 0.$$

Thus

$$r = \frac{-b \pm \sqrt{b^2 - 4 \cdot 4 \cdot 9}}{2 \cdot 4} = \frac{-b \pm \sqrt{b^2 - 144}}{8} \Rightarrow r = \frac{-b \pm \sqrt{b^2 - 144}}{8}.$$

- (ii) **Classification and general solutions (for $b \geq 0$).**

The discriminant is $\Delta = b^2 - 144$.

- Overdamped (decaying, no oscillation): $\Delta > 0 \iff b > 12$. Roots r_1, r_2 are real and distinct:

$$r_{1,2} = \frac{-b \pm \sqrt{b^2 - 144}}{8}.$$

$$\boxed{\text{General solution: } y(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}, \quad (b > 12).}$$

- Critically damped: $\Delta = 0 \iff b = 12$. Double root $r = -\frac{12}{8} = -\frac{3}{2}$.

$$\boxed{\text{General solution: } y(t) = (C_1 + C_2 t) e^{-3t/2}, \quad (b = 12).}$$

- Underdamped (decaying with oscillation): $\Delta < 0 \iff 0 \leq b < 12$.

Write $\sqrt{144 - b^2} =: \omega_d$ so the complex roots are

$$r = -\frac{b}{8} \pm i \frac{\sqrt{144 - b^2}}{8} = -\frac{b}{8} \pm i\omega, \quad \omega = \frac{\sqrt{144 - b^2}}{8}.$$

General (real) solution: $y(t) = e^{-bt/8}(A \cos(\omega t) + B \sin(\omega t)), \quad 0 \leq b < 12, \omega = \frac{\sqrt{144 - b^2}}{8}$
--

- (iii) **New equation when $m = 2$ and equilibrium stretch 5 m under gravity.**

According to Hooke's law, the force (mg) due to the spring is proportional to the stretched length at equilibrium:

Equilibrium condition $kx_{eq} = mg$ with $m = 2, g = 10, x_{eq} = 5$ gives

$$k \cdot 5 = 2 \cdot 10 \Rightarrow k = 4 \text{ (N/m)}.$$

Including gravity, the ODE becomes

$$2\ddot{y} + b\dot{y} + 4y = 2g = 20,$$

or equivalently

$$2\ddot{y} + b\dot{y} + 4y = 20.$$

A particular (equilibrium) solution is $y_p = \frac{20}{4} = 5$ (consistent with the given stretch). The general solution is the equilibrium plus the homogeneous solution. The homogeneous characteristic equation is

$$2r^2 + br + 4 = 0,$$

so the homogeneous roots are

$$r = \frac{-b \pm \sqrt{b^2 - 32}}{4}.$$

Thus the general solution is

$y(t) = 5 + C_1 e^{r_1 t} + C_2 e^{r_2 t}, \quad r_{1,2} = \frac{-b \pm \sqrt{b^2 - 32}}{4}.$

(If $b^2 < 32$ write the homogeneous part as $e^{-bt/4}(A \cos(\tilde{\omega}t) + B \sin(\tilde{\omega}t))$ with $\tilde{\omega} = \frac{\sqrt{32 - b^2}}{4}$.)

Practice Problems

Exercise 4.3.3

Solve the following equations:

1. $2y'' - 5y' - 3y = 0$

2. $y'' - 5y' + 6y = 0$

3. $y'' + 9y = 0$

4. $y''' + y'' - y' - y = 0$ ³

³ 1 is a root of the characteristic polynomial

5. Given that 1 is a root of the characteristic polynomial, solve $y''' + 3y'' - 4y = 0$.

6. Solve:

$$4y'' + 4y' + 17y = 0, \quad y(0) = -1, \quad y'(0) = 2.$$

7. Solve:

$$y'' - 3y' + 2y = 0, \quad y(0) = 2, \quad y'(0) = 1.$$

8.

$$y'' + 2y' + 5y = 0, \quad y(0) = 0, \quad y'(0) = 2.$$

9.

$$y'' - 4y' + 4y = 0, \quad y(0) = 3, \quad y'(0) = 0.$$

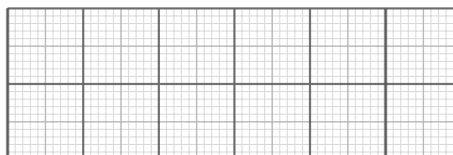
Exercise 4.3.4

The complex-valued function⁴ $f(t)$ depends on real variable t :

⁴ Notation $e^x = \exp(x)$

$$f(t) = \exp[(-1 + i2\pi)t]$$

- Compute the real part of the function $\operatorname{Re} f(t)$ and the imaginary part $\operatorname{Im} f(t)$.
- Sketch the real and imaginary parts for $0 < t < 3$. Make sure to accurately represent growth/decay of each function, their value at $t = 0$, and points at which the functions intersect with the horizontal axis.
- For what values t is the function $f(t)$ purely real (imaginary part is 0) and for what values is it purely imaginary (real part is 0)?



Exercise 4.3.5

Compute solutions to the following linear homogeneous ODEs. In all cases, use real-valued formulas.

(a) $y'' + 4y = 0$

(b) $y'' + 2y' + y = 0$

Exercise 4.3.6

An unforced response of an oscillator can be modeled using the ODE

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = 0,$$

where m is the mass, c is the damping coefficient, and k the spring stiffness.

- For $m = 4$ and $k = 9$ and undetermined $c \geq 0$, find the roots of the characteristic equation.
- Find the ranges of values of c such that the general solution is: (i) decaying and shows no oscillation, (ii) decaying and has oscillations.

- (iii) Choose one value for each range you determined and write out the formula for the general solution.

Exercise 4.3.7

Calculate real and imaginary parts of the following functions ($t \in \mathbb{R}$ always):

- | | | |
|-------------------------------|-------------------------|--------------------------------|
| (a) $(2 + i) \cos(t)$ | (c) $\exp(i2t)$ | (e) $\exp(3t - i2t)$ |
| (b) $3i[\cos(t) - i \sin(t)]$ | (d) $(2 + i) \exp(i2t)$ | (f) $5e^{(2+i\frac{\pi}{4})t}$ |

Exercise 4.3.8

Given $y'' - 2y' + 2y = 0$.

- (a) Verify that $y = c_1 e^x \cos(x) + c_2 e^x \sin(x)$ is a general solution to the given ODE.
 (b) Find c_1 and c_2 so that $y(0) = 1$, and $y'(\pi) = 0$.
 (c) Use Wronskian to verify that $e^x \cos(x)$ and $e^x \sin(x)$ are linearly independent.

Exercise 4.3.9

Find the general (or particular) solution and determine if the solution will exhibit growth or decay.

- | | |
|---------------------------|-------------------------------------|
| (a) $y'' + 8y' + 16y = 0$ | (c) $y'' + 2y' - 8y = 0, y(0) = 7,$ |
| (b) $y'' + 2y' + 2y = 0$ | $y'(0) = 2$ |

Exercise 4.3.10

For each of the following pairs of numbers:

- find quadratic equations with real coefficients that have the following roots,
- describe (oscillating or not, growing/decaying/neither) the shape of solutions of a 2nd order linear ODE whose characteristic equation would be given by the quadratic you just found.

- | | | | |
|-------------|-------------|---------------|----------------|
| (a) $-2, 3$ | (c) $2, 2$ | (e) $2 \pm i$ | (g) $A \pm iB$ |
| (b) $0, 2$ | (d) $\pm i$ | (f) A, B | |

where A, B are both real constants.

Exercise 4.3.11

For each of the functions below

- Give an example of a ODE whose solution is given by the function above.
- Give the initial conditions $y(0)$ and $y'(0)$ of the function.

- | | |
|--------------------------------|----------------------------------|
| (a) $y(x) = 2e^{-3x} + 3e^x$ | (c) $y(x) = xe^{-2x} - 3e^{-2x}$ |
| (b) $y(x) = \sin(x) + \cos(x)$ | (d) $y(x) = e^{-x} \cos(2x)$ |

Exercise 4.3.12

A mass of 4kg is attached to a spring whose spring constant is 16 kg/s². What is the period of the motion of the mass when it is allowed to bounce freely?

Exercise 4.3.13

A mass of 20kg is attached to a spring. If the frequency of its free motion is $2/\pi$ cycles/s, what is the spring constant k ? What is the frequency of free motion if the original mass is replaced with an 80kg mass?

Section 4.4: Method of Undetermined Coefficients - Superposition

While characteristic equations are extremely useful for solving homogeneous ODEs, they do not help much when solving the inhomogeneous subproblem.

Undetermined coefficients is a technique for making **educated guesses** about **candidates** for the inhomogeneous solutions. These candidates have undetermined parameters in them that are computed by plugging the candidate into the DiffEq.

Rely as little as possible on tables of undetermined coefficients. Instead focus on three principles:

- If the input function is a sum of terms, then we can solve inhomogeneous equations with each of the terms in the sum in isolation, and add their solutions together.
- A good candidate for an inhomogeneous solution is formed by a linear combination of derivatives of the input function.
- If the candidate for inhomogeneous solution matches a component in the *homogeneous* solution, multiply the candidate by the independent variable (repeat if necessary).

In a particular case of undamped oscillators, the third principle above corresponds to the **resonance**: growth of the output despite bounded input.

Consider a linear nonhomogeneous differential equation with constant coefficients:

$$L[y] = y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1y' + a_0y = g(x),$$

where L is a linear differential operator with constant coefficients.

Idea

- Solve the **homogeneous equation** $L[y] = 0$ to get the complementary solution y_c .
- Find a **particular solution** y_p by guessing a form similar to $g(x)$ (with undetermined coefficients).
- The general solution is $y(x) = y_c + y_p$.

Superposition Principle

If

$$L[y] = g_1(x) + g_2(x) + \cdots + g_k(x),$$

then

$$y_p = y_{p1} + y_{p2} + \cdots + y_{pk},$$

where each y_{pi} corresponds to a particular solution of $L[y] = g_i(x)$ for $i = 1, 2, \dots, k$.

Example 4.4.1. Solve $y'' - 3y' + 2y = 2x^2 - 3x + 5$

Solution. Homogeneous part:

$$r^2 - 3r + 2 = 0 \quad \Rightarrow \quad (r - 1)(r - 2) = 0.$$

Right-hand side $g(x)$	Trial particular solution $y_p(x)$
$P_n(x)$	General polynomial of degree n : $a_0 + a_1x + \cdots + a_nx^n$
$e^{\alpha x}$	$Ae^{\alpha x}$
$P_n(x)e^{\alpha x}$	Polynomial of degree n times $e^{\alpha x}$: $(a_0 + a_1x + \cdots + a_nx^n)e^{\alpha x}$
$\cos(\beta x)$ or $\sin(\beta x)$	$A \cos(\beta x) + B \sin(\beta x)$
$P_n(x) \cos(\beta x)$	Polynomial of degree n times sines and cosines: $(a_0 + \cdots + a_nx^n) \cos(\beta x) + (b_0 + \cdots + b_nx^n) \sin(\beta x)$
$P_n(x)e^{\alpha x} \cos(\beta x)$	Polynomial of degree n times $e^{\alpha x}$ times sines and cosines
Adjustment Rule	If any term of $y_p(x)$ is a solution of the homogeneous equation, multiply the entire trial solution by the smallest power of x until no overlap remains.

So $y_c = C_1e^x + C_2e^{2x}$.

Example 4.4.2. Find the general solution of $y'' - 3y' + 2y = 2 \sin 3x$.

Example 4.4.3. Solve $y'' - 3y' + 2y = 4x - 5 + 6xe^{2x}$.

Example 4.4.4 (A Glitch in the system). Solve $y'' - 5y' + 4y = 8e^x$.

Example 4.4.5 (Another Glitch in the system). Solve $y'' - 2y' + y = e^x$.

Example 4.4.6. (a) Explain: **What is resonance**, from physical and mathematical standpoints.

(b) What are the possible values of k in the equation below that result in resonance?

$$2 \frac{d^2x}{dt^2} + 4kx(t) = \sin(2t) + e^{3t}$$

(c) Form the **candidate for the forced solution** for the resonant case (that is, when k is set to the value in which resonance occurs). Show your work. (No need to find values of the coefficients, leave them as $A, B, C, \dots$, just forming the candidate is enough.)

Table 4.1: Table of Undetermined Coefficients with adjustment rule for candidate solutions.

Note:

$P_n(x)$ is a polynomial of deg n . Any representation of cos applies for sin too.

Solution:

- (a) **Physical view.** Resonance occurs when a system is driven at one of its natural frequencies. The driving force repeatedly supplies energy at the same frequency that the system naturally oscillates.

Mathematical view. Resonance occurs when a forcing term *matches the frequency one of the homogeneous solutions* (e.g. $e^{\lambda t}$ or $\cos(\omega t), \sin(\omega t)$).

Consider

$$2x'' + 4kx(t) = \sin(2t) + e^{3t}.$$

Divide by 2:

$$x'' + 2kx = \frac{1}{2} \sin(2t) + \frac{1}{2} e^{3t}.$$

For the homogeneous part, **Characteristic equation:**

$$r^2 + 2k = 0 \implies r = \pm \sqrt{-2k}.$$

When $k > 0$, we have that $r = \pm i\sqrt{2k}$, and so 'the' homogeneous solution is

$$x_h = C_1 \cos(\sqrt{2k}t) + C_2 \sin(\sqrt{2k}t).$$

When $k \leq 0$, we have that $r = \pm \sqrt{2|k|}$, and so 'the' homogeneous solution is

$$x_h = C_1 e^{\sqrt{2|k|}t} + C_2 e^{-\sqrt{2|k|}t}$$

- (b) **Finding the values of k that produce resonance.**

- Resonance with the $\sin(2t)$ term: The forcing frequency is 2. Resonance occurs when the natural frequency equals 2,

$$\sqrt{2k} = 2 \implies 2k = 4 \implies k = 2.$$

- Resonance with the e^{3t} term: An exponential forcing e^{3t} resonates if $r = 3$ is a root of the characteristic equation, i.e.

$$r^2 + 2k = 0 \text{ with } r = 3 \implies 9 + 2k = 0 \implies k = -\frac{9}{2}.$$

(For $k = -\frac{9}{2}$ the characteristic roots are $r = \pm 3$, so e^{3t} is a homogeneous mode.)

Resonant values: $k = 2$ (resonance with $\sin 2t$), $k = -\frac{9}{2}$ (resonance with e^{3t}).

- (c) **Candidates for the forced (particular) solution in the resonant cases.**

- (i) For $k = 2$: the homogeneous solution contains $\cos 2t, \sin 2t$. Thus the $\sin 2t$ forcing is resonant. The usual nonresonant trial for $\sin 2t$ is $A \cos 2t + B \sin 2t$, so in the resonant case we multiply by t . The exponential forcing e^{3t} is nonresonant at $k = 2$. Hence a suitable particular-candidate is

$$x_p(t) = t(A \cos 2t + B \sin 2t) + Ce^{3t}, \quad (k = 2).$$

- (ii) For $k = -\frac{9}{2}$: the homogeneous solution contains e^{3t}, e^{-3t} , and the e^{3t} forcing is resonant. Thus the sinusoidal forcing $\sin 2t$ is nonresonant in this case. The usual trial for e^{3t} is De^{3t} , so in the resonant case multiply by t . Hence a suitable candidate for the particular solution is

$$x_p(t) = E \cos 2t + F \sin 2t + t(Ge^{3t}), \quad (k = -\frac{9}{2}).$$

Example 4.4.7. Solve

$$y'' - 3y' + 2y = e^x + \sin x.$$

Solution:

For $g(x) = e^x$: since e^x already appears in y_c , guess

$$y_{p1} = Axe^x.$$

Substitute:

$$y'_{p1} = Ae^x + Axe^x, \quad y''_{p1} = 2Ae^x + Axe^x.$$

So

$$y''_{p1} - 3y'_{p1} + 2y_{p1} = (2Ae^x + Axe^x) - 3(Ae^x + Axe^x) + 2(Axe^x).$$

Simplify:

$$= (2A - 3A)e^x + (A - 3A + 2A)xe^x = -Ae^x.$$

Set equal to e^x : $-A = 1 \Rightarrow A = -1$. So $y_{p1} = -xe^x$.

For $g(x) = \sin x$: guess

$$y_{p2} = A \cos x + B \sin x.$$

Substitute:

$$y'_{p2} = -A \sin x + B \cos x, \quad y''_{p2} = -A \cos x - B \sin x.$$

So

$$y''_{p2} - 3y'_{p2} + 2y_{p2} = (-A \cos x - B \sin x) - 3(-A \sin x + B \cos x) + 2(A \cos x + B \sin x).$$

Simplify:

$$= (-A - 3B + 2A) \cos x + (-B + 3A + 2B) \sin x = (A - 3B) \cos x + (3A + B) \sin x.$$

Match with $\sin x$: system

$$A - 3B = 0, \quad 3A + B = 1.$$

From first: $A = 3B$. Substitute: $3(3B) + B = 1 \Rightarrow 10B = 1 \Rightarrow B = \frac{1}{10}, A = \frac{3}{10}$.

So

$$y_{p2} = \frac{3}{10} \cos x + \frac{1}{10} \sin x.$$

General solution:

$$y(x) = C_1 e^x + C_2 e^{2x} - xe^x + \frac{3}{10} \cos x + \frac{1}{10} \sin x.$$

Practice Problems

Exercise 4.4.1

A particular solution to a second-order ODE is given by

$$y(x) = 2e^{-3x} + 3e^x$$

- Give the initial conditions $y(0)$ and $y'(0)$ of the function above.
- Give an example of a ODE whose solution is given by the function above.

Exercise 4.4.2

Calculate general solutions to the following linear ODEs, and particular solutions to IVPs where appropriate.

(a) $\ddot{x} - 2\dot{x} - 3x = 0, \quad x(0) = 0, \dot{x}(0) = 1$

(b) $\ddot{y} - 5\dot{y} + 6y = 0, \quad y(0) = 1, \dot{y}(0) = -1$

(c) $y'''(x) - 3y''(x) + 2y'(x) = 0$

Exercise 4.4.3

Determine the particular solution of $y'' - y' + 25y = 5x^3e^{-x} - 7e^{-x}$

Exercise 4.4.4

Determine the particular solution of $y'' + 4y = x \cos x$.

Exercise 4.4.5

Solve the following using the superposition method:

1. $y'' + y = 5 \cos(2x)$

2. $y'' - y = xe^x$

3. $y'' + 4y' + 4y = e^{-2x} + x$

Exercise 4.4.6

For each of the following, determine the homogeneous solution and the **form** (do not solve) of a particular non-homogeneous solution:

(a) $y'' + 4y = 2x^2$

(c) $y'' + 4y = \sin(2x)$

(b) $y'' + 4y = 10 + \sin(x)$

(d) $y'' + 4y = e^x \sin(2x)$

Exercise 4.4.7

For the differential equation

$$y'' + 9y = x + \sin(x)$$

(i) Solve

(a) Homog. Subproblem

(b) Inhomog. Subproblem

(ii) General solution using part (i).

(iii) Particular solution that satisfies $y(0) = 1, y'(0) = 0$.

Exercise 4.4.8

A mass of 24kg, attached to the end of a spring of unknown coefficient k , stretches it 4cm under influence of gravity ($g \approx 10\text{ms}^{-2}$) when the mass is not moving.

(i) Write the differential equation for the system.

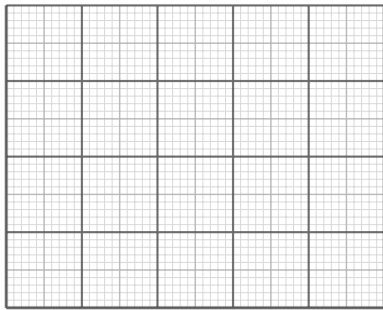
- (ii) Compute the equilibrium solution for the equation and determine spring coefficient from it.
- (iii) Initially, the mass is released from rest from a point 3cm above the equilibrium position. Find the particular solution of the differential equation.
- (iv) In a second scenario, a periodic force $\cos(\Omega t)$ is applied to the mass in addition to gravity. What is the value of frequency Ω which excites resonance in the motion of the weight?

SAGEMATH Question 4.4.9

$$\ddot{x} + 4x = \sin(\Omega t), \quad x(0) = p_0, \dot{x}(0) = v_0 \quad (4.1)$$

by changing the line 10 appropriately. ⁵

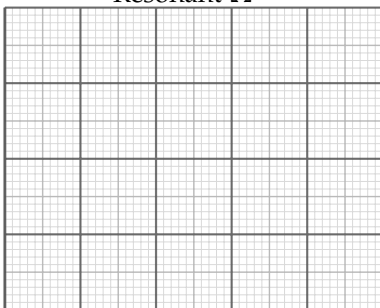
- (a) Copy the sketch for $\Omega = 1$ into the graph below. ⁶



- (b) Describe how the graphs are different compared to the graphs without $\sin(\Omega t)$ added (you can look back to Week 5).
- (c) Are these new curves growing, decaying, or neither? Are they oscillating or not?
- (d) Write down the characteristic equation and find its roots. What is the value of the *resonant frequency* of this oscillator?
- (e) Set Ω equal to the *resonant frequency* of the differential equation. Then copy the graph into the space on the right.

Resonant $\Omega =$

Resonant $\Omega =$



Are the curves growing, decaying, or neither?

⁵ To represent symbol Ω you can add a variable `Omega` in the same fashion as `mass`, `damping`, etc. are defined. Set $\Omega = 1$ initially.

⁶ Set `xmax=20` in lines 22 and 28 and `ymin=-10` and `ymax=20` at the end of the code to show you the full extent of solution curves.

Exercise 4.4.10

In each of the following equations, undetermined coefficient candidate for the inhomogeneous solution is incorrectly proposed.

- Proceed with the incorrect candidate and **show how the solution process fails**.
- **Propose the correct undetermined coefficients candidate**.
- **Find the coefficients** of the inhomogeneous solution.

(a) $y'' + 2y' + 5y = x^2$, incorrect candidate: $y_p(x) = Ax^2 + B$

(b) $y'' + 9y = \sin(x)$, incorrect candidate: $y_p(x) = A \sin(x)$

(c) $y'' + 9y = \sin(3x)$, incorrect candidate: $y_p(x) = A \cos(3x) + B \sin(3x)$

Exercise 4.4.11

For each of the following, determine the homogeneous solution and the **form** of a particular non-homogeneous solution:

(a) $y'' + 3y' + 2y = -10x^2$

(d) $y'' + 4y' + 13y = 2xe^{-2x} \sin(3x)$

(b) $y'' + 3y' + 2y = e^{4x}$

(e) $y'' + 2y' + 2y = 5e^{-x} \cos(x)$

(c) $y'' + 2y' = x^2 + 5x$

(f) $y'' + 6y' + 9y = x^3e^{-3x}$

Exercise 4.4.12

Using the method of undetermined coefficients to find the general solution:

(i) $y'' + 3y' + 2y = -10x^2$

(ii) $y'' + 3y' + 2y = e^{4x}$

(iii) $y'' + y' - 2y = 20 \sin(2x)$

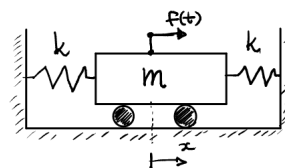
(iv) $y''' - 2y'' + y' = 2 - 24e^x + 40e^{5x}$, $y(0) = 1/2$, $y'(0) = 5/2$, $y''(0) = -9/2$.

Exercise 4.4.13

A weight of 8kg hung from a spring stretches the spring 3cm. The mass-spring system is additionally encumbered by air resistance, which is proportional to velocity with coefficient 1kg/s. The mass moves under the influence of gravity ($g \approx 10\text{ms}^{-2}$) and an external force of $10 \cos(4t)\text{kgms}^{-2}$ (t is seconds after the release). If the weight is pulled down 1m from its equilibrium position and given an initial upward velocity of 2m/s, write an initial value problem for the position of the object at time t seconds and solve.

Exercise 4.4.14

The figure in the margin shows a cart strapped by two elastic springs to walls. Sketch a free-body diagram for the cart and use it to write down the 2nd order ODE for position variable x using Newton's laws of motion. Leave $f(t)$ as an unknown force function.



Derivation:

Sum of forces (positive right):

$$m\ddot{x} = f(t) + F_1 + F_2.$$

Using Hooke's law with equilibrium at $x = 0$:

$$F_1 = -k_1x, \quad F_2 = -k_2x.$$

Therefore

$$m\ddot{x} = f(t) - (k_1 + k_2)x \implies$$

$$m\ddot{x} + (k_1 + k_2)x = f(t).$$

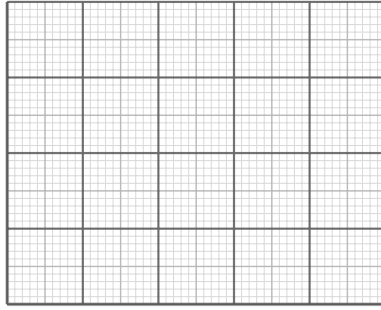
⁷ Code for mass-damper-spring oscillator:
<https://goo.gl/p9EVdC>

SAGEMATH Question 4.4.15

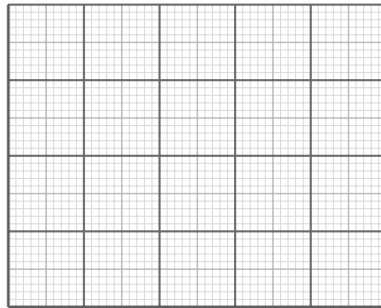
The **SAGEMATH** code in the link⁷ computes solutions to the mass-damper-spring mechanical oscillator evolving from initial position p_0 and initial velocity v_0 , given by equation:

$$m\ddot{x} + c\dot{x} + kx = 0, \quad x(0) = p_0, \dot{x}(0) = v_0 \quad (4.2)$$

- (a) Copy the sketch into the graph below. Pay special attention to the points in which either of the functions intersects with the horizontal.



- (b) What is the position curve doing when the velocity curve intersects the horizontal?
- (c) Are these curves growing, decaying (transient), or neither? Are they oscillating or not?
- (d) Slowly increase damping c by modifying the code until the curves stop oscillating. Try to find the smallest positive c that stops the oscillations. Then copy the graph into the space on the right and note the value of this *critical damping* c .



Critical $c =$

Exercise 4.4.16

Verify that $y(x) = e^x \sin(x)$ satisfies $y''(x) - 2y'(x) + 2y(x) = 0$.

Exercise 4.4.17

Given a linear ODE $y'' = y' + 12y$.

- (a) Is this equation homogeneous or not? How do you know?
- (b) Calculate the general solution of the ODE.
- (c) Calculate c_1 and c_2 so that $y(0) = 1$, and $y'(\pi) = 0$.

Section 4.5: Method of Undetermined Coefficients: Annihilator

we will not cover this approach in class, but you may review and use it.

The **annihilator method** uses differential operators to systematically find particular solutions.

Steps

1. Write the equation as $L[y] = g(x)$.
2. Find a differential operator A (called the annihilator) such that $Ag(x) = 0$.
3. Apply A to both sides: $AL[y] = 0$, giving a higher-order homogeneous equation.
4. Solve for the general solution of $AL[y] = 0$.
5. Extract the particular solution from the part not included in the original homogeneous solution.

Example 4.5.1. Solve

$$y'' - y = e^{2x}.$$

Solution.

Step 1: Homogeneous part:

$$r^2 - 1 = 0 \Rightarrow r = \pm 1.$$

So $y_c = C_1 e^x + C_2 e^{-x}$.

Step 2: RHS is $g(x) = e^{2x}$. Annihilator is $(D - 2)$, since $(D - 2)e^{2x} = 0$.

Step 3: Apply $(D - 2)$ to both sides:

$$(D - 2)(D^2 - 1)y = 0.$$

Step 4: Characteristic equation:

$$(r - 2)(r^2 - 1) = 0 \Rightarrow (r - 2)(r - 1)(r + 1) = 0.$$

So roots: 2, 1, -1. General solution:

$$y(x) = C_1 e^x + C_2 e^{-x} + C_3 e^{2x}.$$

Step 5: Since $C_1 e^x + C_2 e^{-x}$ are already solutions of original homogeneous part, the new term $C_3 e^{2x}$ is the particular solution.

Thus the general solution is

$$y(x) = C_1 e^x + C_2 e^{-x} + C_3 e^{2x}.$$

Example 4.5.2. Solve

$$y'' + y = \sin x.$$

Solution. Homogeneous part:

$$r^2 + 1 = 0 \Rightarrow r = \pm i.$$

So $y_c = C_1 \cos x + C_2 \sin x$.

RHS $g(x) = \sin x$. Annihilator: $(D^2 + 1)$.

Apply to both sides:

$$(D^2 + 1)(D^2 + 1)y = 0.$$

Characteristic equation:

$$(r^2 + 1)^2 = 0,$$

roots $r = \pm i$ of multiplicity 2. General solution:

$$y(x) = (C_1 + C_2x) \cos x + (C_3 + C_4x) \sin x.$$

The terms $C_1 \cos x + C_2 \sin x$ come from homogeneous solution, while $C_3x \sin x + C_4x \cos x$ represent the particular solution.

So general solution is

$$y(x) = C_1 \cos x + C_2 \sin x + C_3x \cos x + C_4x \sin x.$$

Exercise 4.5.1

Use the annihilator method to solve:

1. $y'' - 4y = 3e^{2x}$
2. $y'' + y = \cos 2x$
3. $y'' - 2y' + y = xe^x$

Section 4.6: Variation of Parameters for Higher-Order Linear ODEs

Example 4.6.1 (Motivating Example). Let's solve a motivating example:

Solve

$$y'' + y = 1.$$

For a first-order linear ODE in standard form

$$y' + p(x)y = q(x)y,$$

let $\{y_h\}$ be a solution of the homogeneous equation

$$y' + p(x)y = 0.$$

The particular solution via variation of parameters is sought as

$$y_p = u(x)y_h(x)$$

For a second-order linear ODE in standard form

$$y'' + p(x)y' + q(x)y = g(x),$$

let $\{y_1, y_2\}$ be a fundamental set of solutions of the homogeneous equation

$$y'' + py' + qy = 0.$$

The particular solution via variation of parameters is sought as

$$y_p = u_1(x)y_1(x) + u_2(x)y_2(x),$$

Variation of Parameters (second order)

For a second-order linear ODE in standard form

$$y'' + p(x)y' + q(x)y = g(x),$$

let $\{y_1, y_2\}$ be a fundamental set of solutions of the homogeneous equation

$$y'' + py' + qy = 0.$$

The particular solution via variation of parameters is sought as

$$y_p = u_1(x)y_1(x) + u_2(x)y_2(x),$$

with the auxiliary conditions

$$\begin{cases} u_1'y_1 + u_2'y_2 = 0, \\ u_1'y_1' + u_2'y_2' = g(x). \end{cases}$$

Solving this 2×2 linear system gives the standard formulas

$$u_1' = -\frac{g y_2}{W}, \quad u_2' = \frac{g y_1}{W},$$

where $W = W(y_1, y_2) = y_1 y_2' - y_1' y_2$ is the Wronskian. Integrate u_1', u_2' to obtain u_1, u_2 and thus y_p .

Below are three worked examples illustrating different typical right-hand sides.

Example 4.6.2. Solve

$$y'' + y = 1.$$

Solution.

Homogeneous basis: $y_1 = \cos x$, $y_2 = \sin x$, $W = 1$ as before.

$$u_1' = -1 \cdot \sin x = -\sin x, \quad u_2' = 1 \cdot \cos x = \cos x.$$

Integrate:

$$u_1 = \int -\sin x \, dx = \cos x, \quad u_2 = \int \cos x \, dx = \sin x.$$

Particular solution:

$$y_p = u_1 \cos x + u_2 \sin x = \cos x \cos x + \sin x \sin x = \cos^2 x + \sin^2 x = 1.$$

General solution:

$$y(x) = C_1 \cos x + C_2 \sin x + 1.$$

(As expected, the constant function 1 is a simple particular solution.)

Example 4.6.3. Solve

$$y'' + y = \sec x, \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Solution.

Homogeneous solutions: $y_1 = \cos x$, $y_2 = \sin x$. Wronskian:

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1.$$

Formulas:

$$u'_1 = -g \frac{y_2}{W} = -\sec x \sin x = -\tan x, \quad u'_2 = g \frac{y_1}{W} = \sec x \cos x = 1.$$

Integrate:

$$u_1 = \int -\tan x \, dx = \ln |\cos x|, \quad u_2 = \int 1 \, dx = x.$$

Particular solution:

$$y_p = u_1 \cos x + u_2 \sin x = \cos x \ln |\cos x| + x \sin x.$$

General solution:

$$y(x) = C_1 \cos x + C_2 \sin x + \cos x \ln |\cos x| + x \sin x, \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Example 4.6.4. Solve

$$y'' - y = e^{2x}.$$

Solution.

Put in standard form $y'' + 0 \cdot y' + (-1)y = e^{2x}$. Homogeneous solutions: $y_1 = e^x$, $y_2 = e^{-x}$. Wronskian:

$$W = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} = -e^x e^{-x} - e^x e^{-x} = -2.$$

Formulas:

$$u'_1 = -\frac{g y_2}{W} = -\frac{e^{2x} e^{-x}}{-2} = \frac{e^x}{2}, \quad u'_2 = \frac{g y_1}{W} = \frac{e^{2x} e^x}{-2} = -\frac{e^{3x}}{2}.$$

Integrate:

$$u_1 = \int \frac{e^x}{2} \, dx = \frac{e^x}{2}, \quad u_2 = \int -\frac{e^{3x}}{2} \, dx = -\frac{e^{3x}}{6}.$$

Particular solution:

$$y_p = u_1 e^x + u_2 e^{-x} = \frac{e^x}{2} e^x - \frac{e^{3x}}{6} e^{-x} = \frac{e^{2x}}{2} - \frac{e^{2x}}{6} = \frac{1}{3} e^{2x}.$$

General solution:

$$y(x) = C_1 e^x + C_2 e^{-x} + \frac{1}{3} e^{2x}.$$

Example 4.6.5. Determine the particular solution of $y'' + 4y = x \cos x$ using the variation of parameters approach.

Consider the n th-order linear nonhomogeneous differential equation

$$y^{(n)} + a_1(t)y^{(n-1)} + a_2(t)y^{(n-2)} + \cdots + a_n(t)y = g(t),$$

where the coefficients $a_j(t)$ and the forcing term $g(t)$ are continuous functions on an interval I .

Step 1: Solve the homogeneous equation

The corresponding homogeneous equation

$$y^{(n)} + a_1(t)y^{(n-1)} + \cdots + a_n(t)y = 0$$

has a fundamental set of solutions $\{y_1(t), y_2(t), \dots, y_n(t)\}$. Thus the general solution of the homogeneous equation is

$$y_h(t) = C_1y_1(t) + C_2y_2(t) + \cdots + C_ny_n(t).$$

Step 2: Assume a particular solution

In Variation of Parameters, we replace the constants C_j by functions $u_j(t)$:

$$y_p(t) = u_1(t)y_1(t) + u_2(t)y_2(t) + \cdots + u_n(t)y_n(t).$$

Step 3: Impose conditions

To ensure uniqueness of the functions $u_j(t)$, impose the auxiliary conditions

$$\begin{aligned} u'_1(t)y_1(t) + u'_2(t)y_2(t) + \cdots + u'_n(t)y_n(t) &= 0, \\ u'_1(t)y'_1(t) + u'_2(t)y'_2(t) + \cdots + u'_n(t)y'_n(t) &= 0, \\ &\vdots \\ u'_1(t)y_1^{(n-2)}(t) + \cdots + u'_n(t)y_n^{(n-2)}(t) &= 0. \end{aligned}$$

This leaves only the final condition involving $y_p^{(n)}$ free to capture the forcing $g(t)$.

Step 4. System of equations

The remaining condition gives

$$u'_1(t)y_1^{(n-1)}(t) + u'_2(t)y_2^{(n-1)}(t) + \cdots + u'_n(t)y_n^{(n-1)}(t) = g(t).$$

Hence we obtain an $n \times n$ linear system in the unknowns $u'_1(t), \dots, u'_n(t)$:

$$\begin{bmatrix} y_1(t) & y_2(t) & \cdots & y_n(t) \\ y'_1(t) & y'_2(t) & \cdots & y'_n(t) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(t) & y_2^{(n-1)}(t) & \cdots & y_n^{(n-1)}(t) \end{bmatrix} \begin{bmatrix} u'_1(t) \\ u'_2(t) \\ \vdots \\ u'_n(t) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ g(t) \end{bmatrix}.$$

Step 5: Use the Wronskian

The matrix above is the *Wronskian* of $\{y_1, \dots, y_n\}$:

$$W(t) = \det \begin{bmatrix} y_1(t) & \cdots & y_n(t) \\ y'_1(t) & \cdots & y'_n(t) \\ \vdots & \ddots & \vdots \\ y_1^{(n-1)}(t) & \cdots & y_n^{(n-1)}(t) \end{bmatrix}.$$

By Cramer's rule, each $u'_j(t)$ is given by

$$u'_j(t) = \frac{\det W_j(t)}{W(t)},$$

where $W_j(t)$ is obtained from $W(t)$ by replacing the j th column with $(0, 0, \dots, 0, g(t))^T$.

Step 6: Integrate

Finally,

$$u_j(t) = \int u'_j(t) dt,$$

and the particular solution is

$$y_p(t) = \sum_{j=1}^n u_j(t) y_j(t).$$

Therefore the general solution of the nonhomogeneous ODE is

$$y(t) = y_h(t) + y_p(t).$$

Section 4.7: Cauchy–Euler Equations

A **Cauchy–Euler equation** (or **equidimensional equation**) is a linear differential equation of the form

$$a_n x^n y^{(n)} + a_{n-1} x^{n-1} y^{(n-1)} + \dots + a_1 x y' + a_0 y = g(x), \quad x > 0,$$

where the coefficients are monomials in x of the same degree as the derivative order.

In the second-order case,

$$a_2 x^2 y'' + a_1 x y' + a_0 y = g(x).$$

Homogeneous Case

Consider

$$a_2 x^2 y'' + a_1 x y' + a_0 y = 0.$$

We look for a solution of the form $y = x^r$, $x > 0$.

Characteristic Equation

Substituting $y = x^r$ gives

$$a_2 r(r-1)x^r + a_1 r x^r + a_0 x^r = 0.$$

Dividing by x^r (valid for $x > 0$):

$$a_2 r(r-1) + a_1 r + a_0 = 0.$$

This quadratic in r is the **indicial equation**.

Solution Forms

1. **Distinct real roots** $r_1 \neq r_2$:

$$y_h(x) = C_1 x^{r_1} + C_2 x^{r_2}.$$

2. **Repeated root** r :

$$y_h(x) = C_1 x^r + C_2 x^r \ln x.$$

3. **Complex roots** $r = \alpha \pm i\beta$:

$$y_h(x) = x^\alpha (C_1 \cos(\beta \ln x) + C_2 \sin(\beta \ln x)).$$

Nonhomogeneous Equations

For

$$a_2x^2y'' + a_1xy' + a_0y = g(x),$$

the general solution is

$$y(x) = y_h(x) + y_p(x),$$

where y_p is a particular solution (often found via the method of undetermined coefficients or variation of parameters).

Example 4.7.1 (Distinct Real Roots). *Solve*

$$x^2y'' - 3xy' + 4y = 0, \quad x > 0.$$

Solution: *Indicial equation:*

$$r(r-1) - 3r + 4 = r^2 - 4r + 4 = (r-2)^2 = 0.$$

Repeated root $r = 2$. So the general solution is

$$y(x) = C_1x^2 + C_2x^2 \ln x.$$

Example 4.7.2 (Complex Roots). *Solve*

$$x^2y'' + xy' + y = 0, \quad x > 0.$$

Solution: *Indicial equation:*

$$r(r-1) + r + 1 = r^2 + 1 = 0.$$

Roots: $r = \pm i$. So

$$y_h(x) = x^0 (C_1 \cos(\ln x) + C_2 \sin(\ln x)).$$

Hence

$$y(x) = C_1 \cos(\ln x) + C_2 \sin(\ln x).$$

Example 4.7.3. *Solve*

$$x^2y'' - xy' - y = x^2, \quad x > 0.$$

Solution:

Indicial equation:

$$r(r-1) - r - 1 = r^2 - 2r - 1 = 0.$$

Roots: $r = 1 \pm \sqrt{2}$. Distinct reals. So

$$y_h(x) = C_1x^{1+\sqrt{2}} + C_2x^{1-\sqrt{2}}.$$

Try a particular solution of the form $y_p = Ax^2$. Compute:

$$y_p' = 2Ax, \quad y_p'' = 2A.$$

Substitute:

$$x^2(2A) - x(2Ax) - Ax^2 = 2Ax^2 - 2Ax^2 - Ax^2 = -Ax^2.$$

We want $-Ax^2 = x^2 \implies A = -1$. So

$$y_p = -x^2.$$

Thus general solution:

$$y(x) = C_1 x^{1+\sqrt{2}} + C_2 x^{1-\sqrt{2}} - x^2.$$

Practice for test — Higher order linear ODEs

There is no homework due in this discussion. Instead, study from the previous weeks and additionally work through problems below. You can use the following step-by-step calculator to help you study: <https://goo.gl/sZEyvL>. Do not rely on it too much, though, as you will not have such help in an exam.

Exercise 4.7.1

Solve each of the following Cauchy–Euler equations:

1. $x^2 y'' - 5xy' + 9y = 0$.
2. $x^2 y'' + 3xy' + y = 0$.
3. $x^2 y'' - 4xy' + 6y = x^2$.
4. $x^2 y'' + 2xy' - 3y = \ln x$.
5. $x^2 y'' - 2xy' + 2y = 0$.

Exercise 4.7.2

Use variation of parameters to find a particular solution and the general solution on an appropriate interval for each ODE:

1. $y'' + y = \ln x$, $x > 0$.
2. $y'' - 4y = e^{2x}$.
3. $y'' + 4y = \sec(2x)$, $x \in (-\frac{\pi}{4}, \frac{\pi}{4})$.
4. $y'' + y = \cos x$ (use variation of parameters even though undetermined coefficients applies).

Exercise 4.7.3

Find the general solution of the following ODE by completing each of the subparts.

$$\frac{1}{5}y''(x) + \frac{4}{5}y'(x) + 4y(x) = 4x^2.$$

- (a) Put the equation into the standard form and circle the input function $f(x)$.
- (b) Find the solution to the homogeneous subproblem.
- (c) According to undetermined coefficients prescription, formulate the **candidate forced (inhomogeneous) solution** for the inhomogeneous subproblem. **Explain why you chose that form.**
- (d) Use the solution candidate to find a forced (inhomogeneous) solution.

- (e) Use the forced solution and the homogeneous solution to formulate the general solution.
- (f) Give an example of an input function $f(x)$ that would result in *resonance* occurring. What would the candidate forced solution look like in that case?

Exercise 4.7.4

Verify that the functions $\cos(\ln(x))$, $\sin(\ln(x))$ are a fundamental solution set for $x^2 y'' + xy' + y = 0$, $x > 0$, and form the general solution for the ODE.

Exercise 4.7.5

For each of the following ODEs, use any technique appropriate to find the **general solution**. Then use the initial condition to find the **particular solution**. Avoid using imaginary numbers in your solution (that is, use cos and sin functions when appropriate).

- (a) $3y'' - 21y' + 30y = 2 + t^2 + t$, $y(0) = 1, y'(0) = 1$
- (b) $3y'' - 30y' + 75y = 6x \sin(x)$, $y(0) = 0, y'(0) = 1$
- (c) $y'' - 4y = x^{-1}e^{2x}$, $y(1) = 1, y'(1) = 1$
- (d) $3y'' - 30y' + 87 = 3 + 6 \cos(3x)$, $y(0) = -1, y'(0) = 0$

Exercise 4.7.6

Which of the following three ODEs models resonance? Find forced (inhomogeneous) solution of each of the three equations and sketch them.

- (a) $y'' + 4y = \sin(x)$
- (b) $y'' + 4y = e^x \sin(2x)$
- (c) $y'' + 4y = \sin(2x)$

Modeling with Higher-Order Differential Equations

Many physical systems are governed by higher-order linear differential equations. Important examples include:

- **Mechanical vibration models** (mass-spring-damper systems),
- **Electrical circuits** (involving inductors, capacitors, and resistors).

These models typically lead to second-order (sometimes higher-order) linear differential equations with constant coefficients. The structure of solutions is determined by the roots of the characteristic polynomial.

Mass-Spring-Damper Systems

Given a physical spring-mass (and possibly damper and external force) problem, follow these steps.

1. **Draw the free-body diagram.** Indicate mass m , spring with stiffness k (force $-kx$), damper with coefficient c (force $-c\dot{x}$), and external force $F(t)$ (positive in $+x$ direction).

2. **Apply Newton's 2nd law:** sum of forces = $m\ddot{x}$. This gives

$$m\ddot{x} + c\dot{x} + kx = F(t).$$

(If there is no damper, set $c = 0$. If no external forcing, set $F(t) = 0$.)

3. **Put into standard form:** divide by m to get

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \frac{F(t)}{m},$$

where

$$\omega_n = \sqrt{\frac{k}{m}} \quad (\text{undamped natural frequency}), \quad \zeta = \frac{c}{2\sqrt{mk}} \quad (\text{damping ratio}).$$

(Equivalently $2\zeta\omega_n = c/m$.)

4. **Classify the system (free motion):**

- $\zeta = 0$: undamped, pure oscillation at ω_n .
- $0 < \zeta < 1$: **underdamped** — decaying oscillations with damped frequency $\omega_d = \omega_n\sqrt{1 - \zeta^2}$.
- $\zeta = 1$: **critically damped** — fastest decay without oscillation.
- $\zeta > 1$: **overdamped** — two real negative exponent rates, no oscillation.

5. **Solve:** find homogeneous solution $x_h(t)$ from the characteristic equation $r^2 + 2\zeta\omega_nr + \omega_n^2 = 0$. If $F(t) \neq 0$, find a particular solution $x_p(t)$ (e.g. undetermined coefficients for sinusoidal forcing, or variation of parameters).

6. **Apply initial conditions:** given $x(0) = x_0$ (often called the amplitude) and $\dot{x}(0) = v_0$, determine constants.

7. **Physical interpretation:** the total solution is $x(t) = x_h(t) + x_p(t)$. Transient x_h decays if $\zeta > 0$; long-time behavior is x_p (steady-state).

Useful closed-form expressions

- Natural frequency: $\omega_n = \sqrt{\frac{k}{m}}$.
- Damping ratio: $\zeta = \frac{c}{2\sqrt{mk}}$.
- Damped frequency (if $0 \leq \zeta < 1$): $\omega_d = \omega_n\sqrt{1 - \zeta^2}$.
- For sinusoidal forcing $F(t) = F_0 \cos(\omega t)$, the steady-state amplitude is

$$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2}} = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}.$$

The steady-state response is

$$x_p(t) = A(\omega) \cos(\omega t - \phi(\omega)), \quad \phi(\omega) = \arctan \frac{c\omega}{k - m\omega^2}.$$

- Resonant-like peak occurs near $\omega \approx \omega_n$ for light damping. For small ζ the peak frequency is approximately $\omega_r \approx \omega_n\sqrt{1 - 2\zeta^2}$ (valid for $\zeta \ll 1$).

Example 4.7.4 (Undamped free: given mass and amplitude). $m = 2 \text{ kg}$, $k = 8 \text{ N/m}$.

The mass is released from rest with initial displacement $x(0) = 0.10 \text{ m}$ and $\dot{x}(0) = 0$. No damping, $c = 0$, no forcing.

Model & solution:

$$m\ddot{x} + kx = 0 \Rightarrow \ddot{x} + \frac{k}{m}x = 0.$$

Compute natural frequency:

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{8}{2}} = \sqrt{4} = 2 \text{ rad/s}.$$

General solution:

$$x(t) = C_1 \cos(\omega_n t) + C_2 \sin(\omega_n t).$$

Initial conditions: $x(0) = C_1 = 0.10$. Velocity:

$$\dot{x}(t) = -\omega_n C_1 \sin(\omega_n t) + \omega_n C_2 \cos(\omega_n t),$$

so $\dot{x}(0) = \omega_n C_2 = 0 \Rightarrow C_2 = 0$. Hence

$$x(t) = 0.10 \cos(2t) \quad (m).$$

Example 4.7.5 (Damped free (underdamped) with initial amplitude). **Problem:** $m = 2 \text{ kg}$, $k = 8 \text{ N/m}$, $c = 1 \text{ Ns/m}$. Initial $x(0) = 0.10 \text{ m}$, $\dot{x}(0) = 0$. No forcing.

Model: $m\ddot{x} + c\dot{x} + kx = 0$. Compute parameters:

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{8}{2}} = 2 \text{ rad/s}.$$

Damping ratio:

$$\zeta = \frac{c}{2\sqrt{mk}} = \frac{1}{2\sqrt{2 \cdot 8}} = \frac{1}{2\sqrt{16}} = \frac{1}{2 \cdot 4} = \frac{1}{8} = 0.125.$$

Since $0 < \zeta < 1$, system is underdamped. Damped frequency:

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} = 2\sqrt{1 - (0.125)^2} = 2\sqrt{1 - 0.015625} = 2\sqrt{0.984375}.$$

Compute $\sqrt{0.984375}$. Note $0.984375 = 63/64$; $\sqrt{63} \approx 7.9372539$ so

$$\sqrt{0.984375} = \frac{\sqrt{63}}{8} \approx \frac{7.9372539}{8} = 0.9921567375.$$

Thus

$$\omega_d \approx 2 \times 0.9921567375 = 1.984313475 \text{ rad/s}.$$

General solution (underdamped form):

$$x(t) = e^{-\zeta\omega_n t} (C_1 \cos(\omega_d t) + C_2 \sin(\omega_d t)).$$

Apply initial conditions. At $t = 0$:

$$x(0) = C_1 = 0.10.$$

Velocity:

$$\dot{x}(t) = -\zeta\omega_n e^{-\zeta\omega_n t} (C_1 \cos \omega_d t + C_2 \sin \omega_d t) + e^{-\zeta\omega_n t} (-C_1 \omega_d \sin \omega_d t + C_2 \omega_d \cos \omega_d t).$$

Evaluate at $t = 0$:

$$\dot{x}(0) = -\zeta\omega_n C_1 + \omega_d C_2 = 0.$$

So

$$C_2 = \frac{\zeta\omega_n C_1}{\omega_d} = \frac{(0.125)(2)(0.10)}{1.984313475} = \frac{0.025}{1.984313475}.$$

Compute $0.025/1.984313475 \approx 0.012597$. Thus $C_2 \approx 0.012597$.

Final solution (numeric form)

$$x(t) = e^{-0.25t} (0.10 \cos(1.984313t) + 0.012597 \sin(1.984313t)) \text{ m}$$

where the exponential envelope is $e^{-\zeta\omega_n t} = e^{-0.125 \cdot 2t} = e^{-0.25t}$. Transient decays with time constant $1/(\zeta\omega_n) = 1/0.25 = 4$ seconds.

Example 4.7.6 (Damped and driven (steady-state amplitude and phase)).

Problem: $m = 1$ kg, $k = 4$ N/m, $c = 0.5$ Ns/m. Forcing $F(t) = F_0 \cos(\omega t)$ with $F_0 = 2$ N and forcing frequency $\omega = 1.5$ rad/s. Find the steady-state amplitude A and phase ϕ .

Model:

$$\ddot{x} + \frac{c}{m} \dot{x} + \frac{k}{m} x = \frac{F_0}{m} \cos(\omega t).$$

Compute parameters:

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{4} = 2 \text{ rad/s}, \quad 2\zeta\omega_n = \frac{c}{m} = 0.5 \Rightarrow \zeta = \frac{0.5}{2 \cdot 2} = 0.125.$$

(So lightly damped.)

Amplitude formula:

$$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2}} = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}.$$

Compute numerator and denominator step-by-step:

$$k - m\omega^2 = 4 - 1 \cdot (1.5)^2 = 4 - 2.25 = 1.75.$$

$$(c\omega)^2 = (0.5 \times 1.5)^2 = (0.75)^2 = 0.5625.$$

$$(k - m\omega^2)^2 = (1.75)^2 = 3.0625.$$

Denominator squared = $3.0625 + 0.5625 = 3.625$. Thus denominator = $\sqrt{3.625}$.

Compute $\sqrt{3.625}$ approximately: $\sqrt{3.625} \approx 1.902627$. Then

$$A = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}} = \frac{2}{1.902627} \approx 1.0516 \text{ (m)}.$$

Phase:

$$\phi(\omega) = \arctan \frac{c\omega}{k - m\omega^2} = \arctan \frac{0.75}{1.75}.$$

Compute ratio $0.75/1.75 = 0.428571$. So $\phi \approx \arctan(0.428571) \approx 0.4049 \text{ rad} \approx 23.2^\circ$.

Steady-state solution:

$$x_p(t) = 1.0516 \cos(1.5t - 0.4049)$$

The full solution is $x(t) = x_h(t) + x_p(t)$, where x_h is the transient homogeneous solution which decays (because $c > 0$), leaving x_p as $t \rightarrow \infty$.

Quick checklist when given a problem

1. Identify $m, k, c, F(t)$. If “amplitude” is quoted, check whether it means initial displacement or forcing amplitude F_0 .
2. Write $m\ddot{x} + c\dot{x} + kx = F(t)$. Divide by m to get standard form.
3. Compute $\omega_n = \sqrt{k/m}$ and $\zeta = c/(2\sqrt{mk})$.
4. Classify damping: $\zeta = 0, 0 < \zeta < 1, \zeta = 1, \zeta > 1$.
5. For free motion: solve characteristic equation and use initial conditions to find constants.
6. For sinusoidal forcing: compute $A(\omega)$ and $\phi(\omega)$ with formulas above; combine with decaying transient.
7. Check units and typical magnitudes (time constants $\tau = 1/(\zeta\omega_n)$, period $T = 2\pi/\omega_d$ when underdamped).

Mass m attached to a spring (stiffness k) and damper (damping constant c). Displacement $x(t)$ from equilibrium. Newton’s second law gives:

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = F(t),$$

where $F(t)$ is an external force.

Cases

1. **Free vibration without damping ($c = 0, F = 0$):**

$$m\ddot{x} + kx = 0.$$

Oscillatory motion with frequency $\omega = \sqrt{\frac{k}{m}}$.

2. **Free vibration with damping ($F = 0$):**

$$m\ddot{x} + c\dot{x} + kx = 0.$$

Behavior depends on discriminant $c^2 - 4mk$:

- Overdamped: $c^2 > 4mk \Rightarrow$ real distinct roots.
- Critically damped: $c^2 = 4mk \Rightarrow$ repeated root.
- Underdamped: $c^2 < 4mk \Rightarrow$ oscillatory decay.

3. **Forced vibration ($F(t) \neq 0$):**

$$m\ddot{x} + c\dot{x} + kx = F(t).$$

General solution = homogeneous (transient) + particular (steady-state).

Practice problems

1. $m = 1$ kg, $k = 9$ N/m, $c = 0$. Release from rest at $x(0) = 0.2$ m. Find $x(t)$.
2. $m = 3$ kg, $k = 27$ N/m, $c = 6$ N s/m. With $x(0) = 0.1$ m and $\dot{x}(0) = 0$, find the solution and state whether it oscillates.
3. $m = 1$, $k = 4$, $c = 0.2$, $F(t) = 1.5 \cos(1.9t)$. Compute the steady-state amplitude and phase.
4. **Resonance:** For $m = 1$, $k = 4$, compute the forcing frequency that maximizes the steady-state amplitude for $c = 0.1$ (approximate numeric answer).

Electrical Series Circuits

RLC Circuit

For a series circuit with resistance R , inductance L , and capacitance C connected to a voltage source $E(t)$, let $q(t)$ be the charge on the capacitor, and $i(t) = q'(t)$ be the current.

Kirchhoff's voltage law gives:

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t).$$

Cases

1. **Free oscillation:**

$$Lq'' + Rq' + \frac{1}{C}q = 0.$$

Same classification as mass-spring-damper: overdamped, critically damped, underdamped.

2. **Forced oscillation:**

$$Lq'' + Rq' + \frac{1}{C}q = E(t).$$

Solution = transient + steady-state.

Analogies Between Mechanical and Electrical Models

Mechanical System	Electrical Circuit
Mass m	Inductance L
Damping c	Resistance R
Spring constant k	$1/C$ (inverse capacitance)
Displacement $x(t)$	Charge $q(t)$
Velocity $\dot{x}(t)$	Current $i(t) = q'(t)$
External force $F(t)$	Source voltage $E(t)$

General Structure of Solutions

1. Write the governing ODE from physical laws (Newton's or Kirchhoff's).
2. Solve the **homogeneous equation** using the characteristic polynomial.
3. Add a **particular solution** if a forcing function is present.
4. Combine to obtain the **general solution**.
5. Apply **initial conditions** (e.g., initial displacement, velocity, charge, current).

Practice Problems**Exercise 4.7.7**

A 2 kg mass is attached to a spring with $k = 8$ N/m and damping constant $c = 4$ Ns/m. Find the general solution of the displacement $x(t)$ when no external force is applied.

Exercise 4.7.8

For a series RLC circuit with $L = 1$ H, $R = 6 \Omega$, and $C = \frac{1}{5}$ F, solve for the charge $q(t)$ when $E(t) = 0$.

Exercise 4.7.9

A spring-mass system with $m = 1$, $k = 9$, $c = 0$ is subject to a forcing function $F(t) = \cos(2t)$. Find the general solution.

Chapter 5: Laplace Transforms

Section 5.1: Introduction

Functions that represent solutions of differential equations can be given a formula (*representation*) in time-domain, where it's very easy to interpret their values/graph them, or in Laplace domain, where it is very easy to manipulate them through derivatives and integrals.

Notice that even though $f(t)$ is a real valued function of a real number, its Laplace transform $F(s)$ is a complex-valued function of a complex-number. That means that $f(t)$ can be visualized as a curve over a real axis, but we run into problems when visualizing $F(s)$. The domain is a complex plane $s \in \mathbb{C}$ but what do we put on "height" axis above this plane?

The value of $F(s)$ is a complex number itself, requiring either real/imaginary or modulus/argument representation, so we have to choose which component to plot as "height".

At this point, it's illustrative to visualize $|F(z)|$ and $\arg F(z)$ separately in some way.

The **SAGEMATH** code <https://goo.gl/2H2uBF> visualizes a function $f(t) = e^{-2t}$ in three different ways: (i) as "usual" in time-domain, (ii) magnitude of its Laplace transform $|F(s)|$ in 3d, (iii) its Laplace transform as brightness ($|F(s)|$) and color ($\arg F(s)$).

If you visit the link you can also rotate the 3d graph, show angle, real, and imaginary parts of $F(s)$.

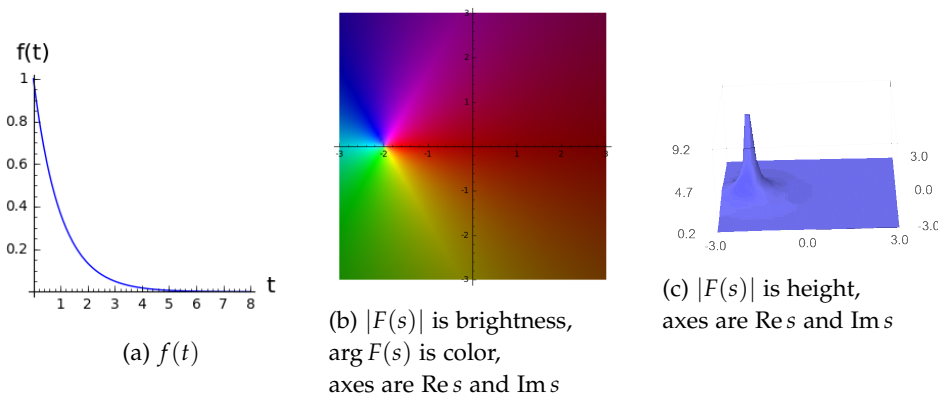


Figure 5.1: $f(t)$ and the corresponding $F(s)$ plotted by <https://goo.gl/2H2uBF>

Notice that the decay rate -2 from $f(t)$ translates into points where $|F(s)|$ grows very large (diverges). At the same time, the color around $s = -2 + 0i$ rotates through the full color spectrum, indicating that its angle (argument) changes around that point.

Definition 5.1.1 (Laplace Transform). The *Laplace transform* of a function $f(t)$, defined for $t \geq 0$, is given by

$$\mathcal{L}\{f(t)\}(s) = F(s) = \int_0^{\infty} e^{-st} f(t) dt, \quad s \in \mathbb{C}.$$

Here, $s = \sigma + i\omega$ is a complex variable, and the integral is assumed to converge.

Definition 5.1.2 (Exponential Order). A function $f(t)$ is said to be of exponen-

tial order α if there exist constants $M, \alpha, T > 0$ such that

$$|f(t)| \leq Me^{\alpha t}, \quad \text{for all } t > T.$$

This condition guarantees convergence of the Laplace transform.

Theorem 5.1.3 (Sufficient Condition for Existence). If $f(t)$ is piecewise continuous on every finite interval $[0, T]$ and of exponential order α , i.e.

$$|f(t)| \leq Me^{\alpha t}, \quad t \geq 0,$$

then the Laplace transform

$$F(s) = \int_0^\infty e^{-st} f(t) dt$$

exists for all s with $\operatorname{Re}(s) > \alpha$.

Theorem 5.1.4 (Boundedness of $F(s)$). If $f(t)$ satisfies the above conditions, then $F(s)$ is bounded for $\operatorname{Re}(s) > \alpha$ and is analytic in that region.

Theorem 5.1.5 (Decay of $F(s)$ as $s \rightarrow \infty$). If $f(t)$ is piecewise continuous on $[0, \infty)$ and of exponential order, then

$$\lim_{s \rightarrow \infty} F(s) = 0.$$

Example 5.1.6. For $f(t) = 1$, we have $F(s) = 1/s \rightarrow 0$ as $s \rightarrow \infty$. For $f(t) = t$, we have $F(s) = 1/s^2 \rightarrow 0$ even faster.

Theorem 5.1.7 (Linearity of Laplace Transform). If $f(t)$ and $g(t)$ have Laplace transforms $F(s)$ and $G(s)$ respectively, then for constants $a, b \in \mathbb{R}$,

$$\mathcal{L}\{af(t) + bg(t)\} = aF(s) + bG(s).$$

Theorem 5.1.8 (First Shifting Theorem (frequency shift)). If $\mathcal{L}\{f(t)\} = F(s)$, then

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a).$$

Example 5.1.9. Compute $\mathcal{L}\{e^{3t} \sin(2t)\}$.

Solution: We know $\mathcal{L}\{\sin(2t)\} = \frac{2}{s^2 + 4}$. By first shifting theorem:

$$\mathcal{L}\{e^{3t} \sin(2t)\} = F(s - 3) = \frac{2}{(s - 3)^2 + 4}.$$

Laplace integral:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

Properties, given that

$$\mathcal{L}\{f(t)\} = F(s), \quad \mathcal{L}\{g(t)\} = G(s):$$

- **Linearity:**
 $\mathcal{L}\{c_1 f(t) + c_2 g(t)\} = c_1 F(s) + c_2 G(s)$
- **s-shift:**
 $\mathcal{L}\{e^{at} f(t)\} = F(s - a)$
- **t-shift:**
 $\mathcal{L}\{\mathcal{U}(t - a) f(t - a)\} = e^{-sa} F(s)$
- **Derivative in t:** If $\mathcal{L}\{y(t)\} = Y(s)$, then
 $\mathcal{L}\{y'(t)\} = sY(s) - y(0),$
 $\mathcal{L}\{y''(t)\} = s^2 Y(s) - sy(0) - y'(0)$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$\delta(t - a)$	e^{-as}
1	$\frac{1}{s}$
$\mathcal{U}(t)$	$\frac{1}{s}$
$\mathcal{U}(t - a)$	$\frac{e^{-as}}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s - a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s - a)^2 + \omega^2}$
$e^{at} \cos(\omega t)$	$\frac{s - a}{(s - a)^2 + \omega^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$

Definition 5.1.10 (Unit Step Function). The unit step function (also called the Heaviside function) is defined as

$$\mathcal{U}(t-a) = \begin{cases} 0, & t < a, \\ 1, & t \geq a, \end{cases}$$

for some constant $a \geq 0$. It represents a switch that turns “on” at $t = a$. In Laplace transform problems, the unit step function is used to represent piecewise-defined or shifted forcing functions.

Theorem 5.1.11 (Second Shifting Theorem (time shift)). If $\mathcal{L}\{f(t)\} = F(s)$ and $\mathcal{U}(t-a)$ is the unit step function, then

$$\mathcal{L}\{\mathcal{U}(t-a)f(t-a)\} = e^{-as}F(s).$$

Example 5.1.12. Compute $\mathcal{L}\{\mathcal{U}(t-2) \cdot (t-2)\}$.

Solution: We know $\mathcal{L}\{t\} = \frac{1}{s^2}$. By second shifting theorem:

$$\mathcal{L}\{\mathcal{U}(t-2)(t-2)\} = e^{-2s} \cdot \frac{1}{s^2}.$$

Example 5.1.13. Find the Laplace transform of the piecewise function

$$f(t) = \begin{cases} 0, & 0 \leq t < 3, \\ 5, & t \geq 3. \end{cases}$$

Solution: We rewrite using the unit step function:

$$f(t) = 5 \mathcal{U}(t-3).$$

Thus,

$$\begin{aligned} \mathcal{L}\{f(t)\} &= 5 \mathcal{L}\{\mathcal{U}(t-3)\} \\ &= \frac{5e^{-3s}}{s}. \end{aligned}$$

$$F(s) = \frac{5e^{-3s}}{s}$$

Example 5.1.14. Find the Laplace transform of

$$f(t) = \begin{cases} t, & 0 \leq t < 1, \\ 1, & t \geq 1. \end{cases}$$

Solution: We rewrite in terms of $\mathcal{U}(t)$:

$$f(t) = t(1 - \mathcal{U}(t-1)) + 1 \cdot \mathcal{U}(t-1).$$

Simplify:

$$f(t) = t - t \mathcal{U}(t-1) + \mathcal{U}(t-1).$$

Laplace integral:

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

Properties, given that

$$\mathcal{L}\{f(t)\} = F(s), \mathcal{L}\{g(t)\} = G(s):$$

- **Linearity:**
 $\mathcal{L}\{c_1 f(t) + c_2 g(t)\} = c_1 F(s) + c_2 G(s)$
- **s-shift:**
 $\mathcal{L}\{e^{at} f(t)\} = F(s-a)$
- **t-shift:**
 $\mathcal{L}\{\mathcal{U}(t-a)f(t-a)\} = e^{-sa} F(s)$
- **Derivative in t:** If $\mathcal{L}\{y(t)\} = Y(s)$, then
 $\mathcal{L}\{y'(t)\} = sY(s) - y(0),$
 $\mathcal{L}\{y''(t)\} = s^2 Y(s) - sy(0) - y'(0)$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$\delta(t-a)$	e^{-as}
1	$\frac{1}{s}$
$\mathcal{U}(t)$	$\frac{1}{s}$
$\mathcal{U}(t-a)$	$\frac{e^{-as}}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s-a)^2 + \omega^2}$
$e^{at} \cos(\omega t)$	$\frac{s-a}{(s-a)^2 + \omega^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$

Take Laplace transform:

$$\begin{aligned} F(s) &= \mathcal{L}\{t\} - \mathcal{L}\{\mathcal{U}(t-1)(t)\} + \mathcal{L}\{\mathcal{U}(t-1)\} \\ &= \frac{1}{s^2} - e^{-s} \mathcal{L}\{t+1\} + \frac{e^{-s}}{s}. \end{aligned}$$

Since $\mathcal{L}\{t+1\} = \frac{1}{s^2} + \frac{1}{s}$, we get

$$F(s) = \frac{1}{s^2} - e^{-s} \left(\frac{1}{s^2} + \frac{1}{s} \right) + \frac{e^{-s}}{s}.$$

Simplify:

$$F(s) = \frac{1}{s^2} - \frac{e^{-s}}{s^2}$$

Theorem 5.1.15 (Differentiation in Time Domain).

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0),$$

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0).$$

Example 5.1.16. Find $\mathcal{L}\{y''(t)\}$ if $y(t)$ satisfies $y(0) = 2, y'(0) = 1$ and $\mathcal{L}\{y(t)\} = Y(s)$.

Solution: By differentiation in time domain:

$$\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0) = s^2Y(s) - 2s - 1.$$

Theorem 5.1.17 (Differentiation in s-Domain).

$$\mathcal{L}\{tf(t)\} = -F'(s).$$

Example 5.1.18. Compute $\mathcal{L}\{t \cos(2t)\}$.

Solution: We know $\mathcal{L}\{\cos(2t)\} = \frac{s}{s^2+4}$. By differentiation in s-domain:

$$\mathcal{L}\{t \cos(2t)\} = -\frac{d}{ds} \frac{s}{s^2+4} = \frac{s^2-4}{(s^2+4)^2}.$$

Theorem 5.1.19 (Integration in Time Domain).

$$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{F(s)}{s}.$$

Example 5.1.20. Compute $\mathcal{L}\left\{\int_0^t e^{2\tau} d\tau\right\}$.

Solution:

$$F(s) = \mathcal{L}\{e^{2t}\} = \frac{1}{s-2} \implies \mathcal{L}\left\{\int_0^t e^{2\tau} d\tau\right\} = \frac{1}{s(s-2)}.$$

Laplace integral:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

Properties, given that

$$\mathcal{L}\{f(t)\} = F(s), \mathcal{L}\{g(t)\} = G(s):$$

- **Linearity:**
 $\mathcal{L}\{c_1f(t) + c_2g(t)\} = c_1F(s) + c_2G(s)$
- **s-shift:**
 $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$
- **t-shift:**
 $\mathcal{L}\{\mathcal{U}(t-a)f(t-a)\} = e^{-sa}F(s)$
- **Derivative in t:** If $\mathcal{L}\{y(t)\} = Y(s)$, then
 $\mathcal{L}\{y'(t)\} = sY(s) - y(0),$
 $\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$\delta(t-a)$	e^{-as}
1	$\frac{1}{s}$
$\mathcal{U}(t)$	$\frac{1}{s}$
$\mathcal{U}(t-a)$	$\frac{e^{-as}}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s-a)^2+\omega^2}$
$e^{at} \cos(\omega t)$	$\frac{s-a}{(s-a)^2+\omega^2}$
$\cosh(at)$	$\frac{s}{s^2-a^2}$
$\sinh(at)$	$\frac{a}{s^2-a^2}$

Theorem 5.1.21 (Multiplication by Exponential / Convolution Property).

$$\mathcal{L}\{f(t) * g(t)\} = F(s)G(s),$$

where $f * g = \int_0^t f(\tau)g(t - \tau) d\tau$ is the convolution of f and g .

Example 5.1.22. Compute $\mathcal{L}\{t * e^{-t}\}$.

Solution:

$$\mathcal{L}\{t\} = \frac{1}{s^2}, \quad \mathcal{L}\{e^{-t}\} = \frac{1}{s+1}.$$

By convolution theorem:

$$\mathcal{L}\{t * e^{-t}\} = \frac{1}{s^2} \cdot \frac{1}{s+1} = \frac{1}{s^2(s+1)}.$$

Definition 5.1.23. The *Dirac delta function*, denoted by $\delta(t - a)$, is not a function in the traditional sense but a generalized function or distribution. It is defined by the following properties:

1. $\delta(t - a) = 0$ for all $t \neq a$,
2. $\int_{-\infty}^{\infty} \delta(t - a) f(t) dt = f(a)$ for any continuous test function $f(t)$.

Intuitively, $\delta(t - a)$ can be thought of as an “infinite spike” at $t = a$ with unit area under the curve.

Theorem 5.1.24 (Laplace Transform of Delta). For $a \geq 0$, the Laplace transform is

$$\mathcal{L}\{\delta(t - a)\} = e^{-as}.$$

Example 5.1.25. Solve the IVP

$$y'' + y = \delta(t - \pi), \quad y(0) = 0, \quad y'(0) = 0.$$

Solution:

Step 1: Laplace transform.

$$(s^2 Y(s)) + Y(s) = e^{-\pi s}.$$

Step 2: Solve for $Y(s)$.

$$Y(s) = \frac{e^{-\pi s}}{s^2 + 1}.$$

Step 3: Inverse Laplace.

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 1}\right\} = \sin t.$$

Step 4: Apply shifting theorem.

$$y(t) = u(t - \pi) \sin(t - \pi).$$

Final Answer:

$$y(t) = u(t - \pi) \sin(t - \pi).$$

Laplace integral:

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

Properties, given that

$$\mathcal{L}\{f(t)\} = F(s), \quad \mathcal{L}\{g(t)\} = G(s):$$

- **Linearity:**
 $\mathcal{L}\{c_1 f(t) + c_2 g(t)\} = c_1 F(s) + c_2 G(s)$
- **s-shift:**
 $\mathcal{L}\{e^{at} f(t)\} = F(s - a)$
- **t-shift:**
 $\mathcal{L}\{\mathcal{U}(t - a) f(t - a)\} = e^{-sa} F(s)$
- **Derivative in t:** If $\mathcal{L}\{y(t)\} = Y(s)$, then
 $\mathcal{L}\{y'(t)\} = sY(s) - y(0),$
 $\mathcal{L}\{y''(t)\} = s^2 Y(s) - sy(0) - y'(0)$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$\delta(t - a)$	e^{-as}
1	$\frac{1}{s}$
$\mathcal{U}(t)$	$\frac{1}{s}$
$\mathcal{U}(t - a)$	$\frac{e^{-as}}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s - a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s - a)^2 + \omega^2}$
$e^{at} \cos(\omega t)$	$\frac{s - a}{(s - a)^2 + \omega^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$

Example 5.1.26. Solve the IVP

$$y' + y = \delta(t - 2), \quad y(0) = 0.$$

Solution:

Step 1: Laplace transform.

$$(sY(s)) + Y(s) = e^{-2s}.$$

$$(s + 1)Y(s) = e^{-2s}.$$

Step 2: Solve for $Y(s)$.

$$Y(s) = \frac{e^{-2s}}{s + 1}.$$

Step 3: Inverse Laplace using shift theorem.

$$\mathcal{L}^{-1}\left\{\frac{1}{s + 1}\right\} = e^{-t}.$$

$$y(t) = u(t - 2)e^{-(t-2)}.$$

Final Answer:

$$y(t) = u(t - 2)e^{-(t-2)}.$$

Theorem 5.1.27 (Initial Value Theorem). Let $F(s) = \mathcal{L}\{f(t)\}$. Then

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s),$$

provided the limit exists.

Theorem 5.1.28 (Final Value Theorem). Let $F(s) = \mathcal{L}\{f(t)\}$. Then

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0^+} sF(s),$$

provided the poles of $sF(s)$ lie strictly in $\text{Re}(s) < 0$.

Example 5.1.29 (Transform of a constant).

$$f(t) = 1, \quad F(s) = \int_0^\infty e^{-st} dt.$$

Compute:

$$F(s) = \left[\frac{-1}{s} e^{-st} \right]_0^\infty = \frac{1}{s}, \quad \text{Re}(s) > 0.$$

Example 5.1.30 (Transform of t).

$$f(t) = t, \quad F(s) = \int_0^\infty te^{-st} dt.$$

Using integration by parts:

$$u = t, \quad dv = e^{-st} dt \quad \Rightarrow \quad du = dt, \quad v = -\frac{1}{s}e^{-st}.$$

Laplace integral:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

Properties, given that

$$\mathcal{L}\{f(t)\} = F(s), \quad \mathcal{L}\{g(t)\} = G(s):$$

- **Linearity:**
 $\mathcal{L}\{c_1 f(t) + c_2 g(t)\} = c_1 F(s) + c_2 G(s)$
- **s-shift:**
 $\mathcal{L}\{e^{at} f(t)\} = F(s - a)$
- **t-shift:**
 $\mathcal{L}\{u(t - a) f(t - a)\} = e^{-sa} F(s)$
- **Derivative in t:** If $\mathcal{L}\{y(t)\} = Y(s)$, then
 $\mathcal{L}\{y'(t)\} = sY(s) - y(0),$
 $\mathcal{L}\{y''(t)\} = s^2 Y(s) - sy(0) - y'(0)$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$\delta(t - a)$	e^{-as}
1	$\frac{1}{s}$
$\mathcal{U}(t)$	$\frac{1}{s}$
$\mathcal{U}(t - a)$	$\frac{e^{-as}}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s - a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s - a)^2 + \omega^2}$
$e^{at} \cos(\omega t)$	$\frac{s - a}{(s - a)^2 + \omega^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$

Then

$$F(s) = \left[-\frac{t}{s} e^{-st} \right]_0^\infty + \int_0^\infty \frac{1}{s} e^{-st} dt.$$

First term vanishes, so

$$F(s) = \frac{1}{s^2}, \quad \operatorname{Re}(s) > 0.$$

Example 5.1.31 (Transform of e^{at}).

$$f(t) = e^{at}, \quad F(s) = \int_0^\infty e^{-st} e^{at} dt = \int_0^\infty e^{-(s-a)t} dt.$$

Provided $\operatorname{Re}(s) > a$, we get

$$F(s) = \frac{1}{s-a}.$$

Practice Problems

Exercise 5.1.1

Compute using the definition of Laplace transform:

1. $\mathcal{L}\{t^2\}$.
2. $\mathcal{L}\{\sin(bt)\}$.
3. $\mathcal{L}\{e^{at} \cos(bt)\}$.
4. $\mathcal{L}\{\mathcal{U}(t-2)\}$.
5. Use the table to find $\mathcal{L}\{3t^2 + 5e^{2t}\}$.
6. Use the first shifting theorem to compute $\mathcal{L}\{e^{3t} \sin(4t)\}$.
7. Use the second shifting theorem to find $\mathcal{L}\{\mathcal{U}(t-2)(t-2)\}$.
8. Use the definition to check the transform of $\cos(bt)$ from the table.

(Hints: use integration by parts and shifting theorems.)

1. Express the piecewise function

$$f(t) = \begin{cases} 2, & 0 \leq t < 4, \\ 0, & t \geq 4, \end{cases}$$

using the unit step function, and find $\mathcal{L}\{f(t)\}$.

2. Compute $\mathcal{L}\{\mathcal{U}(t-5) \sin(t-5)\}$.

3. Let

$$f(t) = \begin{cases} t^2, & 0 \leq t < 2, \\ 0, & t \geq 2. \end{cases}$$

Rewrite in terms of unit step functions and find $F(s)$.

Laplace integral:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

Properties, given that

$$\mathcal{L}\{f(t)\} = F(s), \mathcal{L}\{g(t)\} = G(s):$$

- **Linearity:**
 $\mathcal{L}\{c_1 f(t) + c_2 g(t)\} = c_1 F(s) + c_2 G(s)$
- **s-shift:**
 $\mathcal{L}\{e^{at} f(t)\} = F(s-a)$
- **t-shift:**
 $\mathcal{L}\{\mathcal{U}(t-a) f(t-a)\} = e^{-sa} F(s)$
- **Derivative in t:** If $\mathcal{L}\{y(t)\} = Y(s)$, then
 $\mathcal{L}\{y'(t)\} = sY(s) - y(0),$
 $\mathcal{L}\{y''(t)\} = s^2 Y(s) - sy(0) - y'(0)$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$\delta(t-a)$	e^{-as}
1	$\frac{1}{s}$
$\mathcal{U}(t)$	$\frac{1}{s}$
$\mathcal{U}(t-a)$	$\frac{e^{-as}}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s-a)^2 + \omega^2}$
$e^{at} \cos(\omega t)$	$\frac{s-a}{(s-a)^2 + \omega^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$

Section 5.2: Inverse Laplace Transforms

Definition 5.2.1 (Inverse Laplace Transform). Given a Laplace transform $F(s)$ of a function $f(t)$, the **inverse Laplace transform** is the operation that recovers $f(t)$ from $F(s)$. It is denoted by

$$f(t) = \mathcal{L}^{-1}\{F(s)\}.$$

Theorem 5.2.2 (Uniqueness of Inverse Laplace Transform). If $F(s)$ is the Laplace transform of a piecewise continuous function $f(t)$ of exponential order, then the inverse Laplace transform exists and is unique almost everywhere.

Properties of the Inverse Transform

- **Linearity:** $\mathcal{L}^{-1}\{aF(s) + bG(s)\} = a\mathcal{L}^{-1}\{F(s)\} + b\mathcal{L}^{-1}\{G(s)\}.$
- **First Shifting Theorem:** $\mathcal{L}^{-1}\{F(s - a)\} = e^{at}f(t).$
- **Second Shifting Theorem:** $\mathcal{L}^{-1}\{e^{-as}F(s)\} = \mathcal{U}(t - a)f(t - a)$, where $\mathcal{U}(t - a)$ is the unit step function.
- **Differentiation in s :** $\mathcal{L}^{-1}\left\{-\frac{d}{ds}F(s)\right\} = tf(t).$

Methods of Finding Inverse Laplace Transforms

1. Using a **reference table** of common Laplace transforms.
2. **Partial fraction decomposition** for rational functions of s .
3. Using the **first and second shifting theorems**.
4. **Convolution theorem:** $\mathcal{L}^{-1}\{F(s)G(s)\} = (f * g)(t) = \int_0^t f(\tau)g(t - \tau) d\tau.$

Example 5.2.3. Find $\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\}.$

Solution:

From the table, $\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$, hence

$$\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} = e^{3t}.$$

Example 5.2.4. Find $\mathcal{L}^{-1}\left\{\frac{2}{s^2+9}\right\}.$

Solution:

From the table, $\mathcal{L}^{-1}\left\{\frac{b}{s^2+b^2}\right\} = \sin(bt)$, so

$$\mathcal{L}^{-1}\left\{\frac{2}{s^2+9}\right\} = \frac{2}{3}\sin(3t).$$

Example 5.2.5. Find $\mathcal{L}^{-1}\left\{\frac{s+1}{s^2+4s+5}\right\}.$

Solution:

Complete the square: $s^2 + 4s + 5 = (s + 2)^2 + 1.$

Rewrite numerator: $s + 1 = (s + 2) - 1.$

Laplace integral:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st}f(t)dt$$

Properties, given that

$$\mathcal{L}\{f(t)\} = F(s), \mathcal{L}\{g(t)\} = G(s):$$

- **Linearity:**
 $\mathcal{L}\{c_1f(t) + c_2g(t)\} = c_1F(s) + c_2G(s)$
- **s-shift:**
 $\mathcal{L}\{e^{at}f(t)\} = F(s - a)$
- **t-shift:**
 $\mathcal{L}\{\mathcal{U}(t - a)f(t - a)\} = e^{-sa}F(s)$
- **Derivative in t :** If $\mathcal{L}\{y(t)\} = Y(s)$, then
 $\mathcal{L}\{y'(t)\} = sY(s) - y(0),$
 $\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$\delta(t - a)$	e^{-as}
1	$\frac{1}{s}$
$\mathcal{U}(t)$	$\frac{1}{s}$
$\mathcal{U}(t - a)$	$\frac{e^{-as}}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{at}\sin(\omega t)$	$\frac{\omega}{(s-a)^2 + \omega^2}$
$e^{at}\cos(\omega t)$	$\frac{s-a}{(s-a)^2 + \omega^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$

Split into two terms:

$$\frac{s+1}{(s+2)^2+1} = \frac{s+2}{(s+2)^2+1} - \frac{1}{(s+2)^2+1}.$$

From the table:

$$\mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+1}\right\} = e^{-2t} \cos t, \quad \mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2+1}\right\} = e^{-2t} \sin t.$$

Hence,

$$\mathcal{L}^{-1}\left\{\frac{s+1}{s^2+4s+5}\right\} = e^{-2t}(\cos t - \sin t).$$

Example 5.2.6. Solve the IVP

$$y' - 3y = 6, \quad y(0) = 2,$$

using Laplace transforms.

Solution. Let $Y(s) = \mathcal{L}\{y(t)\}(s)$.

Taking Laplace transforms and using $\mathcal{L}\{y'\} = sY(s) - y(0)$ gives

$$\begin{aligned} \mathcal{L}\{y'\} - 3\mathcal{L}\{y\} &= \mathcal{L}\{6\} \\ (sY(s) - y(0)) - 3Y(s) &= \frac{6}{s} \\ (s-3)Y(s) - 2 &= \frac{6}{s} \\ (s-3)Y(s) &= \frac{6}{s} + 2 = \frac{6+2s}{s} \\ Y(s) &= \frac{6+2s}{s(s-3)} = \frac{2s+6}{s(s-3)}. \end{aligned}$$

Perform partial fraction decomposition: seek A, B with

$$\frac{2s+6}{s(s-3)} = \frac{A}{s} + \frac{B}{s-3}.$$

Equating numerators,

$$2s+6 = A(s-3) + Bs = (A+B)s - 3A.$$

Compare coefficients:

$$\begin{aligned} A+B &= 2, \\ -3A &= 6 \Rightarrow A = -2, \quad B = 2 - A = 4. \end{aligned}$$

Thus

$$Y(s) = -\frac{2}{s} + \frac{4}{s-3}.$$

Taking inverse Laplace termwise yields

$$\begin{aligned} y(t) &= -2\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + 4\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} \\ &= -2 + 4e^{3t}. \end{aligned}$$

$$\boxed{y(t) = -2 + 4e^{3t}}$$

Laplace integral:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

Properties, given that

$$\mathcal{L}\{f(t)\} = F(s), \quad \mathcal{L}\{g(t)\} = G(s):$$

- **Linearity:**
 $\mathcal{L}\{c_1 f(t) + c_2 g(t)\} = c_1 F(s) + c_2 G(s)$
- **s-shift:**
 $\mathcal{L}\{e^{at} f(t)\} = F(s-a)$
- **t-shift:**
 $\mathcal{L}\{\mathcal{U}(t-a) f(t-a)\} = e^{-sa} F(s)$
- **Derivative in t:** If $\mathcal{L}\{y(t)\} = Y(s)$, then
 $\mathcal{L}\{y'(t)\} = sY(s) - y(0),$
 $\mathcal{L}\{y''(t)\} = s^2 Y(s) - sy'(0) - y''(0)$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$\delta(t-a)$	e^{-as}
1	$\frac{1}{s}$
$\mathcal{U}(t)$	$\frac{1}{s}$
$\mathcal{U}(t-a)$	$\frac{e^{-as}}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s-a)^2 + \omega^2}$
$e^{at} \cos(\omega t)$	$\frac{s-a}{(s-a)^2 + \omega^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$

Example 5.2.7. Solve the IVP:

$$y'' - 4y' + 4y = 0, \quad y(0) = 1, \quad y'(0) = 0,$$

using Laplace transforms.

Solution.

Let $Y(s) = \mathcal{L}\{y(t)\}(s)$.

Taking Laplace transforms and using

$$\mathcal{L}\{y''\} = s^2Y(s) - sy(0) - y'(0) \text{ and } \mathcal{L}\{y'\} = sY(s) - y(0) \text{ gives}$$

$$\begin{aligned} \mathcal{L}\{y''\} - 4\mathcal{L}\{y'\} + 4\mathcal{L}\{y\} &= 0 \\ (s^2Y(s) - s \cdot 1 - 0) - 4(sY(s) - 1) + 4Y(s) &= 0 \\ (s^2 - 4s + 4)Y(s) - s + 4 &= 0 \\ (s - 2)^2Y(s) &= s - 4 \\ Y(s) &= \frac{s - 4}{(s - 2)^2}. \end{aligned}$$

Rewrite the numerator to prepare for inversion:

$$\begin{aligned} \frac{s - 4}{(s - 2)^2} &= \frac{(s - 2) - 2}{(s - 2)^2} = \frac{s - 2}{(s - 2)^2} - \frac{2}{(s - 2)^2} \\ &= \frac{1}{s - 2} - \frac{2}{(s - 2)^2}. \end{aligned}$$

Take inverse Laplace termwise:

$$\begin{aligned} y(t) &= \mathcal{L}^{-1}\left\{\frac{1}{s - 2}\right\} - 2\mathcal{L}^{-1}\left\{\frac{1}{(s - 2)^2}\right\} \\ &= e^{2t} - 2te^{2t} \\ &= (1 - 2t)e^{2t}. \end{aligned}$$

$y(t) = (1 - 2t)e^{2t}$

Laplace integral:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

Properties, given that

$$\mathcal{L}\{f(t)\} = F(s), \quad \mathcal{L}\{g(t)\} = G(s):$$

- **Linearity:**
 $\mathcal{L}\{c_1f(t) + c_2g(t)\} = c_1F(s) + c_2G(s)$
- **s-shift:**
 $\mathcal{L}\{e^{at}f(t)\} = F(s - a)$
- **t-shift:**
 $\mathcal{L}\{\mathcal{U}(t - a)f(t - a)\} = e^{-sa}F(s)$
- **Derivative in t:** If $\mathcal{L}\{y(t)\} = Y(s)$, then
 $\mathcal{L}\{y'(t)\} = sY(s) - y(0),$
 $\mathcal{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$\delta(t - a)$	e^{-as}
1	$\frac{1}{s}$
$\mathcal{U}(t)$	$\frac{1}{s}$
$\mathcal{U}(t - a)$	$\frac{e^{-as}}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s - a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s - a)^2 + \omega^2}$
$e^{at} \cos(\omega t)$	$\frac{s - a}{(s - a)^2 + \omega^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$

Example 5.2.8. Solve the IVP:

$$y'' + y = \sin x, \quad y(0) = 0, \quad y'(0) = 1.$$

Solution using Laplace:

Taking the Laplace transform of the IVP $y'' + y = \sin x$ with $y(0) = 0, y'(0) = 1$:

$$\begin{aligned}
s^2 Y(s) - 1 + Y(s) &= \frac{1}{s^2 + 1} \\
(s^2 + 1)Y(s) &= 1 + \frac{1}{s^2 + 1} = \frac{s^2 + 2}{s^2 + 1} \\
Y(s) &= \frac{s^2 + 2}{(s^2 + 1)^2} \\
&= \frac{s^2 + 1 + 1}{(s^2 + 1)^2} \\
&= \frac{1}{s^2 + 1} + \frac{1}{(s^2 + 1)^2}
\end{aligned}$$

Using inverse Laplace transforms and the convolution theorem for the second term:

$$\begin{aligned}
y(x) &= \mathcal{L}^{-1} \left(\frac{1}{s^2 + 1} \right) + \mathcal{L}^{-1} \left(\frac{1}{(s^2 + 1)^2} \right) \\
&= \sin x + (\sin x * \sin x) \\
&= \sin x + \left(\frac{1}{2} \sin x - \frac{x}{2} \cos x \right) \\
&= \frac{3}{2} \sin x - \frac{x}{2} \cos x
\end{aligned}$$

Example 5.2.9. Solve the IVP (first-order, nonhomogeneous):

$$y' + 2y = e^{-t}, \quad y(0) = 0,$$

using Laplace transforms.

Take Laplace transforms:

$$\begin{aligned}
\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} &= \mathcal{L}\{e^{-t}\} \\
(sY(s) - y(0)) + 2Y(s) &= \frac{1}{s + 1} \\
(s + 2)Y(s) - 0 &= \frac{1}{s + 1} \\
Y(s) &= \frac{1}{(s + 2)(s + 1)}.
\end{aligned}$$

Partial fractions: $\frac{1}{(s + 1)(s + 2)} = \frac{A}{s + 1} + \frac{B}{s + 2}.$

$$\begin{aligned}
1 &= A(s + 2) + B(s + 1) \Rightarrow A + B = 0, \quad 2A + B = 1 \\
&\Rightarrow A = 1, \quad B = -1.
\end{aligned}$$

$$\begin{aligned}
Y(s) &= \frac{1}{s + 1} - \frac{1}{s + 2} \\
\Rightarrow y(t) &= \mathcal{L}^{-1}\{Y(s)\} = e^{-t} - e^{-2t}.
\end{aligned}$$

$$y(t) = e^{-t} - e^{-2t}$$

Laplace integral:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

Properties, given that

$$\mathcal{L}\{f(t)\} = F(s), \quad \mathcal{L}\{g(t)\} = G(s):$$

- **Linearity:**
 $\mathcal{L}\{c_1 f(t) + c_2 g(t)\} = c_1 F(s) + c_2 G(s)$
- **s-shift:**
 $\mathcal{L}\{e^{at} f(t)\} = F(s - a)$
- **t-shift:**
 $\mathcal{L}\{\mathcal{U}(t - a) f(t - a)\} = e^{-sa} F(s)$
- **Derivative in t:** If $\mathcal{L}\{y(t)\} = Y(s)$, then
 $\mathcal{L}\{y'(t)\} = sY(s) - y(0),$
 $\mathcal{L}\{y''(t)\} = s^2 Y(s) - sy(0) - y'(0)$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$\delta(t - a)$	e^{-as}
1	$\frac{1}{s}$
$\mathcal{U}(t)$	$\frac{1}{s}$
$\mathcal{U}(t - a)$	$\frac{e^{-as}}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s - a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s - a)^2 + \omega^2}$
$e^{at} \cos(\omega t)$	$\frac{s - a}{(s - a)^2 + \omega^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$

Example 5.2.10. Solve the IVP (second-order, sinusoidal forcing):

$$y'' + y = \cos t, \quad y(0) = 0, \quad y'(0) = 0,$$

using Laplace transforms.

Take Laplace transforms:

$$\mathcal{L}\{y''\} + \mathcal{L}\{y\} = \mathcal{L}\{\cos t\}$$

$$(s^2 Y(s) - sy(0) - y'(0)) + Y(s) = \frac{s}{s^2 + 1}$$

$$(s^2 + 1)Y(s) - 0 = \frac{s}{s^2 + 1}$$

$$Y(s) = \frac{s}{(s^2 + 1)^2}.$$

$$\text{Use known transform: } \mathcal{L}\{t \sin t\} = \frac{2s}{(s^2 + 1)^2}.$$

$$\therefore \frac{s}{(s^2 + 1)^2} = \frac{1}{2} \mathcal{L}\{t \sin t\}.$$

$$\Rightarrow y(t) = \frac{1}{2} t \sin t.$$

$$\boxed{y(t) = \frac{1}{2} t \sin t}$$

Example 5.2.11. Solve the IVP (second-order, exponential forcing):

$$y'' - y = e^t, \quad y(0) = 0, \quad y'(0) = 1,$$

using Laplace transforms.

Take Laplace transforms:

$$\mathcal{L}\{y''\} - \mathcal{L}\{y\} = \mathcal{L}\{e^t\}$$

$$(s^2 Y(s) - sy(0) - y'(0)) - Y(s) = \frac{1}{s - 1}$$

$$(s^2 - 1)Y(s) - 1 = \frac{1}{s - 1} \quad (\text{since } y(0) = 0, \quad y'(0) = 1)$$

$$(s^2 - 1)Y(s) = 1 + \frac{1}{s - 1} = \frac{s}{s - 1}$$

$$Y(s) = \frac{s}{(s - 1)(s^2 - 1)} = \frac{s}{(s - 1)^2(s + 1)}.$$

Partial fractions: seek $\frac{s}{(s - 1)^2(s + 1)} = \frac{A}{s - 1} + \frac{B}{(s - 1)^2} + \frac{C}{s + 1}.$

$$s = A(s - 1)(s + 1) + B(s + 1) + C(s - 1)^2.$$

$$\text{Expand and equate coefficients: } (A + C)s^2 + (B - 2C)s + (-A + B + C) = s.$$

$$\Rightarrow A + C = 0, \quad B - 2C = 1, \quad -A + B + C = 0.$$

$$\text{Solve: } C = -A, \quad -A + B - A = 0 \Rightarrow B = 2A, \quad 2A - 2(-A) = 1 \Rightarrow 4A = 1 \Rightarrow A = \frac{1}{4}.$$

$$B = \frac{1}{2}, \quad C = -\frac{1}{4}.$$

Laplace integral:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

Properties, given that

$$\mathcal{L}\{f(t)\} = F(s), \quad \mathcal{L}\{g(t)\} = G(s):$$

- **Linearity:**
 $\mathcal{L}\{c_1 f(t) + c_2 g(t)\} = c_1 F(s) + c_2 G(s)$
- **s-shift:**
 $\mathcal{L}\{e^{at} f(t)\} = F(s - a)$
- **t-shift:**
 $\mathcal{L}\{\mathcal{U}(t - a) f(t - a)\} = e^{-sa} F(s)$
- **Derivative in t:** If $\mathcal{L}\{y(t)\} = Y(s)$, then
 $\mathcal{L}\{y'(t)\} = sY(s) - y(0),$
 $\mathcal{L}\{y''(t)\} = s^2 Y(s) - sy(0) - y'(0)$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$\delta(t - a)$	e^{-as}
1	$\frac{1}{s}$
$\mathcal{U}(t)$	$\frac{1}{s}$
$\mathcal{U}(t - a)$	$\frac{e^{-as}}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s - a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s - a)^2 + \omega^2}$
$e^{at} \cos(\omega t)$	$\frac{s - a}{(s - a)^2 + \omega^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$

$$Y(s) = \frac{1}{4} \frac{1}{s-1} + \frac{1}{2} \frac{1}{(s-1)^2} - \frac{1}{4} \frac{1}{s+1}.$$

$$\Rightarrow y(t) = \frac{1}{4}e^t + \frac{1}{2}te^t - \frac{1}{4}e^{-t}.$$

$$y(t) = e^t \left(\frac{1}{4} + \frac{1}{2}t \right) - \frac{1}{4}e^{-t}$$

Solving IVPs with Unit Step Forcing Functions

Example 5.2.12. Solve the IVP:

$$y'' + y = \mathcal{U}(t - \pi), \quad y(0) = 0, \quad y'(0) = 0.$$

Solution:

Step 1: Laplace transform of both sides.

$$\mathcal{L}\{y''\} + \mathcal{L}\{y\} = \mathcal{L}\{\mathcal{U}(t - \pi)\},$$

$$(s^2 Y(s) - sy(0) - y'(0)) + Y(s) = \frac{e^{-\pi s}}{s}.$$

Substitute $y(0) = 0, y'(0) = 0$:

$$(s^2 + 1)Y(s) = \frac{e^{-\pi s}}{s}.$$

Step 2: Solve for $Y(s)$.

$$Y(s) = \frac{e^{-\pi s}}{s(s^2 + 1)}.$$

Step 3: Inverse Laplace.

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2 + 1)}\right\} = 1 - \cos t.$$

Hence,

$$y(t) = \mathcal{U}(t - \pi)(1 - \cos(t - \pi)).$$

Final Answer:

$$y(t) = \mathcal{U}(t - \pi)(1 - \cos(t - \pi))$$

Example 5.2.13. Solve the IVP:

$$y'' + 4y = \mathcal{U}(t - 1), \quad y(0) = 0, \quad y'(0) = 0.$$

Solution:

Step 1: Laplace transform.

$$(s^2 Y(s) - sy(0) - y'(0)) + 4Y(s) = \frac{e^{-s}}{s},$$

$$(s^2 + 4)Y(s) = \frac{e^{-s}}{s}.$$

Laplace integral:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

Properties, given that

$$\mathcal{L}\{f(t)\} = F(s), \quad \mathcal{L}\{g(t)\} = G(s):$$

- **Linearity:**
 $\mathcal{L}\{c_1 f(t) + c_2 g(t)\} = c_1 F(s) + c_2 G(s)$
- **s-shift:**
 $\mathcal{L}\{e^{at} f(t)\} = F(s - a)$
- **t-shift:**
 $\mathcal{L}\{\mathcal{U}(t - a) f(t - a)\} = e^{-sa} F(s)$
- **Derivative in t:** If $\mathcal{L}\{y(t)\} = Y(s)$, then
 $\mathcal{L}\{y'(t)\} = sY(s) - y(0),$
 $\mathcal{L}\{y''(t)\} = s^2 Y(s) - sy(0) - y'(0)$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$\delta(t - a)$	e^{-as}
1	$\frac{1}{s}$
$\mathcal{U}(t)$	$\frac{1}{s}$
$\mathcal{U}(t - a)$	$\frac{e^{-as}}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s - a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s - a)^2 + \omega^2}$
$e^{at} \cos(\omega t)$	$\frac{s - a}{(s - a)^2 + \omega^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$

Step 2: Solve for $Y(s)$.

$$Y(s) = \frac{e^{-s}}{s(s^2 + 4)}.$$

Step 3: Inverse Laplace.

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2 + 4)}\right\} = \frac{1}{4}(1 - \cos 2t).$$

Step 4: Apply shifting theorem.

$$y(t) = \mathcal{U}(t-1) \cdot \frac{1}{4}(1 - \cos 2(t-1)).$$

Final Answer:

$$y(t) = \frac{1}{4} \mathcal{U}(t-1)(1 - \cos 2(t-1))$$

Example 5.2.14. Solve the IVP:

$$y'' + y' = \mathcal{U}(t-2), \quad y(0) = 0, \quad y'(0) = 1.$$

Solution:

Step 1: Laplace transform.

$$(s^2 Y(s) - sy(0) - y'(0)) + (sY(s) - y(0)) = \frac{e^{-2s}}{s},$$

$$(s^2 + s)Y(s) - 1 = \frac{e^{-2s}}{s}.$$

Step 2: Solve for $Y(s)$.

$$Y(s) = \frac{1}{s(s+1)} + \frac{e^{-2s}}{s^2(s+1)}.$$

Step 3: Inverse Laplace of first term.

$$\frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1} \implies \mathcal{L}^{-1} = 1 - e^{-t}.$$

Step 4: Inverse Laplace of second term.

$$\frac{1}{s^2(s+1)} = \frac{1}{s^2} - \frac{1}{s} + \frac{1}{s+1}.$$

Inverse Laplace:

$$\mathcal{L}^{-1} = t - 1 + e^{-t}.$$

Apply shifting theorem:

$$\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^2(s+1)}\right\} = \mathcal{U}(t-2)((t-2) - 1 + e^{-(t-2)}).$$

Step 5: Combine.

$$y(t) = (1 - e^{-t}) + \mathcal{U}(t-2)((t-3) + e^{-(t-2)}).$$

Final Answer:

$$y(t) = 1 - e^{-t} + \mathcal{U}(t-2)((t-3) + e^{-(t-2)})$$

Laplace integral:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$$

Properties, given that

$$\mathcal{L}\{f(t)\} = F(s), \quad \mathcal{L}\{g(t)\} = G(s):$$

- **Linearity:**
 $\mathcal{L}\{c_1 f(t) + c_2 g(t)\} = c_1 F(s) + c_2 G(s)$
- **s-shift:**
 $\mathcal{L}\{e^{at} f(t)\} = F(s-a)$
- **t-shift:**
 $\mathcal{L}\{\mathcal{U}(t-a) f(t-a)\} = e^{-sa} F(s)$
- **Derivative in t:** If $\mathcal{L}\{y(t)\} = Y(s)$, then
 $\mathcal{L}\{y'(t)\} = sY(s) - y(0),$
 $\mathcal{L}\{y''(t)\} = s^2 Y(s) - sy(0) - y'(0)$

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$\delta(t-a)$	e^{-as}
1	$\frac{1}{s}$
$\mathcal{U}(t)$	$\frac{1}{s}$
$\mathcal{U}(t-a)$	$\frac{e^{-as}}{s}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s-a}$
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$
$e^{at} \sin(\omega t)$	$\frac{\omega}{(s-a)^2 + \omega^2}$
$e^{at} \cos(\omega t)$	$\frac{s-a}{(s-a)^2 + \omega^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$

Example 5.2.15. Solve the IVP

$$y'' + 4y = \sin(t-1) \mathcal{U}(t-1), \quad y(0) = 0, \quad y'(0) = 0.$$

Solution:

Step 1: Laplace transform.

$$(s^2 Y(s)) + 4Y(s) = e^{-s} \frac{1}{s^2 + 1}.$$

$$Y(s) = \frac{e^{-s}}{(s^2 + 4)(s^2 + 1)}.$$

Step 2: Partial fractions.

$$\frac{1}{(s^2 + 4)(s^2 + 1)} = \frac{1}{3} \left(\frac{1}{s^2 + 1} - \frac{1}{s^2 + 4} \right).$$

Step 3: Inverse Laplace (without shift).

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s^2 + 4)(s^2 + 1)} \right\} = \frac{1}{3} \left(\sin t - \frac{1}{2} \sin 2t \right).$$

Step 4: Apply shift theorem.

$$y(t) = \mathcal{U}(t-1) \frac{1}{3} \left(\sin(t-1) - \frac{1}{2} \sin(2(t-1)) \right).$$

Final Answer:

$$y(t) = \frac{1}{3} \mathcal{U}(t-1) \left(\sin(t-1) - \frac{1}{2} \sin(2(t-1)) \right).$$

Time-shifts, steps, and impulse responses

Laplace transform is extremely useful in resolving problems that feature **piecewise**-defined functions.

The main tool for this is the time-shift rule. More advanced courses will further develop your knowledge of this part, here we will only familiarize ourselves with basic mechanics of time-shifts.

Dirac impulse $\delta(t)$ and Heaviside step $\mathcal{U}(t)$ are two functions¹ that you perhaps haven't seen before. They are featured prominently in engineering analysis of dynamical systems, as they allow an engineer to investigate the dynamics of the system without detailed modeling of the physical components inside (so-called "black-box modeling").

Solutions of ODEs with step/impulse inputs and zero initial conditions are called **step/impulse responses**.

¹ Dirac δ is technically *not* a function, but rather a *distribution* or a *generalized* function.

However, using Laplace transform we can work with it as if it was a function.

Exercise 5.2.1

Define (in words), sketch, and give Laplace transforms of:

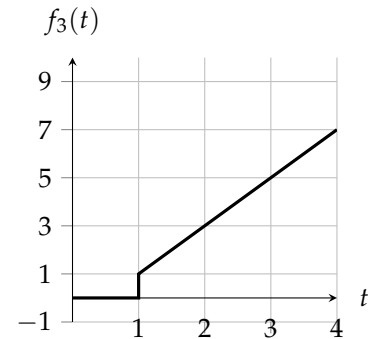
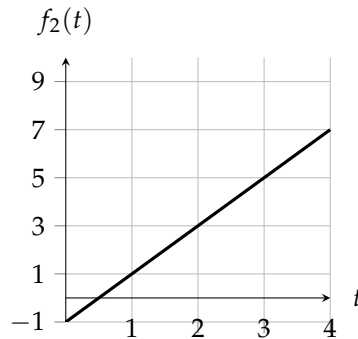
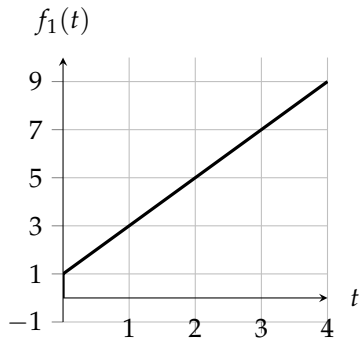
(a) Dirac impulse $\delta(t)$

(b) Heaviside step $\mathcal{U}(t)$

Using derivative rule, compute $\frac{d}{dt} \mathcal{U}(t)$, assuming $\mathcal{U}(0) = 0$. Does the answer make intuitive sense?

Exercise 5.2.2

Use the table of Laplace transforms and the shift theorem (if appropriate) to convert the following functions into Laplace domain.

**Exercise 5.2.3**

Sketch the function $g(t)$ and then use the Laplace transform to solve the IVP $y'(t) + 2y(t) = g(t)$, $y(0) = 1$ in the following two cases:

(a) $g(t) = t - 2$

(b) $g(t) = (t - 2)\mathcal{U}(t - 2)$

Exercise 5.2.4

Solve the IVP and **sketch** the solution of

$$y''(t) + 4y(t) = g(t), \quad y(0) = y'(0) = 0$$

for

(a) $g(t) = \delta(t)$

(c) $g(t) = \delta(t - 1)$

(b) $g(t) = \mathcal{U}(t)$

(d) $g(t) = \mathcal{U}(t - 3)$

Exercise 5.2.5

Convert the following functions from Laplace to time domain or vice versa, as appropriate. Check your work using <https://goo.gl/sRsbrC>.

(a) $f(t) = 3e^{-3t} + 2$

(b) $f(t) = \cos(3t) - 2t$

(c) $f(t) = t - 1$

(d) $f(t) = (t - 1)\mathcal{U}(t - 1)$

(e) $f(t) = \begin{cases} 0 & t < 0 \\ 2t - 2 & t \geq 0 \end{cases}$

(f) $F(s) = \frac{2}{(s+5)(s^2+25)}$

(practice the cover-up method)

(g) $F(s) = \frac{s}{s^2+4s+5}$

(h) $F(s) = \frac{s}{s^2+4s+5}e^{-s}$

(i) $F(s) = \frac{1}{(s-2)^3}$

(j) $F(s) = \frac{s}{s^2+4s+3}e^{-s}$

(k) $F(s) = \frac{2s^2+3s+1}{(s-4)(s+5)(s-2)(s+1)s}$ (practice the cover-up method)

Section 5.3: Solving ODEs using Laplace Transform

Exercise 5.3.1

Solve the following IVPs using the Laplace transform:

- $y'' + 4y' + 5y = e^{-2(t-1)}, \quad y(0) = 0, y'(0) = 1$
- $y'' + 4y' + 5y = e^{-2(t-1)} \mathcal{U}(t-1), \quad y(0) = 0, y'(0) = 1$
- $y' + 4y = \mathcal{U}(t-2) - \mathcal{U}(t-3), \quad y(0) = 0$
- $y' - 3y = \delta(t-2), \quad y(0) = 0$
- $y'' + 2y' = \delta(t-1), \quad y(0) = 0, y'(0) = 1$

Exercise 5.3.2

- Find $\mathcal{L}^{-1} \left\{ \frac{3s+7}{s^2+6s+10} \right\}$.
- Find $\mathcal{L}^{-1} \left\{ \frac{5}{(s+2)^2+16} \right\}$.
- Use the second shifting theorem to find $\mathcal{L}^{-1} \left\{ e^{-3s} \frac{1}{s^2} \right\}$.
- Compute $\mathcal{L}^{-1} \left\{ \frac{2s+1}{s^2-4s+13} \right\}$.
- Solve $y' + 2y = e^{-x}, \quad y(0) = 0$.
- Solve $y'' + 9y = 0, \quad y(0) = 2, y'(0) = 3$.
- Solve $y'' - 2y' + 5y = 0, \quad y(0) = 1, y'(0) = 0$.
- Solve $y'' + 2y' + 2y = e^{-x}, \quad y(0) = 0, y'(0) = 1$.
- $y'' + 9y = \mathcal{U}(t-2), \quad y(0) = 0, y'(0) = 0$.
- $y' + 2y = \mathcal{U}(t-1) \cos(t-1), \quad y(0) = 0$.
- $y'' + y = e^t \mathcal{U}(t-3), \quad y(0) = 0, y'(0) = 1$.
- $y' - y = \mathcal{U}(t-4), \quad y(0) = 2$.

SAGEMATH Question 5.3.3

Modify the line 6 in <https://goo.gl/2H2uBF> to visualize $f(t) = e^{-t} \cos(2t)$ and use the outputs to answer the following questions.

- $F(s) =$
- Sketch the graph of absolute value $|F(s)|$.
- Use the 3d and 2d color graph² to determine distinguished values where $|F(s)|$
 Blows up: Goes to zero:
- By hand, calculate the (complex) roots of the denominator of $F(s)$. Roots =
 How do they relate to your answer in previous item?

² If you find these graphs interesting, there are many more in the gallery at <http://www.visual.wegert.com/>.

Exercise 5.3.4

Use the integral Laplace formula ³ to compute $\mathcal{L}\{7t^2\}$ (hint: use repeated integration-by-parts).

³ Laplace transforms:

$$\mathcal{L}\{f\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$\mathcal{L}^{-1}\{F\} = \int_0^{\infty} e^{st} F(s) ds$$

Exercise 5.3.5

Use table conversions⁴ to compute the following Laplace transforms:

⁴ The Laplace table is in the appendix.

(a) $\mathcal{L}\{e^{-3t} + 4 + 2t\}$

(b) $\mathcal{L}\{3\sin(4t)\}$

(c) $\mathcal{L}\{3e^{3t}\sin(4t)\}$

Exercise 5.3.6

Use table conversions⁵ to compute the following inverse Laplace transforms:

⁵ The Laplace table is in the appendix.

(a) $\mathcal{L}^{-1}\left\{\frac{3}{s+2}\right\}$

(b) $\mathcal{L}^{-1}\left\{\frac{2}{s^2+9}\right\}$

(c) $\mathcal{L}^{-1}\left\{\frac{s+5}{s^2+2s+5}\right\}$

Exercise 5.3.7

Solve the following IVPs using the Laplace transform⁶:

⁶ Make sure to use the derivative rule: $\mathcal{L}\{y'(x)\} = s\mathcal{L}\{y(x)\} - y(0)$

(a) $y'(x) + 2y(x) = e^{-3x}, \quad y(0) = 5$

(b) $y''(x) + 2y'(x) + 5y(x) = 0 \quad y(0) = 0, y'(0) = 1$

Exercise 5.3.8

Find the Laplace transforms using the table and rules:

(a) $\mathcal{L}\{5t^4 - 2e^{5t}\}$

(b) $\mathcal{L}\{\cos(5t) + \sin(2t)\}$

Exercise 5.3.9

Find the inverse Laplace transforms using the table and rules:

(a) $\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\}$

(b) $\mathcal{L}^{-1}\left\{\frac{1}{s^2} - \frac{1}{s} + \frac{1}{s+2} - \frac{1}{s-2}\right\}$

(c) $\mathcal{L}^{-1}\left\{\frac{10s}{s^2+16}\right\}$

(d) $\mathcal{L}^{-1}\left\{\frac{2s-4}{(s^2+s)(s+2)}\right\}$

Exercise 5.3.10

Use the Laplace transform to solve the given IVPs:

(a) $y'' + 5y' + 4y = 0, y(0) = 1, y'(0) = 0.$

(b) $y'' + 9y = e^t, y(0) = 0, y'(0) = 0.$

You can use this online Laplace transform calculator <https://goo.gl/sRsbrC> to check your work while studying.

You can also modify the **SAGEMATH** snippet to check your results for inverse Laplace transform: <https://goo.gl/afyBY8>.

Chapter 6: Systems of First Order Linear ODEs

Section 6.1: Introduction

Many problems in science and engineering are naturally modeled as systems of first-order differential equations. Instead of one equation in one unknown, we deal with multiple equations involving multiple unknown functions.

Several connected linear ODEs can concisely be written using matrix algebra. Studying the matrices involved in the system allows us to better understand the solutions to ODEs in qualitative sense (meaning we can sketch their phase portrait) and even write down formulas for their solutions.

In discussion, we will review the Gauss elimination, forming matrix ODEs, and computation of eigenvalues and eigenvectors.¹

¹ You can use <https://goo.gl/YQhNkW> to check your steps while studying.

Working with $N \times N$ linear systems

In principle, the only computational difference in working with matrix ODEs of dimensions larger than 3 is that the formulas for computing 2×2 and 3×3 determinant have to be replaced with the more general *determinant expansion* formula. Other than that, we still get N eigenvalues as roots of N -degree characteristic polynomial.

Since factoring polynomials by formulas becomes more difficult² as we go beyond degree 2.

² Abel's Impossibility Theorem says factoring by formula is impossible for degrees > 5 (quintic).

For this reason, we'll see that matrices that fall under several special structures (symmetric, skew-symmetric) allow us to infer if their eigenvalues are real/complex before computing the eigenvalues. Other classes of matrices (block-triangular) allow us to simplify the characteristic polynomial or even read-off eigenvalues without any additional calculation

While these techniques are essential for $N \times N$ matrices, they can speed up your work with 2×2 and 3×3 matrices.

Exercise 6.1.1

Compute determinants of the two given matrices by the expansion algorithm.³

³ Strive to expand along rows/columns that make the computation as short as possible.

(a)
$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & -1 & 0 & 3 \\ -1 & 2 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix}$$

$$(b) \begin{bmatrix} 3 & 2 & 0 & 1 \\ 0 & -1 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

Exercise 6.1.2

Identify the block structure in the following matrices and use it to compute the eigenvalues.

$$(a) \begin{bmatrix} 3 & 2 & 0 & 1 \\ 0 & -1 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$(b) \begin{bmatrix} -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 \\ \ln(2) & -2 & 1 & 0 & 0 \\ 0 & 0 & \pi & 4 & 1 \\ 0 & 1 & 1 & 0 & 3 \end{bmatrix}$$

Exercise 6.1.3

Compute general solutions of the following matrix ODEs⁴. Use the trigonometric form of solutions, where appropriate.

$$(a) \frac{d}{dt} \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$$

⁴ Use the block-structure of matrices to simplify eigenvalue calculations. Eigenvectors are computed as usual, without shortcuts.

$$(b) \quad \frac{d}{dt} \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} -3 & 1 & 0 \\ -4 & -3 & 0 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$$

Definition 6.1.1. A system of n first-order differential equations is a collection of equations of the form

$$\frac{dx_1}{dt} = f_1(t, x_1, x_2, \dots, x_n), \quad \frac{dx_2}{dt} = f_2(t, x_1, x_2, \dots, x_n), \quad \dots, \quad \frac{dx_n}{dt} = f_n(t, x_1, x_2, \dots, x_n),$$

which we may write compactly as

$$\mathbf{x}'(t) = \mathbf{f}(t, \mathbf{x}(t)),$$

where $\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$.

Definition 6.1.2. A system is called **linear** if it can be written as

$$\mathbf{x}'(t) = A(t)\mathbf{x}(t) + \mathbf{g}(t),$$

where $A(t)$ is an $n \times n$ matrix (possibly depending on t) and $\mathbf{g}(t)$ is a vector function of t .

If $\mathbf{g}(t) \equiv \mathbf{0}$, the system is **homogeneous**. Otherwise, it is **nonhomogeneous**.

Theorem 6.1.3 (Existence and Uniqueness). Suppose that each $f_i(t, x_1, \dots, x_n)$ is continuous in a region containing $(t_0, \mathbf{x}_0)$ and satisfies a Lipschitz condition in $\mathbf{x}$. Then the IVP

$$\mathbf{x}'(t) = \mathbf{f}(t, \mathbf{x}(t)), \quad \mathbf{x}(t_0) = \mathbf{x}_0,$$

has a unique solution defined in some interval containing t_0 .

Matrix Form and Solutions

For linear systems with constant coefficients:

$$\mathbf{x}'(t) = A\mathbf{x}(t),$$

where A is an $n \times n$ constant matrix, the general solution is expressed in terms of the **matrix exponential**:

$$\mathbf{x}(t) = e^{At}\mathbf{c}, \quad \mathbf{c} = \mathbf{x}(0).$$

Definition 6.1.4. The **matrix exponential** is defined by the power series

$$e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \dots$$

Methods of Solution

1. **Eigenvalue-Eigenvector Method:** For constant coefficient systems, find eigenvalues and eigenvectors of A to construct solutions of the form $\mathbf{x}(t) = e^{\lambda t}\mathbf{v}$.
2. **Fundamental Matrix:** Construct a matrix $\Phi(t)$ whose columns are linearly independent solutions. The general solution is

$$\mathbf{x}(t) = \Phi(t)\mathbf{c}.$$

3. **Variation of Parameters:** For nonhomogeneous systems

$$\mathbf{x}'(t) = A\mathbf{x}(t) + \mathbf{g}(t),$$

one can use

$$\mathbf{x}(t) = \Phi(t) \int \Phi(t)^{-1} \mathbf{g}(t) dt.$$

Applications

- (a) Population models (predator-prey systems).
- (b) Electrical circuits with multiple components.
- (c) Coupled spring-mass systems.
- (d) Chemical reaction dynamics.

Section 6.2: Homogeneous Linear Systems of Differential Equations

Definition 6.2.1. A system of n first-order differential equations is called **homogeneous linear with constant coefficients** if it can be written in matrix form as

$$\mathbf{x}'(t) = A\mathbf{x}(t),$$

where A is an $n \times n$ constant matrix and $\mathbf{x}(t)$ is an n -vector of unknown functions.

General Solution The solution of such systems is based on the **matrix exponential**:

$$\mathbf{x}(t) = e^{At}\mathbf{c},$$

where $\mathbf{c}$ is a constant vector determined by initial conditions.

Method of Eigenvalues and Eigenvectors

1. Find the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ of A by solving

$$\det(A - \lambda I) = 0.$$

2. For each eigenvalue λ , find the corresponding eigenvector $\mathbf{v}$.
3. Each pair $(\lambda, \mathbf{v})$ gives a solution of the form

$$\mathbf{x}(t) = e^{\lambda t}\mathbf{v}.$$

4. If the eigenvalues are distinct, the general solution is a linear combination of these solutions.

Cases

- **Distinct Real Eigenvalues:** If A has n distinct real eigenvalues $\lambda_1, \dots, \lambda_n$ with eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$, then

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + \dots + c_n e^{\lambda_n t} \mathbf{v}_n.$$

- **Complex Eigenvalues:** If $\lambda = \alpha \pm i\beta$ with eigenvector $\mathbf{v} = \mathbf{p} + i\mathbf{q}$, then real solutions can be expressed as

$$e^{\alpha t} (c_1 (\mathbf{p} \cos \beta t - \mathbf{q} \sin \beta t) + c_2 (\mathbf{p} \sin \beta t + \mathbf{q} \cos \beta t)).$$

- **Repeated Eigenvalues:** If an eigenvalue has algebraic multiplicity greater than its geometric multiplicity, generalized eigenvectors are used. In this case, solutions include terms of the form

$$t^k e^{\lambda t} \mathbf{v},$$

where k depends on the size of the Jordan block.

Fundamental Matrix A **fundamental matrix** $\Phi(t)$ is an $n \times n$ matrix whose columns are linearly independent solutions of the system.

$$\mathbf{x}(t) = \Phi(t)\mathbf{c}, \quad \Phi(0) = I.$$

Phase Plane Analysis (2-D Systems)

Phase portraits for 2d linear systems can be sketched with the help of eigenvalues and eigenvectors of the system matrices. Additionally, general solutions for these systems are found again using eigenvalues and eigenvectors.

For a 2×2 system

$$\mathbf{x}'(t) = A\mathbf{x}(t), \quad A \in \mathbb{R}^{2 \times 2},$$

the qualitative behavior of solutions can be classified by the eigenvalues of A :

- Real, distinct, same sign $\implies$ node (stable if negative, unstable if positive).
- Real, distinct, opposite signs $\implies$ saddle point.
- Real, repeated $\implies$ proper or improper node.
- Complex with negative real part $\implies$ stable spiral.
- Complex with positive real part $\implies$ unstable spiral.
- Purely imaginary $\implies$ center.
- The superposition principle holds: linear combinations of solutions are again solutions.
- Homogeneous linear systems always have the trivial solution $\mathbf{x}(t) \equiv \mathbf{0}$.
- Initial conditions $\mathbf{x}(t_0) = \mathbf{x}_0$ uniquely determine the constants in the general solution.

Section 6.3: Nonhomogeneous Linear Systems of Differential Equations

Definition 6.3.1. A system of n first-order differential equations is called *nonhomogeneous linear with constant coefficients* if it can be written in matrix form as

$$\mathbf{x}'(t) = A\mathbf{x}(t) + \mathbf{f}(t),$$

where A is an $n \times n$ constant matrix, $\mathbf{x}(t)$ is an n -vector of unknown functions, and $\mathbf{f}(t)$ is a given non-zero vector function of t .

General Solution The general solution of a nonhomogeneous linear system is the sum of:

$$\mathbf{x}(t) = \mathbf{x}_h(t) + \mathbf{x}_p(t),$$

where

- $\mathbf{x}_h(t)$ is the solution to the associated homogeneous system $\mathbf{x}' = A\mathbf{x}$,
- $\mathbf{x}_p(t)$ is a particular solution of the nonhomogeneous system.

Methods for Finding $\mathbf{x}_p(t)$

1. **Method of Undetermined Coefficients:** Applicable when $\mathbf{f}(t)$ is a linear combination of functions whose derivatives are of the same type (e.g., polynomials, exponentials, sines, cosines). Assume a trial solution of the same form as $\mathbf{f}(t)$, and solve for unknown coefficients.

2. **Variation of Parameters:** General method applicable for any $\mathbf{f}(t)$. If $\Phi(t)$ is a fundamental matrix of the homogeneous system, then a particular solution is

$$\mathbf{x}_p(t) = \Phi(t) \int \Phi^{-1}(t) \mathbf{f}(t) dt.$$

Example of General Solution Structure

For a 2×2 system:

$$\begin{cases} x_1' = 3x_1 + 4x_2 + e^{2t}, \\ x_2' = -x_1 + x_2 + \sin t, \end{cases}$$

- Solve homogeneous system: $\mathbf{x}_h' = A\mathbf{x}_h$, $A = \begin{bmatrix} 3 & 4 \\ -1 & 1 \end{bmatrix}$.
- Find particular solution $\mathbf{x}_p(t)$ using undetermined coefficients or variation of parameters.
- General solution: $\mathbf{x}(t) = \mathbf{x}_h(t) + \mathbf{x}_p(t)$.
- The superposition principle does **not** apply to nonhomogeneous solutions alone; it applies only to the homogeneous part.
- The particular solution $\mathbf{x}_p(t)$ can be chosen in many ways; any choice that satisfies $\mathbf{x}_p' = A\mathbf{x}_p + \mathbf{f}(t)$ is valid.
- Initial conditions $\mathbf{x}(t_0) = \mathbf{x}_0$ determine the constants in the homogeneous solution.

Section 6.4: Matrix Exponential

Definition 6.4.1. For a square matrix $A \in \mathbb{R}^{n \times n}$, the **matrix exponential** e^{At} is defined by the power series

$$e^{At} = \sum_{k=0}^{\infty} \frac{(At)^k}{k!} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \cdots,$$

where I is the $n \times n$ identity matrix.

Properties

1. $e^0 = I$.
2. $\frac{d}{dt}e^{At} = Ae^{At} = e^{At}A$.
3. If A and B commute ($AB = BA$), then $e^{A+B} = e^A e^B$.
4. $e^{At}e^{A(-t)} = I$ (inverse property).
5. If A is diagonalizable, $A = PDP^{-1}$, then

$$e^{At} = Pe^{Dt}P^{-1}, \quad e^{Dt} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t}).$$

Matrix Exponential and Linear Systems For the homogeneous linear system

$$\mathbf{x}'(t) = A\mathbf{x}(t), \quad \mathbf{x}(0) = \mathbf{x}_0,$$

the solution can be written as

$$\mathbf{x}(t) = e^{At} \mathbf{x}_0.$$

Computing e^{At}

1. **Diagonalizable** A : $A = PDP^{-1} \implies e^{At} = Pe^{Dt}P^{-1}$.
2. **Jordan Form**: If A is not diagonalizable, find its Jordan canonical form $A = PJP^{-1}$, then

$$e^{At} = Pe^{Jt}P^{-1}, \quad e^{Jt} \text{ computed block-wise.}$$
3. **Using Laplace Transform**: If $\mathbf{x}' = A\mathbf{x}$, taking Laplace transforms gives

$$sX(s) - \mathbf{x}_0 = AX(s) \implies X(s) = (sI - A)^{-1} \mathbf{x}_0.$$

Inverting gives $\mathbf{x}(t) = e^{At} \mathbf{x}_0$.

Example 6.4.2. Solve $\mathbf{x}' = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \mathbf{x}$, $\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Solution: The matrix is upper-triangular, so

$$e^{At} = \begin{bmatrix} e^{2t} & \int_0^t e^{2(t-\tau)} 1 \cdot e^{3\tau} d\tau \\ 0 & e^{3t} \end{bmatrix} = \begin{bmatrix} e^{2t} & e^{3t} - e^{2t} \\ 0 & e^{3t} \end{bmatrix}.$$

Then

$$\mathbf{x}(t) = e^{At} \mathbf{x}_0 = \begin{bmatrix} e^{2t} \\ 0 \end{bmatrix} + \begin{bmatrix} e^{3t} - e^{2t} \\ e^{3t} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} e^{2t} \\ 0 \end{bmatrix}.$$

Problems Involving Matrices, matrix ODEs, eigenvalues, and eigenvectors

A second-order differential equation

$$y'' + p(t)y' + q(t)y = g(t)$$

can be rewritten as a system of two first-order equations by introducing a new variable

$$x_1 = y, \quad x_2 = y'.$$

Then,

$$x_1' = x_2, \quad x_2' = -q(t)x_1 - p(t)x_2 + g(t).$$

This system can be written in vector form:

$$\mathbf{x}' = A(t)\mathbf{x} + \mathbf{f}(t), \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Example 6.4.3. Convert the IVP

$$y'' - 3y' + 2y = 0, \quad y(0) = 1, \quad y'(0) = 0$$

into a first-order system.

Solution:

$$\begin{aligned}
 x_1 &= y, & x_2 &= y', \\
 x_1' &= x_2, & x_2' &= 3x_2 - 2x_1, \\
 \text{System form: } \mathbf{x}' &= \begin{bmatrix} 0 & 1 \\ -2 & 3 \end{bmatrix} \mathbf{x}, & \mathbf{x}(0) &= \begin{bmatrix} 1 \\ 0 \end{bmatrix}.
 \end{aligned}$$

Example 6.4.4. Convert the IVP

$$y'' + 4y = \sin t, \quad y(0) = 0, \quad y'(0) = 1$$

into a first-order system.

Solution:

$$\begin{aligned}
 x_1 &= y, & x_2 &= y', \\
 x_1' &= x_2, & x_2' &= -4x_1 + \sin t, \\
 \text{System form: } \mathbf{x}' &= \begin{bmatrix} 0 & 1 \\ -4 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ \sin t \end{bmatrix}, & \mathbf{x}(0) &= \begin{bmatrix} 0 \\ 1 \end{bmatrix}.
 \end{aligned}$$

Exercise 6.4.1

Given is the system of algebraic equations

$$\begin{aligned}
 x + 3y + z &= 1 \\
 1 + z + 2y &= 0. \\
 y + z + x &= 1
 \end{aligned}$$

- (a) Convert the system into a matrix equation $A\mathbf{v} = \mathbf{b}$, where $\mathbf{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is the vector of unknowns.

- (b) Without finding solutions, perform a calculation that can determine if this matrix equation has a *unique* solution vector?

- (c) Use Gauss elimination to find the value or a general formula for the solution vector.

Exercise 6.4.2

Given is the system of linear ODEs:

$$\dot{x} = 2y - 3x + 4$$

$$\dot{y} = 3x + z$$

$$\dot{z} = -z$$

- (a) Find the matrix $\mathbf{M}$ and vector $\underline{b}$, such that the matrix differential equation

$$\dot{\underline{v}} = \mathbf{M}\underline{v} + \underline{b} \quad (4.1)$$

is equivalent to the system, using vector of states⁵ $\underline{v}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$.

⁵ *State variable*, or simply *state*, is another term for the *dependent variable*

- (b) If the initial values of some solution⁶ are $[x(0) \ y(0) \ z(0)]^\top = [1 \ -1 \ 2]$, what is the initial rate of change $[\dot{x}(0) \ \dot{y}(0) \ \dot{z}(0)]$ of that solution? Use matrix multiplication to answer this question.

⁶ We use transpose notation to save vertical space, that is $[x(0) \ y(0) \ z(0)]^\top = \begin{bmatrix} x(0) \\ y(0) \\ z(0) \end{bmatrix}$ and similarly elsewhere.

- (c) Find one (or more) fixed points of this system using Gauss elimination.⁷

⁷ Hint: what is the value of $\dot{\underline{v}}$ for equilibrium solutions?

- (d) Verify that the vector $\underline{v}(t) = \begin{bmatrix} e^{-t} \\ -2+e^{-t} \\ -4e^{-t} \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ -4 \end{bmatrix} e^{-t}$ is a solution to the ODE (??), by plugging it into both sides of the ODE.⁸

⁸ To take a derivative of a vector, take derivatives of each of its components.

Exercise 6.4.3

Compute eigenvalues and eigenvectors of the following matrices

(a) $A = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix},$

(b) $A = \begin{bmatrix} -3 & 1 \\ -4 & -3 \end{bmatrix},$

Exercise 6.4.4

If matrix A is 3×4 and B is 2×3 , determine whether AB and BA matrices can be formed. If one can form the product, determine its dimension.

Exercise 6.4.5

(a) Form either AB or BA (whichever is possible) for matrices

$$A = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & -2 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}$$

(b) Given vectors $v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $w = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ find the matrix products $v^\top w$ and $vw^\top$ where $\cdot^\top$ operation denotes the matrix transpose.

Exercise 6.4.6

For each of the following linear systems:

- (i) Rewrite the system in matrix form.
- (ii) Use the determinant calculation to check if the system has a single (unique) solution.
- (iii) Use Gauss elimination to find the general form of the solution vector.

(a) $2x + 6y + z = 7$ $-x - 2y + z = 1$ $5x + 7y - 4z = 9$	(b) $2z + y + x = 0$ $2 + x - y = 0$ $1 + y + z = 0$
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Exercise 6.4.7

Find all fixed points of the system of ODEs using Gauss elimination:

$$\begin{aligned} \dot{x} &= 3x + y + z - 5 \\ \dot{y} &= -6x - 2y + 5z + 3 \\ \dot{z} &= 9x + 3y - 11z - 1 \end{aligned}$$

Exercise 6.4.8

Use **eigenvector equation** and matrix multiplication to find which of the three vectors is an eigenvector of the given matrix (there may be more or one, or there may be none). At the same time, determine the associated eigenvalues.⁹

$$M = \begin{bmatrix} 2 & -3 & 0 \\ 0 & -1 & 0 \\ -1 & 5 & 1 \end{bmatrix}, \quad v_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -7 \\ 0 \\ -7 \end{bmatrix}, \quad v_3 = \begin{bmatrix} -7 \\ 7 \\ 7 \end{bmatrix}.$$

⁹ Note: do not compute the characteristic polynomial. Plug the candidates into the eigenvector equation to see if they satisfy it.

Exercise 6.4.9

Given a system of ODEs

$$\begin{aligned}\frac{dx}{dt} &= x - 3y \\ \frac{dy}{dt} &= -2x + 2y\end{aligned}$$

with $x(0) = 1, y(0) = 4$.

- (i) Convert the system of ODEs into a matrix ODE.
- (ii) Verify that the vectors $\begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{-t}$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{4t}$ are two solutions of the given system.
- (iii) Find the eigenvalues and eigenvectors of the system matrix.

Exercise 6.4.10

1. Convert $y'' + 2y' - 3y = 0$ into a first-order system.
2. Convert $y'' + y' + y = e^t$ into a first-order system.
3. Convert $y'' - 5y' + 6y = \cos t$ with $y(0) = 0, y'(0) = 1$ into a first-order system.
4. Convert $y'' + 9y = 0$ with $y(0) = 2, y'(0) = -1$ into a first-order system.
5. Convert $y'' + 4y' + 4y = t^2$ into a first-order system.

Practice Problems(Phase portraits and general solutions of systems)**Exercise 6.4.11**

Given the system

$$\frac{d}{dt} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ -5 & 4 \end{bmatrix} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

- (a) Find the eigenvalues and the eigenvectors of the system matrix.

- (b) Is the fixed point at the origin stable or not? Do solutions oscillate or not? Explain your answer.

- (c) Find the particular solution that satisfies $x(0) = y(0) = 1$. If eigenvalues are complex, use real-valued formulas (involving sin and cos).

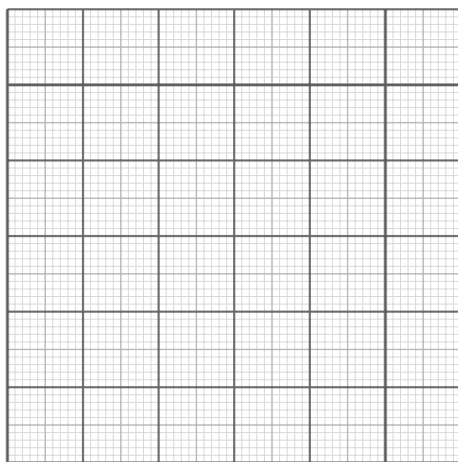
Exercise 6.4.12

Given the system

$$\frac{d}{dt} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

(a) Find eigenvalues and eigenvectors of the system matrix.

(b) Sketch the phase portrait. **Annotate your graph** so it is clear what information you used to sketch it.



(c) Is the fixed point at the origin stable or not? Explain your answer.

(d) Write out the general solution. If eigenvalues are complex, use real-valued solution form.

Exercise 6.4.13

1. Solve the system $\mathbf{x}' = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \mathbf{x}$ with initial condition $\mathbf{x}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ using the matrix exponential.
2. Compute e^{At} for $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$ and verify that it satisfies the differential equation $\mathbf{x}' = A\mathbf{x}$.
3. Solve $\mathbf{x}' = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} \mathbf{x}$ with $\mathbf{x}(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ using diagonalization, if possible.
4. Find e^{At} for the Jordan block $A = \begin{bmatrix} 5 & 1 \\ 0 & 5 \end{bmatrix}$ and use it to solve $\mathbf{x}' = A\mathbf{x}$, $\mathbf{x}(0) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$.
5. Solve the nonhomogeneous system $\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} \sin t \\ \cos t \end{bmatrix}$ with $\mathbf{x}(0) = \mathbf{0}$ using the variation of constants formula with the matrix exponential.
6. Verify that $e^{At}e^{A(-t)} = I$ for $A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$.
7. For the system $\mathbf{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \mathbf{x}$, find e^{At} using the eigenvalue method and solve for $\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

Exercise 6.4.14

For each system

$$\frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = A \begin{bmatrix} x \\ y \end{bmatrix}$$

given by matrices below,

- classify the type and the stability of the equilibrium at the origin,
- sketch a 2-D phase portrait using eigenvalues and eigenvectors, and
- write out the general solution.

$$(a) \quad A = \begin{bmatrix} 5 & 3 \\ -1 & 1 \end{bmatrix},$$

$$(b) \quad A = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix},$$

$$(c) \quad A = \begin{bmatrix} -3 & 1 \\ -4 & -3 \end{bmatrix},$$

$$(d) \quad A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

Exercise 6.4.15

Given is the second order differential equation

$$2\ddot{y}(t) + c\dot{y}(t) + 8y(t) = 0,$$

where c is a parameter.

- Write the ODE as the first-order system of differential equations.¹⁰
- Compute the eigenvalues of the obtained system (they will depend on c).
- Find the range of values of c that makes the equilibrium at origin *stable* and that results in oscillating trajectories (spiral phase portrait).

¹⁰ Introduce variables $x_1 = y$ and $x_2 = \dot{y}$.

Exercise 6.4.16

1. Solve the system

$$\begin{cases} x_1' = 3x_1 + 4x_2, \\ x_2' = -4x_1 + 3x_2, \end{cases}$$

with initial condition $\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

2. Solve

$$\mathbf{x}' = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

using eigenvalues and eigenvectors.

3. Solve the system

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

and find the general solution.

4. Solve

$$\mathbf{x}' = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix},$$

using the **matrix exponential** or **Jordan form**.

5. Find the general solution of the system

$$x_1' = 2x_1 + 3x_2, \quad x_2' = -x_1 + 4x_2,$$

and classify the type of equilibrium at the origin.

6. Solve the nonhomogeneous system

$$\mathbf{x}' = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} e^t \\ t \end{bmatrix}, \quad \mathbf{x}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

using the **variation of constants formula**.

7. Solve

$$x' = \begin{bmatrix} 4 & 0 \\ 1 & 2 \end{bmatrix} x,$$

and discuss whether the solution grows, decays, or oscillates.

8. Solve the system

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{bmatrix} 2 \\ -1 \end{bmatrix},$$

and interpret the solution geometrically.

9. Find the general solution of

$$\mathbf{x}' = \begin{bmatrix} 3 & 1 \\ -4 & -1 \end{bmatrix} \mathbf{x}.$$

10. Solve

$$\mathbf{x}' = \begin{bmatrix} 1 & 1 \\ -2 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} \cos t \\ \sin t \end{bmatrix}, \quad \mathbf{x}(0) = \mathbf{0},$$

using the **matrix exponential**.